

UMC 0.11 um Logic and Mixed-Mode AI Advanced Enhancement 1.2V/3.3V Diode SPICE Model Document

**Version 0.8 Phase 1
2017/08/07**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.1_p1	2010/03/04	Ho Cheng Hu	Original release New DSM version definition from 2009/11/29.
0.2_p1	2010/07/20	Ho Cheng Hu	Version up to 0.2 based on L110AE Si data.
0.3_p1	2010/11/30	Laney Fan	1. Version up to 0.3 based on HS/IO scalable diode data 2. Add DNW/Psub and PW/DNW diode models.
0.4_p1	2011/09/27	Laney Fan	1. Correct the rs equations for HS n+/pw & p+/nw scalable diode model 2. Add corner parameters for nw/psub diode model
0.5_p1	2012/08/09	Laney Fan	1. Multiply 0.9 of w and l in the subckt for shrink methodology 2. Add newdio=1 in eldo models. 3. Revise ikp as ikr for spectre models
0.6_p1	2012/10/17	Laney Fan	1. Correct from multiply 0.9 by w and l to multiply 0.9*0.9 by area
0.7_p1	2016/11/23	Macro Chang	1. Optimize leakage current and break down voltage
0.8_p1	2017/08/07	Henry JC Chen Macro Chang	1. Fix the format of shrink methodology in diode model. 2. Modify parameters of NW/Psub leakage current for high temperature.

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor AI Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

Table 1-2 DIODE device

DIODE (twin well)	N+ to Pwell	P+ to Nwell	Nwell to PSub	Pwell to Deep Nwell
Model Name	dion_l110e	diop_l110e	dionw_l110e	diopw_l110e
DIODE (triple well)	N+ to Buried Pwell	Buried Pwell to Deep Nwell	Deep Nwell to Psub	
Model Name	dion_33_l110e	diop_33_l110e	diodnw_l110e	

1.3 Model Usage

The compact model is provided in various formats. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2014.09 SP2
Cadence SPECTRE version 15.1.0.644.isr13 64bit
Mentor Graphics ELDO version 15.4

For Synopsys HSPICE

- Copy ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
.lib "ModelFileName.lib" CaseName
, where CaseName could be TT, FF, and SS.
- Define device condition by 'Xxx' statement for sub-circuit format model, or 'Dxx' statement for non sub-circuit format model.
Example1:(for sub-circuit format model)
X1 Va Vc DeviceName AREA=1E-10 PJ=4E-5
Example2:(for non sub-circuit format model)
D1 Va Vc DeviceName AREA=1E-10 PJ=4E-5

For Cadence Spectre

- Copy ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
SetOption1 options scale=0.9
include "ModelFileName.lib.scs" section=CaseName
, where CaseName could be TT, FF, and SS.
- Define device condition by 'Xxx' statement for sub-circuit format model, or 'Dxx' statement for non sub-circuit format model.
x1 (Va Vc) DeviceName area=1e-10 pj=4e-5
Example2:(for non sub-circuit format model)
d1 (Va Vc) DeviceName area=1e-10 pj=4e-5

For Mentor Graphics Eldo

- Copy ModelFileName.lib.eldo into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
.lib "ModelFileName.lib.eldo" CaseName
, where CaseName could be TT, FF, and SS.

- Define device condition by 'Xxx' statement for sub-circuit format model, or 'Dxx' statement for non sub-circuit format model.

Example1:(for sub-circuit format model)

X1 Va Vc DeviceName AREA=1E-10 PJ=4E-5

Example2:(for non sub-circuit format model)

D1 Va Vc DeviceName AREA=1E-10 PJ=4E-5

The available diodes are:

- a. 1.2V N+ to P-Well Diode : dion_l110e
- b. 1.2V P+ to N-Well Diode : diop_l110e
- c. 3.3V N+ to P-Well Diode : dion_33_l110e
- d. 3.3V P+ to N-Well Diode : diop_33_l110e
- e. 1.2V N-Well to P-Sub Diode : dionw_l110e
- f. DNW to P-Sub Diode : diodnw_l110e
- g. PW to DNW Diode : diopw_l110e

ModelFileName: l110ae_diode_v081

Geometry range:

This is a scalable model, which uses actual geometry that AREA and PJ in the unit of meter^2 and meter respectively. The geometry range should follow TLR document.

Geometry range (Before 90% shrink):

HS1.2V DION
0.5um<=WDES<=14.3um
5um<=LDES<=25um

HS1.2V DIOP
0.5um<=WDES<=14.3um
5um<=LDES<=25um

IO 3.3V DION
0.5um<=WDES<=14.3um
5um<=LDES<=25um

IO 3.3V DIOP
0.5um<=WDES<=14.3um
5um<=LDES<=25um

DIONW

Area = $1.77\text{E-}7 \text{ m}^2$

Pj = $4.93\text{E-}3 \text{ m}$

DIODNW

Area = $3.2\text{E-}8 \text{ m}^2$

$3.2\text{E-}4 \text{ m} \leq \text{Pj} \leq 6.4\text{E-}3 \text{ m}$

DIOPW

Area = $3.2\text{E-}8 \text{ m}^2$

Pj = $3.2\text{E-}4 \text{ m}$

Temperature range:

$-50 \text{ }^{\circ}\text{C} \sim +125 \text{ }^{\circ}\text{C}$

2 1.2V DION Diode Model Verification

2.1 Silicon Wafer for Modeling

The information of the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for DION

	Dion diode model
Mask ID	BSW0FA
Lot ID	D566R.1
Wafer Number	#1

This model is characterized as the following geometry and temperature ranges depicted. Device behavior beyond these ranges may not be well described by the model. This model are extracted by devices with these two dimensions :

Junction Area Devices :

Area = $5.0\text{e-}8 \text{ m}^2$

Perimeter = $2.0\text{e-}3 \text{ m}$

Junction Edge Devices :

Area = $3.6\text{e-}8 \text{ m}^2$

Perimeter = $3.64\text{e-}2 \text{ m}$

2.2 Testing Conditions

The model is verified to measurement data under following bias conditions

Forward and reverse diode IV curve:

Table 2-2 Forward and reverse diode IV measurement conditions

Forward Diode

	Start	Stop	Step
$V_J(V)$	0	1	0.05

Reverse Diode

	Start	Stop	Step
$V_J(V)$	0	-15	-0.1

Diode junction CV:

Table 2-3 Diode junction CV measurement conditions

V_{BIAS} [V]	VH: pwell & VL: n+ From 0 to 1.32 V in 0.05 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50, 0, 25, 85, 125

2.3 1.2V DION model verification

2.3.1 1.2V DION model verification at room temperature (25 °C)

➤ *Diode junction CV curve at 25 °C.*

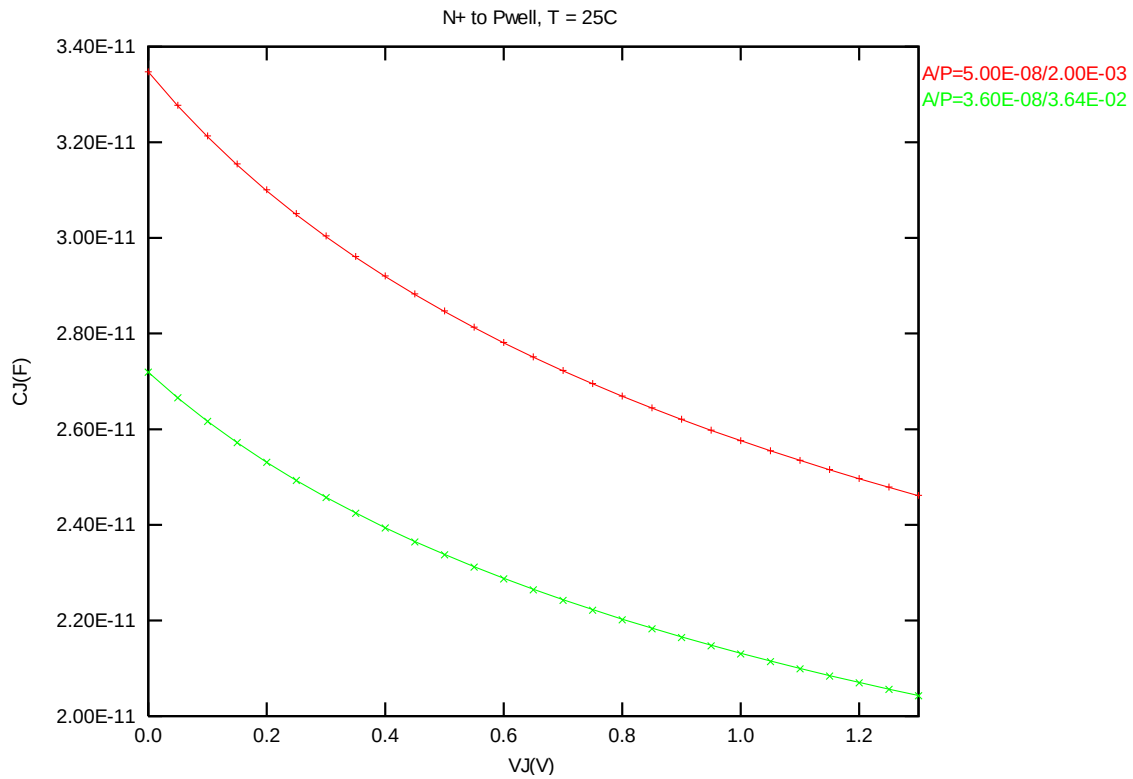


Figure 2-1 Diode junction CV curve at 25 °C

➤ **Forward IV curve at 25 °C.**

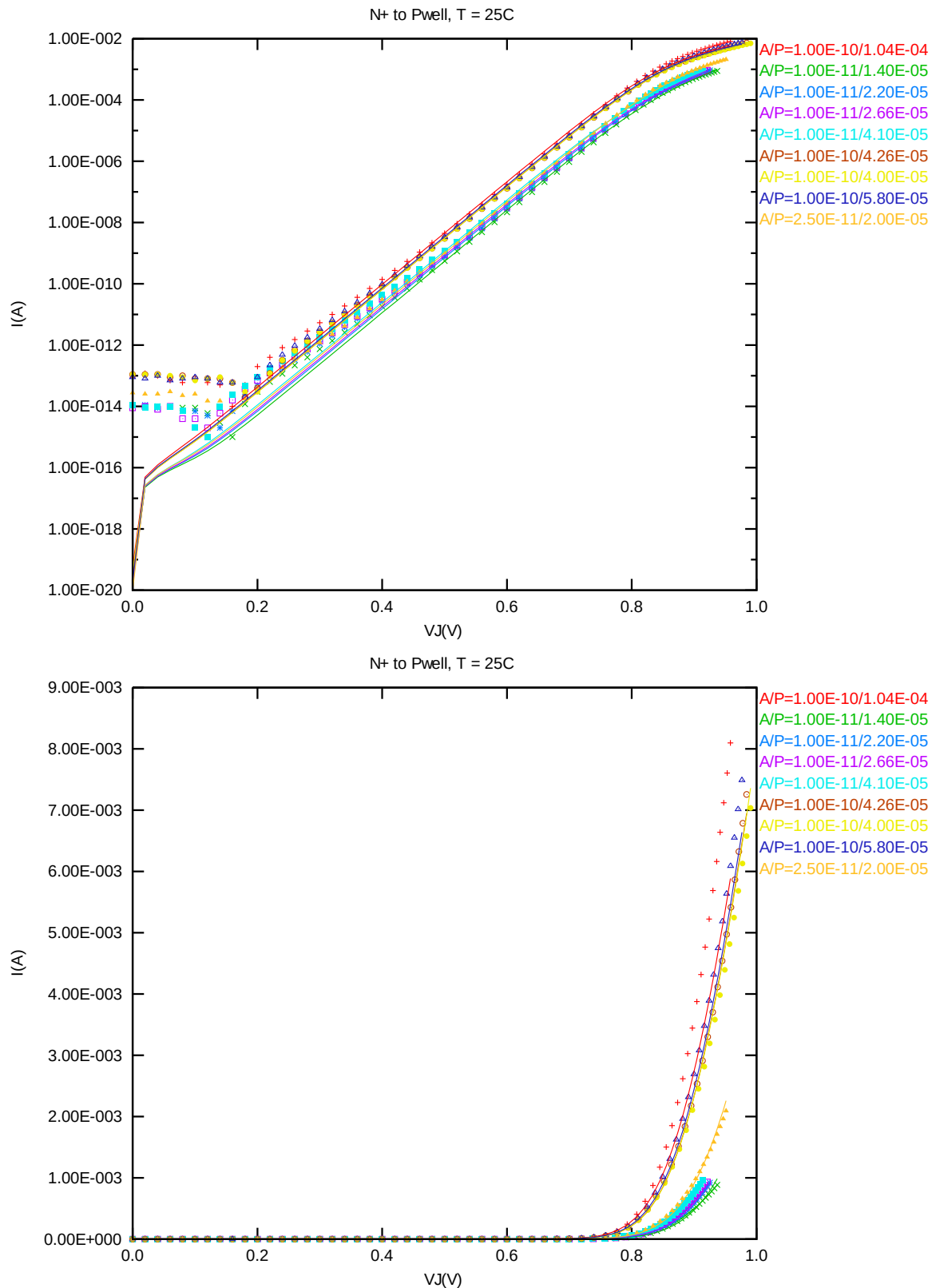


Figure 2-2 Forward IV curve at 25 °C

➤ **Reverse IV curve at 25 °C.**

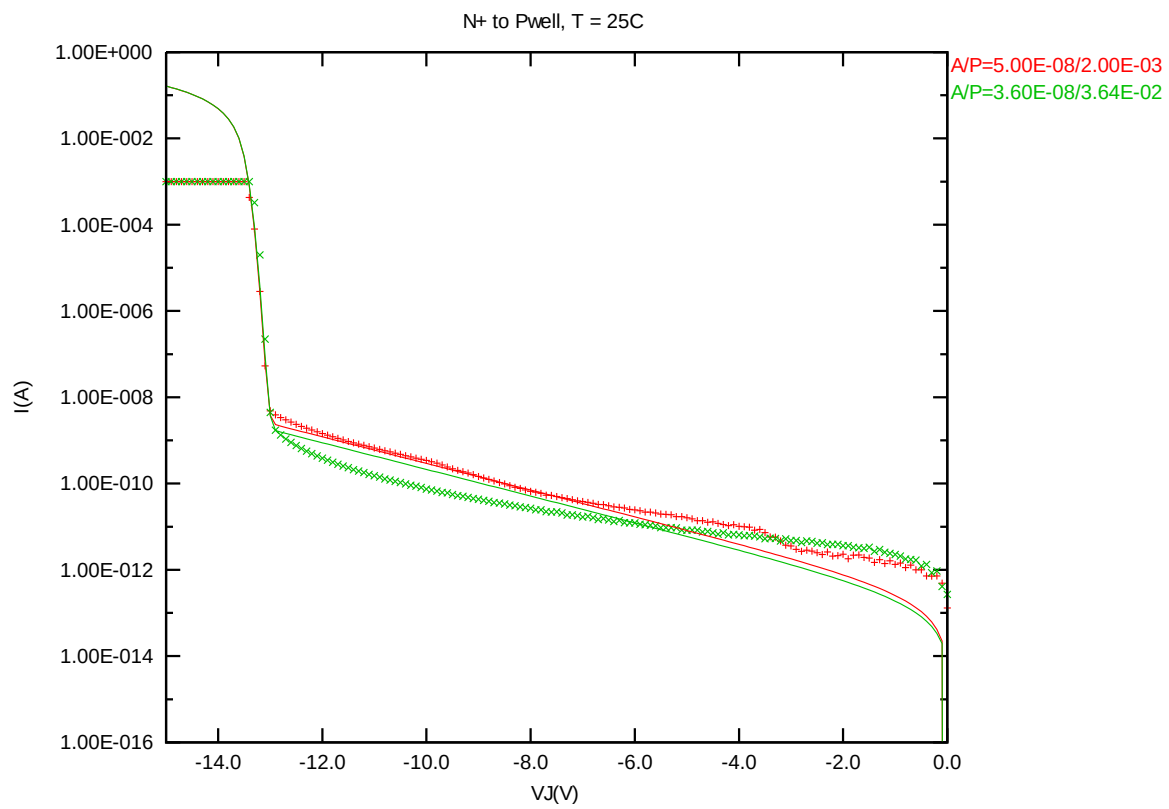


Figure 2-3 Reverse IV curve at 25 °C

2.3.2 1.2V DION model verification at 125 °C

➤ Diode junction CV curve at 125 °C.

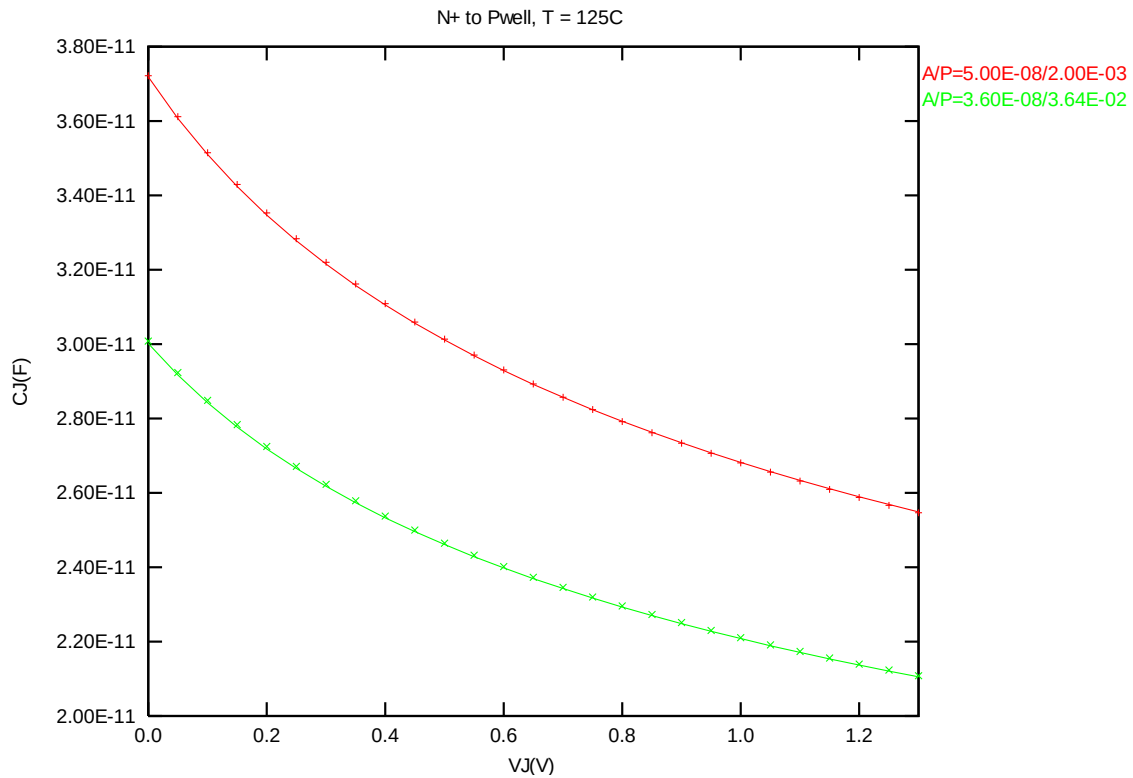
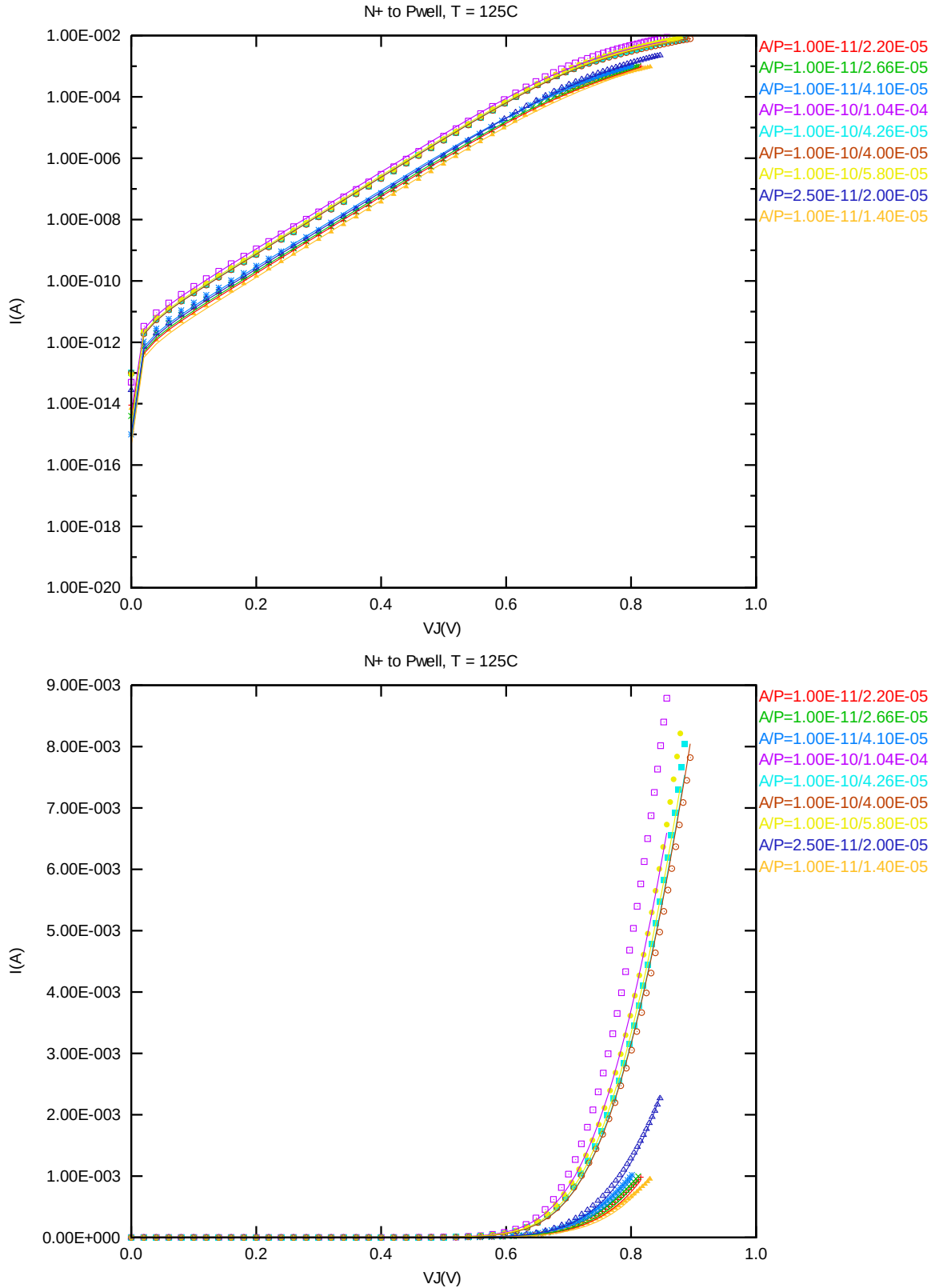


Figure 2-4 Diode junction CV curve at 125 °C

➤ **Forward IV curve at 125 °C.**



➤ **Reverse IV curve at 125 °C.**

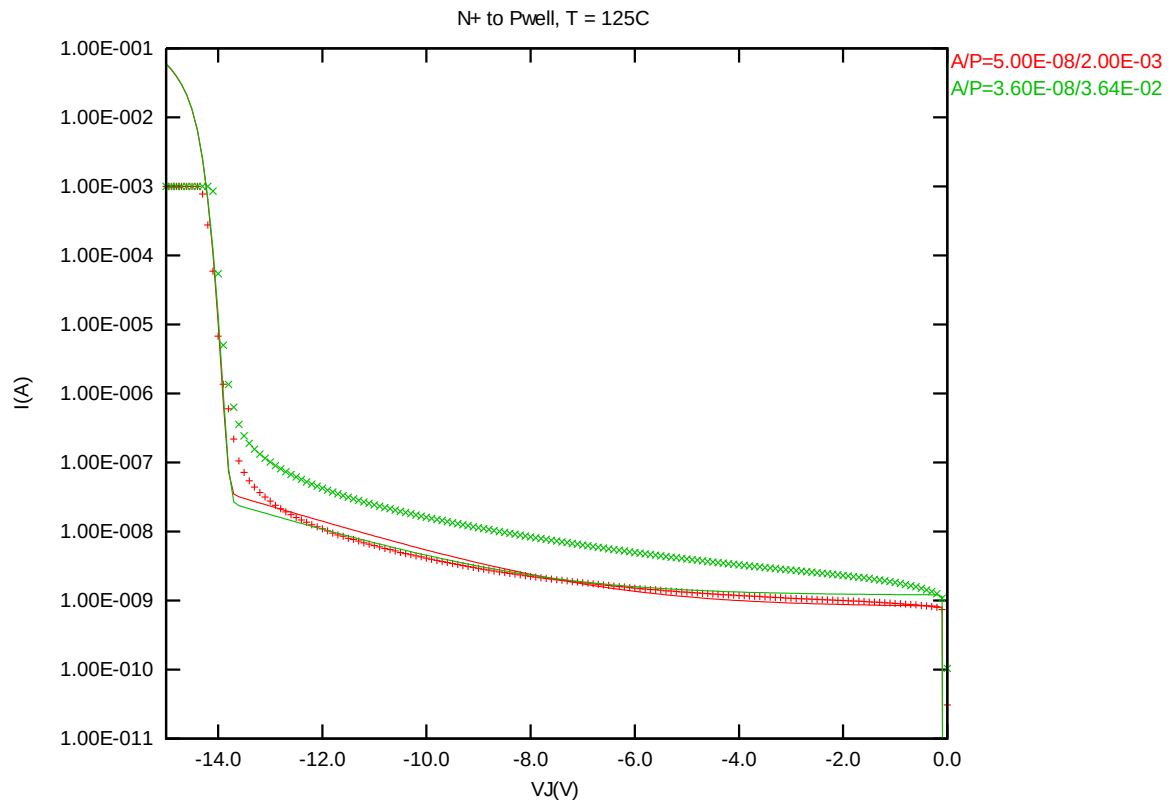


Figure 2-6 Reverse IV curve at 125 °C

2.3.3 1.2V DION model verification at 85 °C

➤ *Diode junction CV curve at 85 °C.*

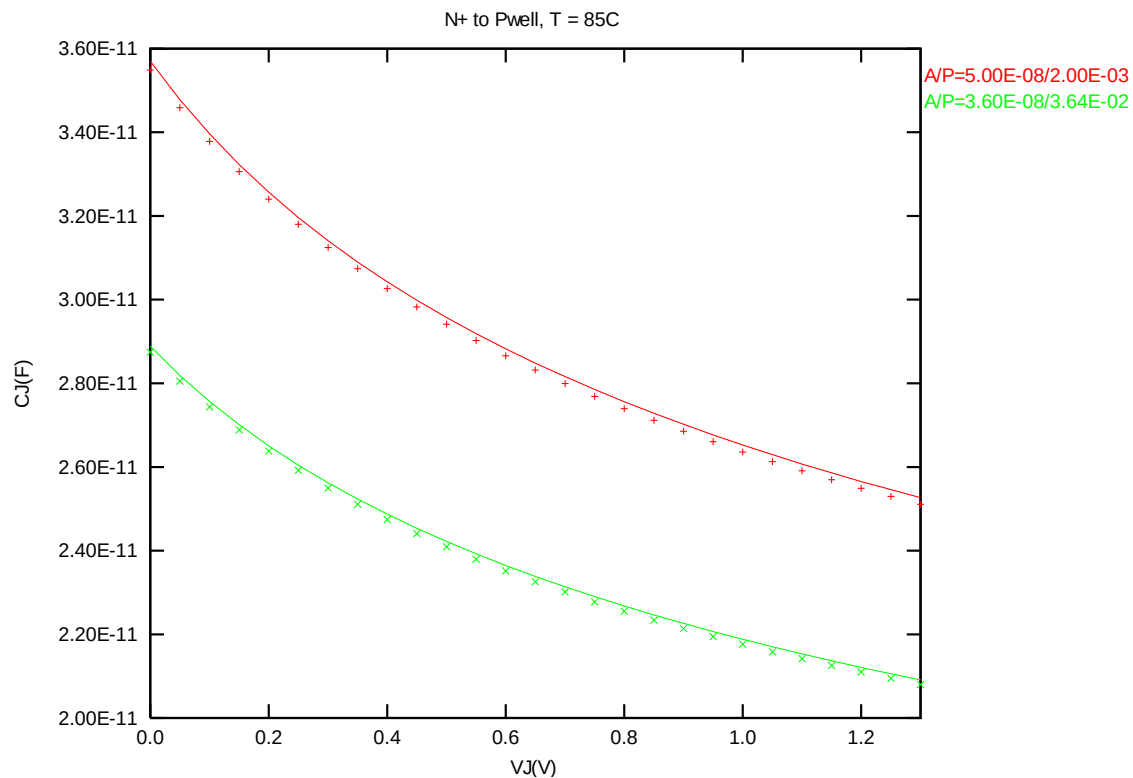


Figure 2-7 Diode junction CV curve at 85 °C

2.3.4 1.2V DION model verification at 0 °C

➤ *Diode junction CV curve at 0 °C.*

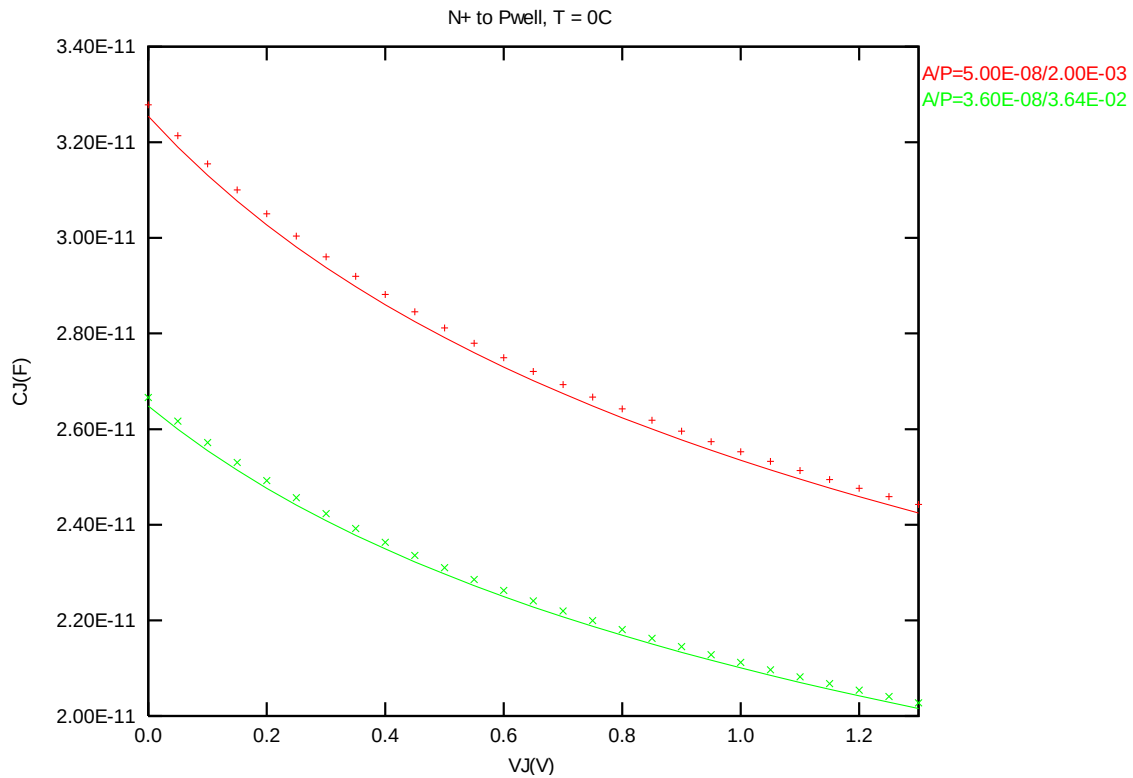


Figure 2-8 Diode junction CV curve at 0 °C

2.3.5 1.2V DION model verification at -50 °C

➤ *Diode junction CV curve at -50 °C.*

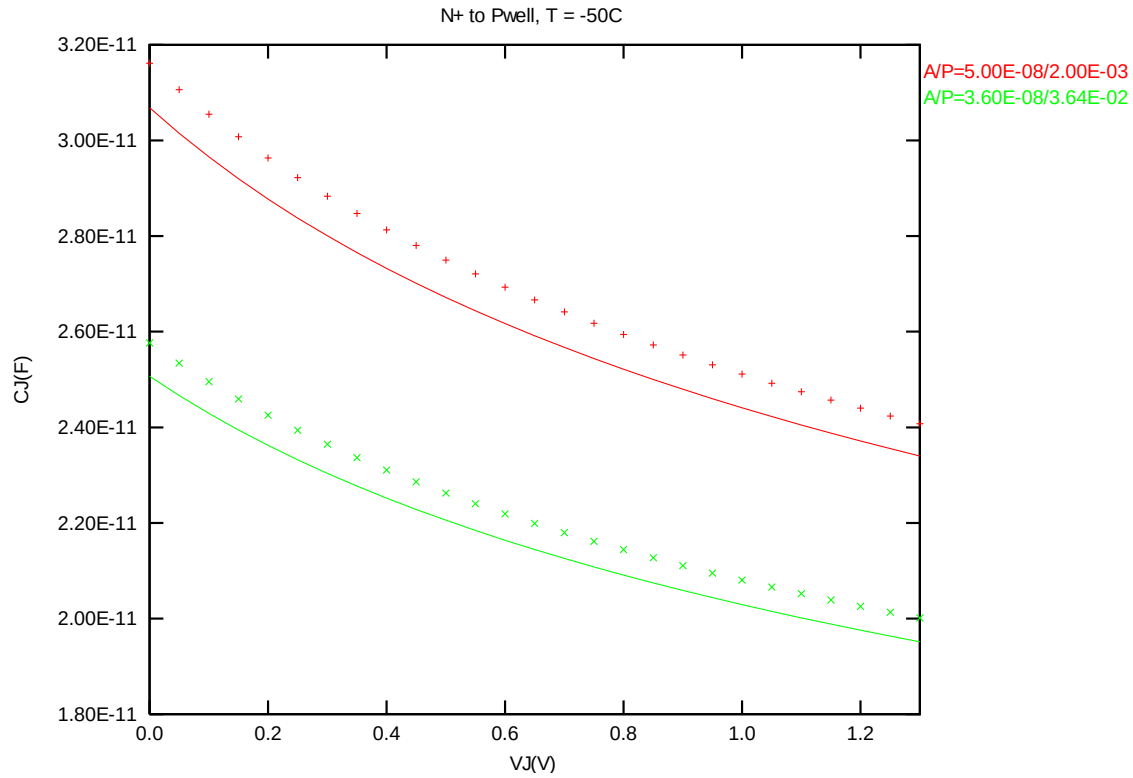


Figure 2-9 Diode junction CV curve at -50 °C

➤ **Forward IV curve at -50 °C.**

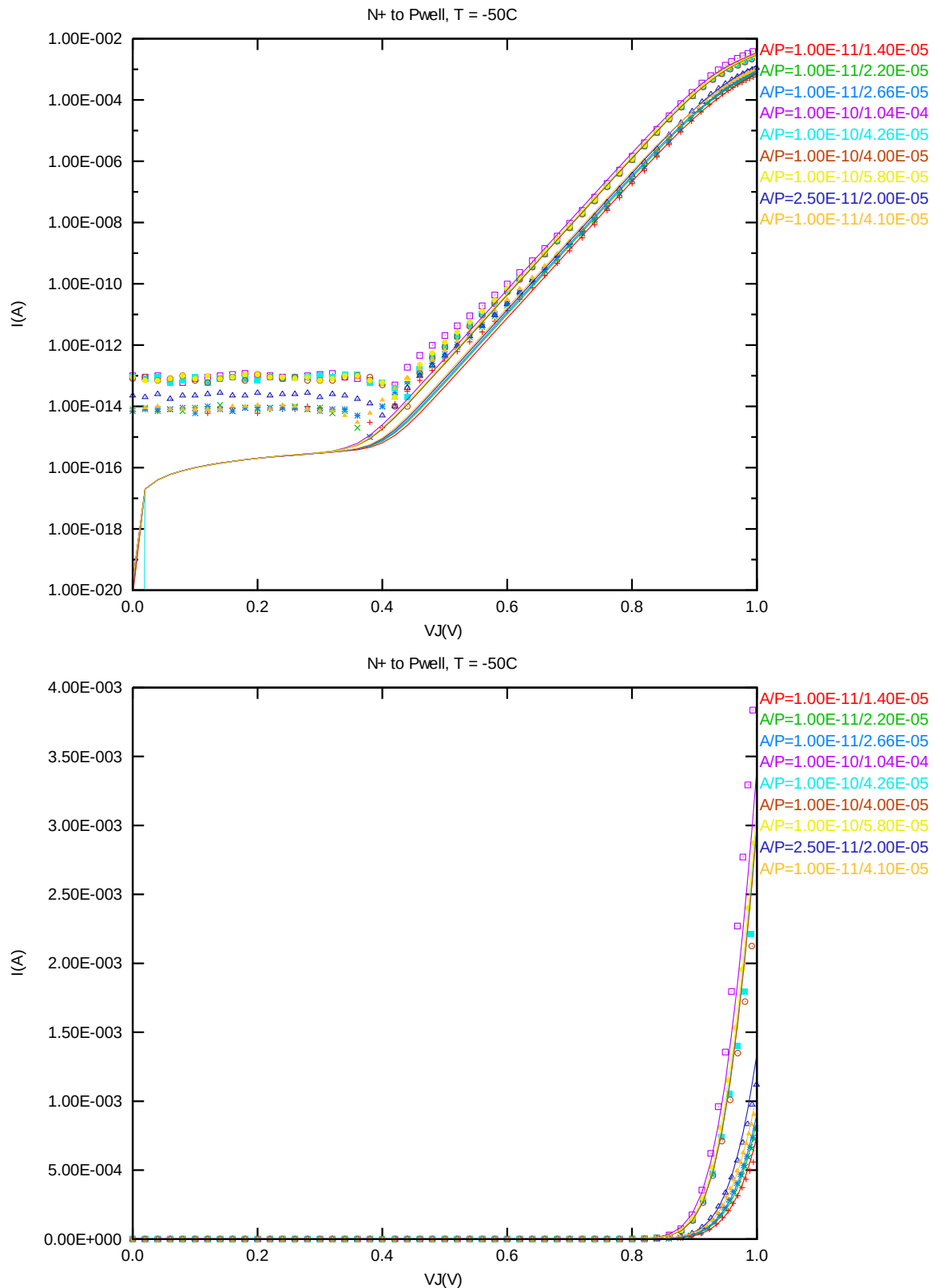


Figure 2-10 Forward IV curve at -50 °C

➤ **Reverse IV curve at -50 °C.**

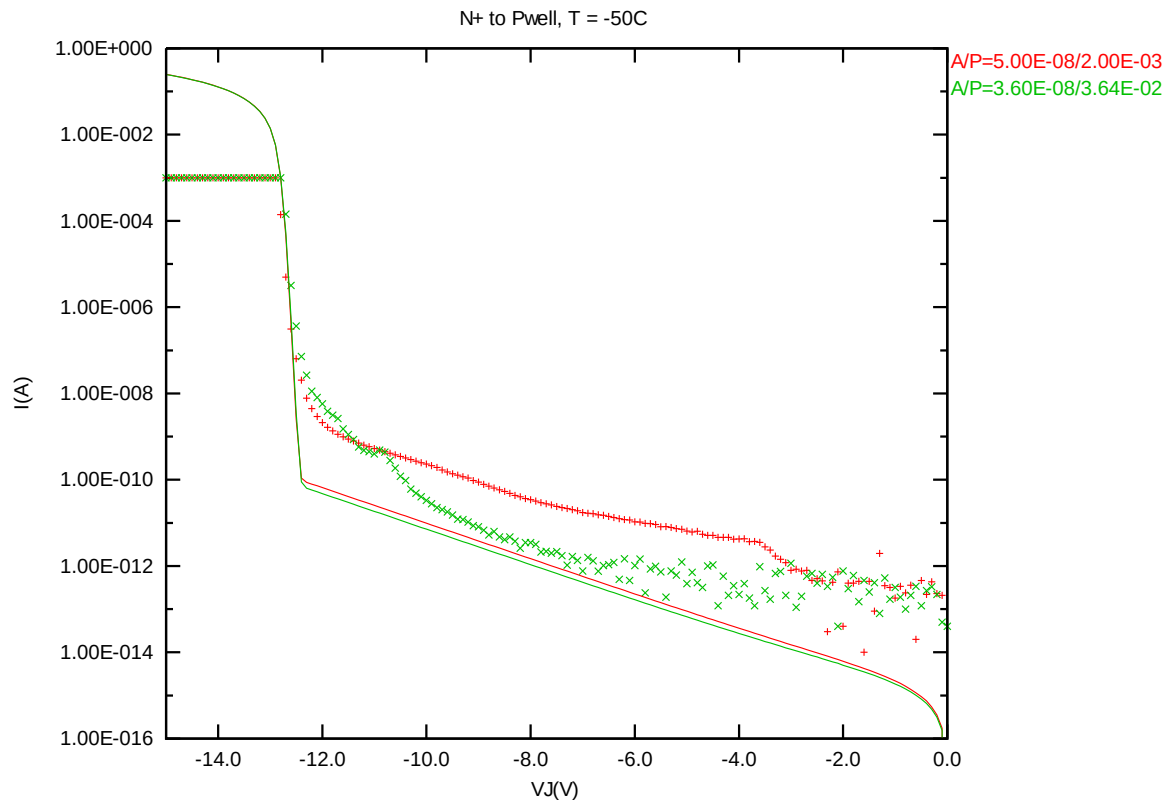


Figure 2-11 Reverse IV curve at -50 °C

2.3.6 1.2V DION model verification for temperature effect

➤ *Diode junction CV curve at 25 °C, 125 °C , 80 °C, 0 °C, -50 °C.*

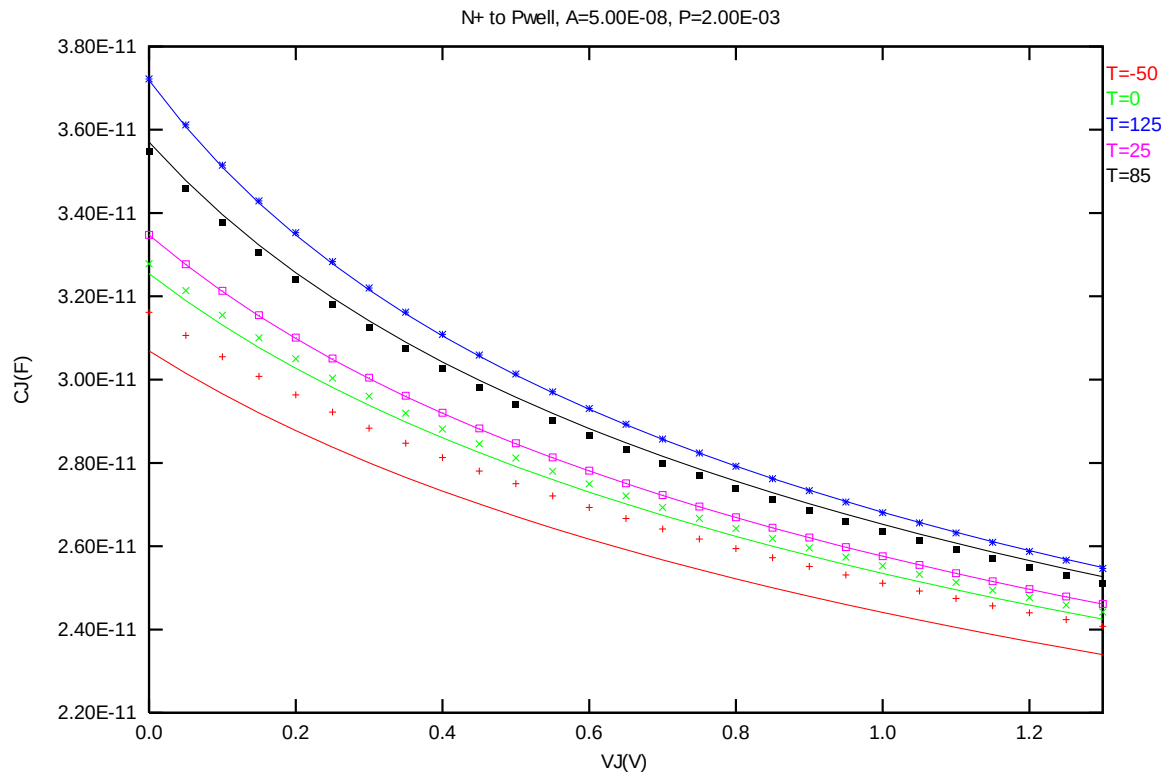


Figure 2-12 Diode junction CV curve at 25 °C, 125 °C , 80 °C, 0 °C, -50 °C.

➤ **Forward IV curve at 25 °C, 125 °C, -50 °C.**

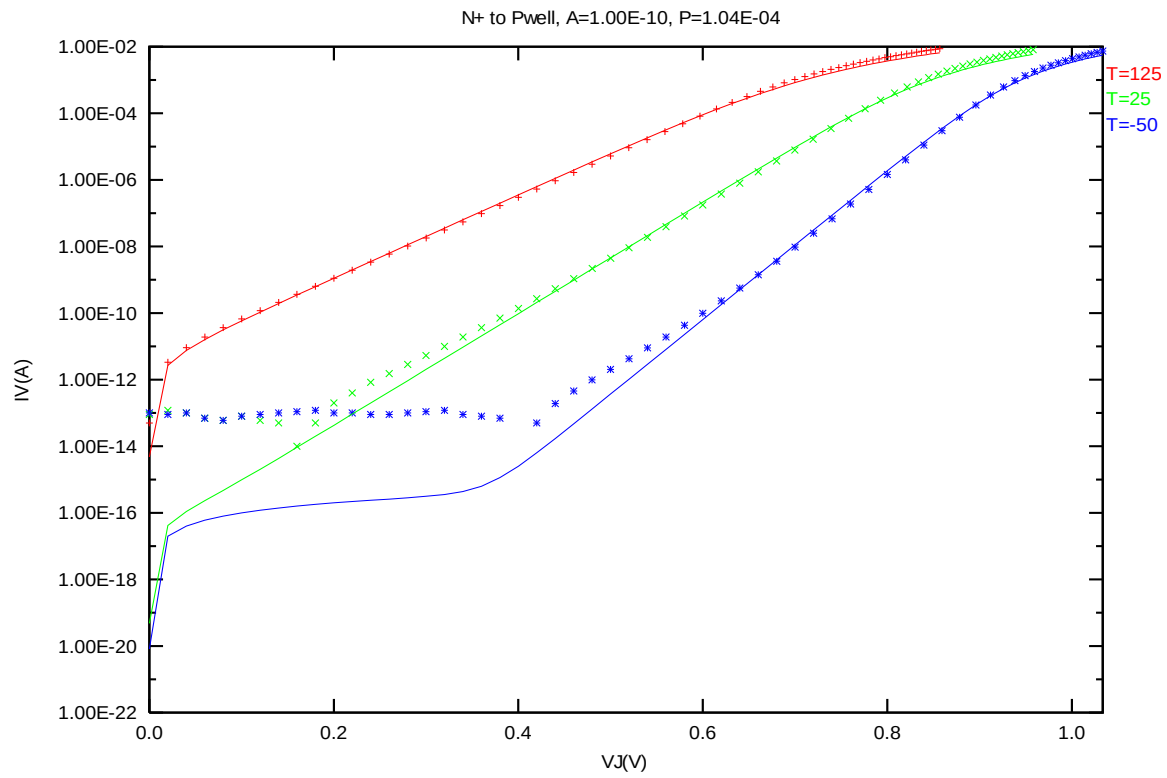


Figure 2-13 Forward IV curve at 25 °C, 125 °C, -50 °C.

➤ **Reverse IV curve at 25 °C, 125 °C, 80 °C, 0 °C, -50 °C.**

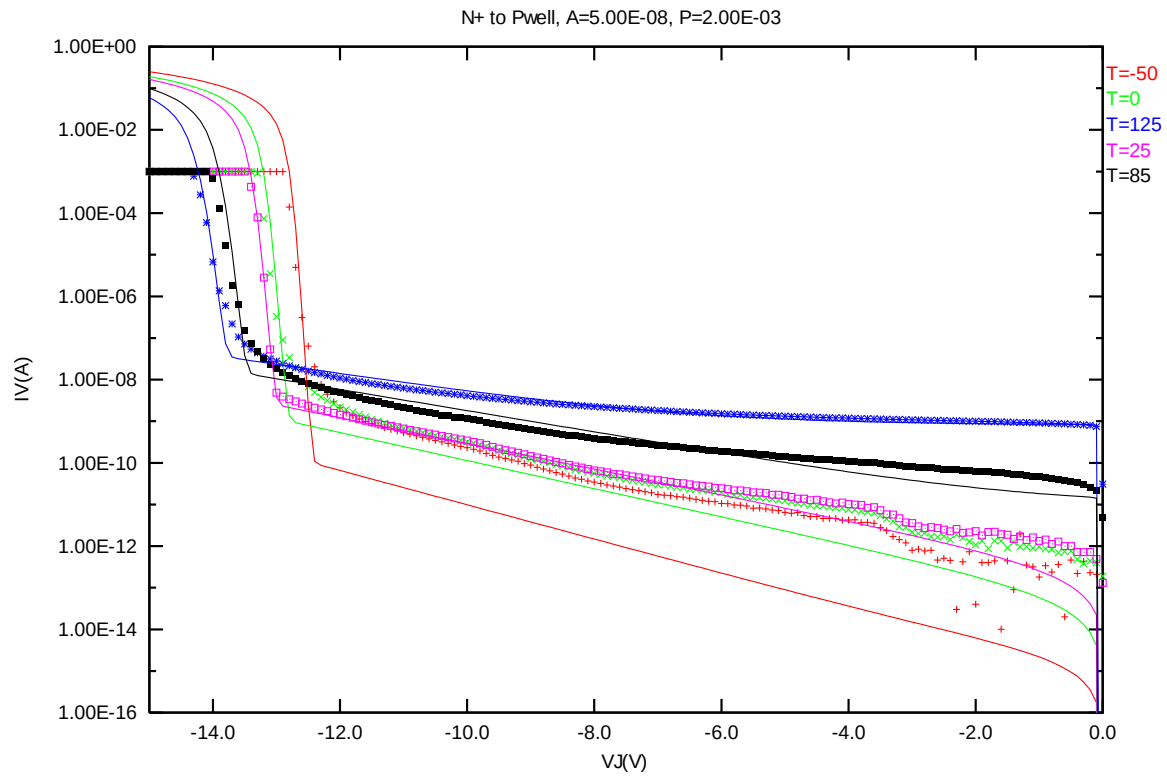


Figure 2-14 Reverse IV curve at 25 °C, 125 °C, 80 °C, 0 °C, -50 °C.

3 1.2V DIOP Diode Model Verification

3.1 Silicon Wafer for Modeling

The information of the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 3-1 Test wafer information for DIOP

	Diop diode model
Mask ID	BSW0FA
Lot ID	D566R.1
Wafer Number	#01

This model is characterized as the following geometry and temperature ranges depicted. Device behavior beyond these ranges may not be well described by the model. This model are extracted by devices with these two dimensions :

Junction Area Devices :

Area = $5.0\text{e-}8 \text{ m}^2$

Perimeter = $2.0\text{e-}3 \text{ m}$

Junction Edge Devices :

Area = $3.6\text{e-}8 \text{ m}^2$

Perimeter = $3.64\text{e-}2 \text{ m}$

3.2 Testing Conditions

The model is verified to measurement data under following bias conditions

Forward and reverse diode IV curve:

Table 3-2 Forward and reverse diode IV measurement conditions

Forward Diode

	Start	Stop	Step
$V_J(V)$	0	1	0.05

Reverse Diode

	Start	Stop	Step
$V_J(V)$	0	-15	-0.1

Diode junction CV:

Table 3-3 Diode junction CV measurement conditions

V_{BIAS} [V]	VH: nwell & VL: p+ From 0 to 1.32 V in 0.05 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50, 0, 25, 85, 125

3.3 1.2V DIOP model verification

3.3.1 1.2V DIOP model verification at room temperature (25 °C)

➤ *Diode junction CV curve at 25 °C.*

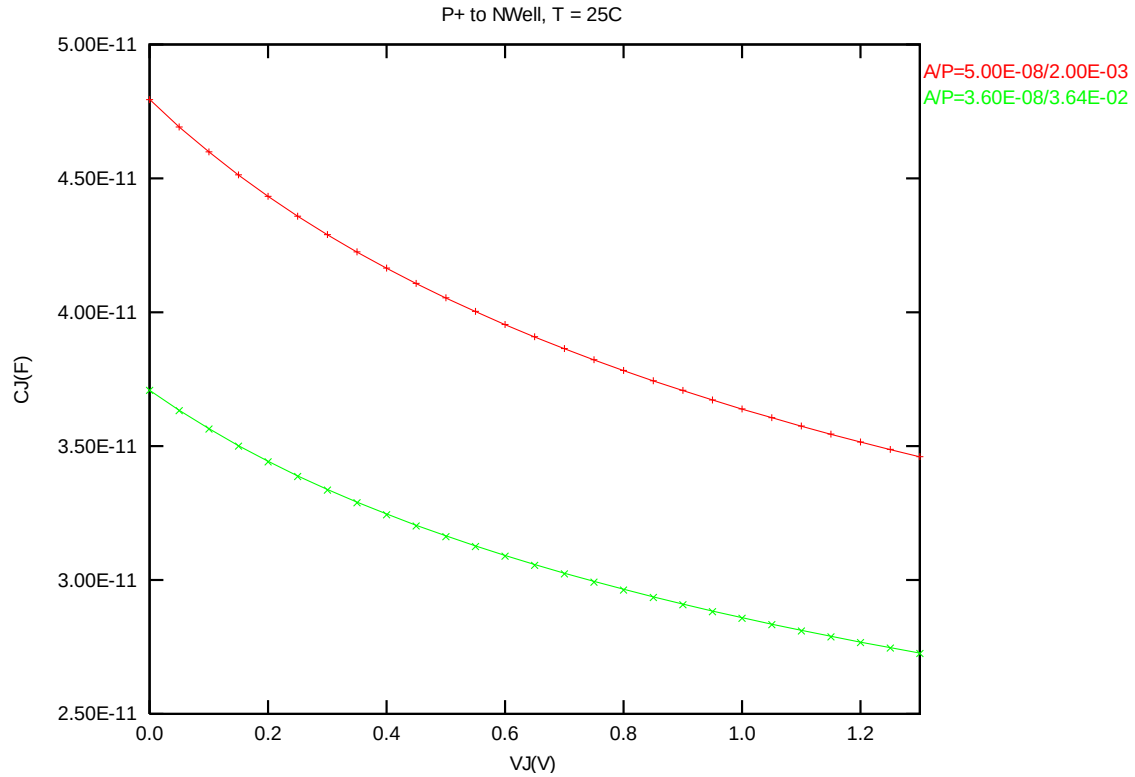
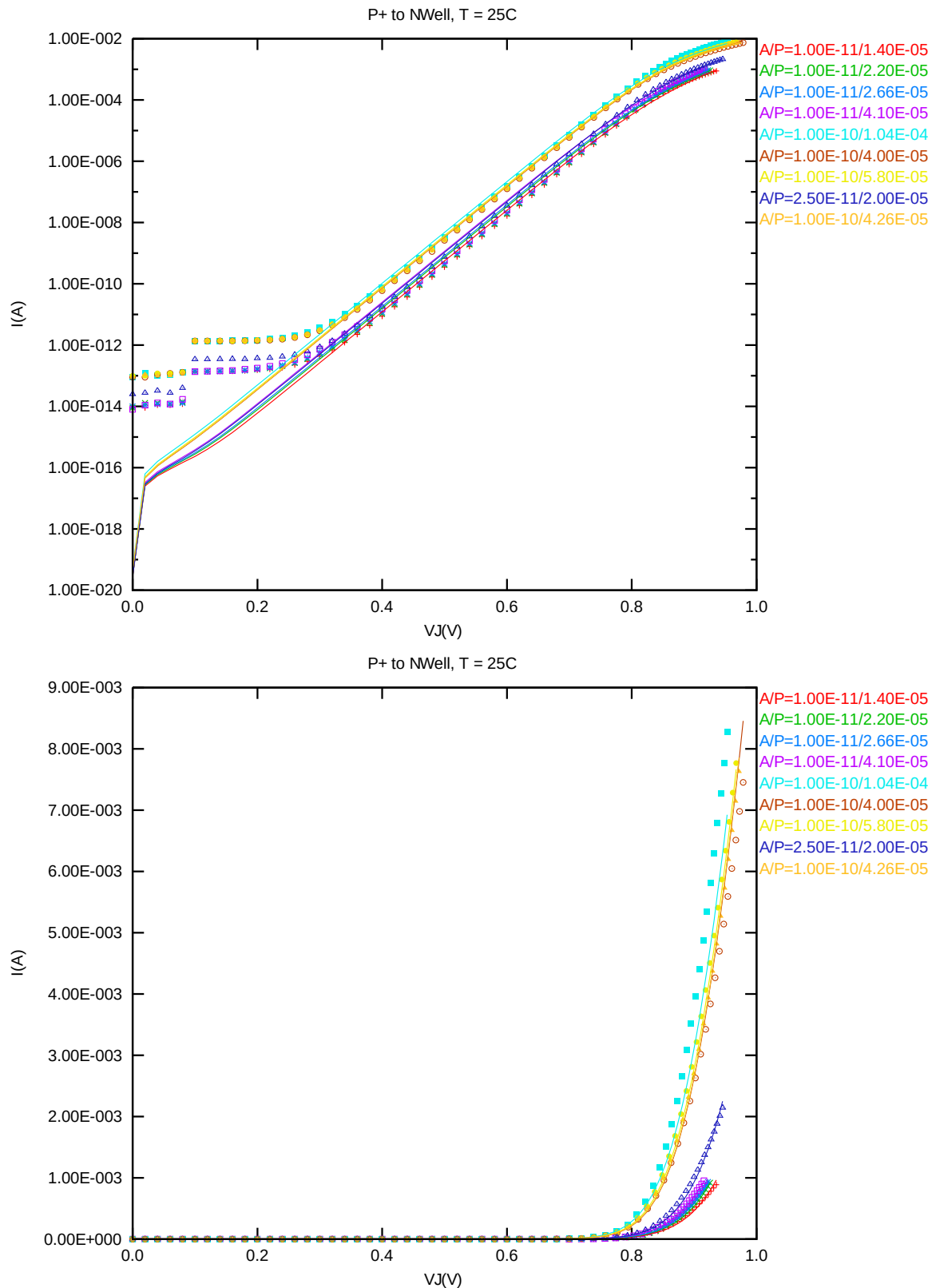


Figure 3-1 Diode junction CV curve at 25 °C

➤ **Forward IV curve at 25 °C.**



➤ **Reverse IV curve at 25 °C.**

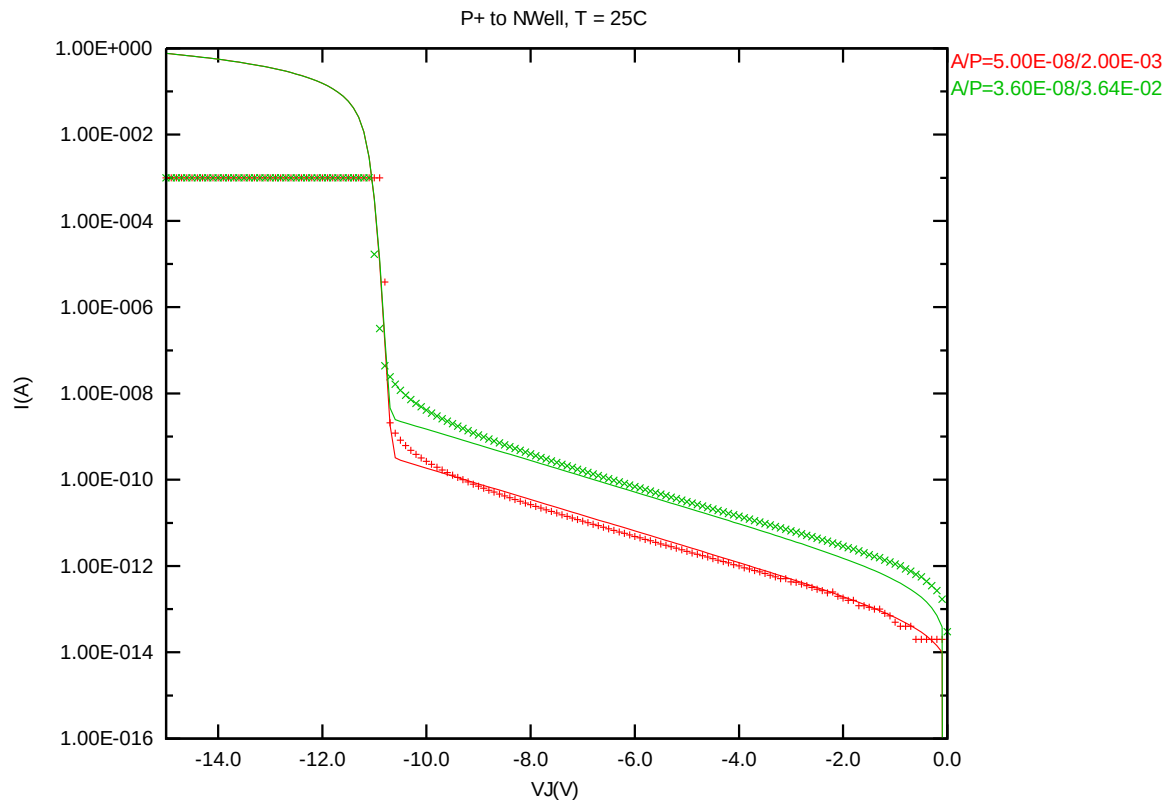


Figure 3-3 Reverse IV curve at 25 °C

3.3.2 1.2V DIOP model verification at 125 °C

➤ *Diode junction CV curve at 125 °C.*

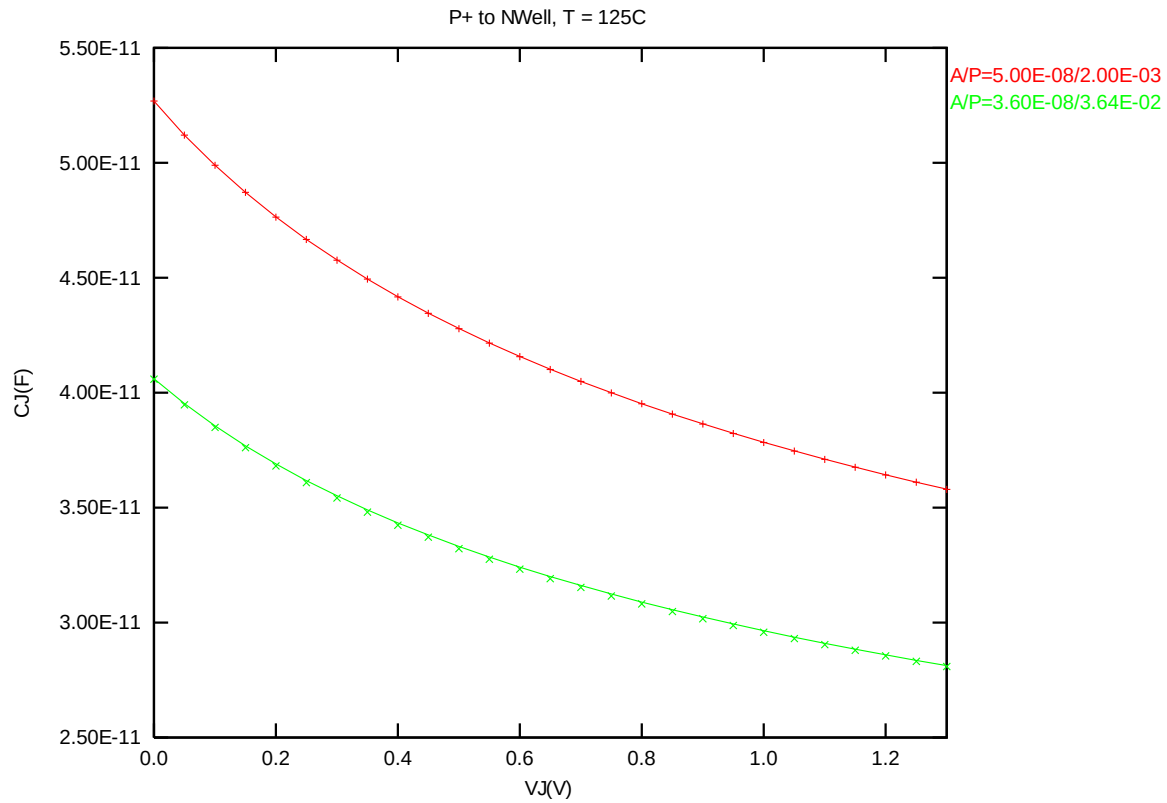
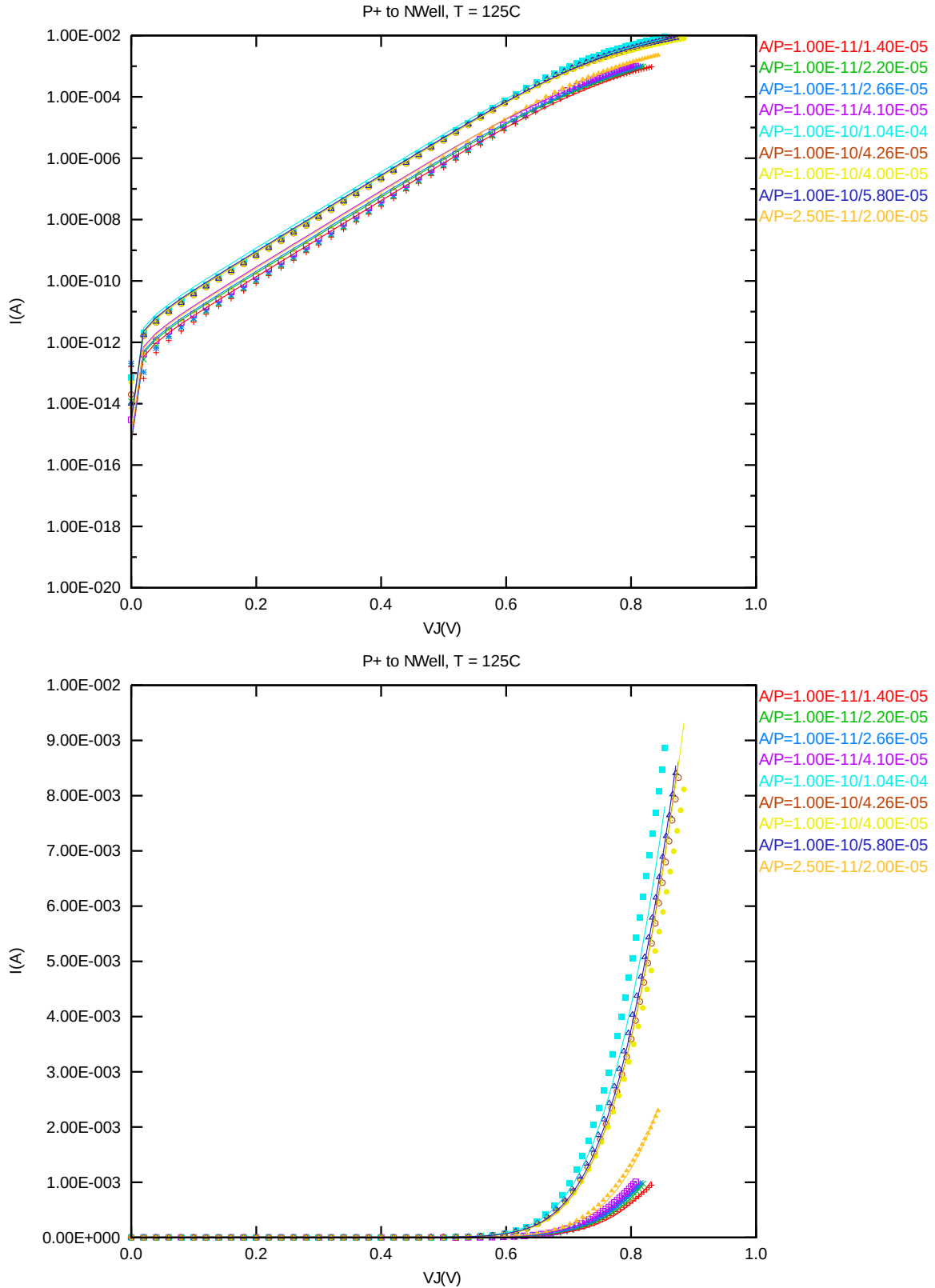


Figure 3-4 Diode junction CV curve at 125 °C

➤ **Forward IV curve at 125 °C.**



➤ **Reverse IV curve at 125 °C.**

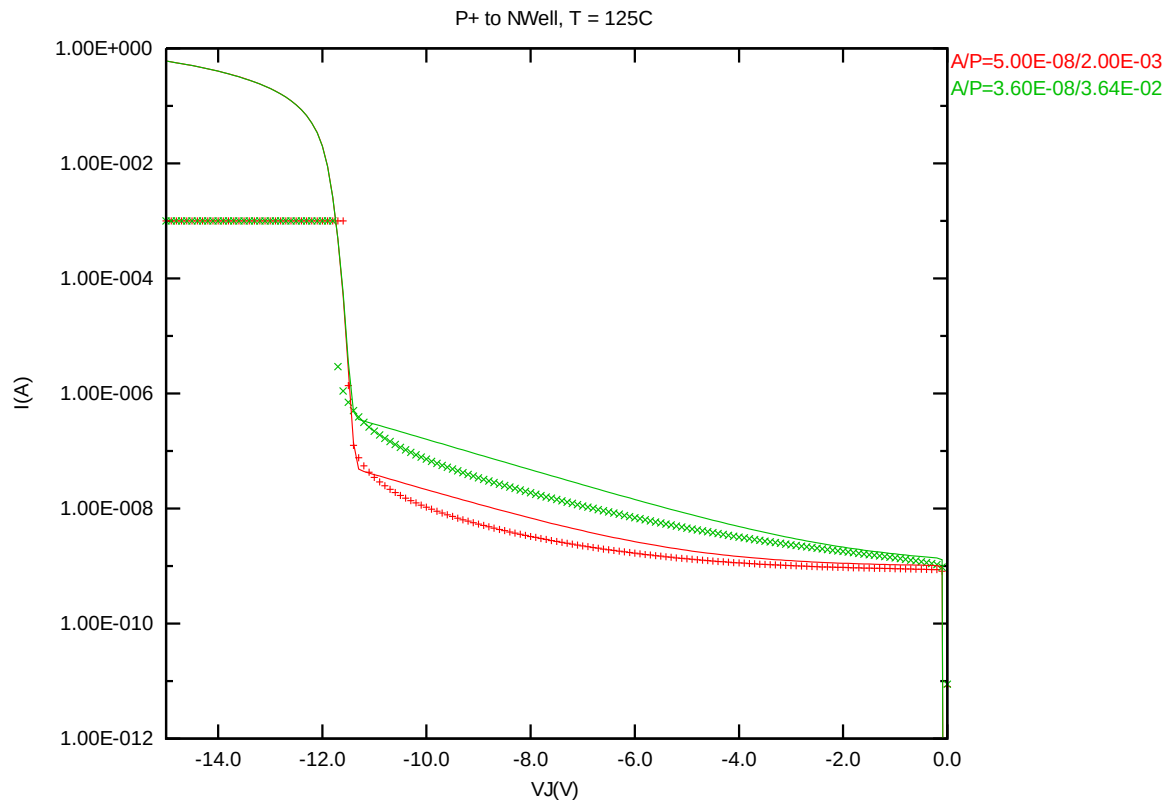


Figure 3-6 Reverse IV curve at 125 °C

3.3.3 1.2V DIOP model verification at 85 °C

➤ *Diode junction CV curve at 85 °C.*

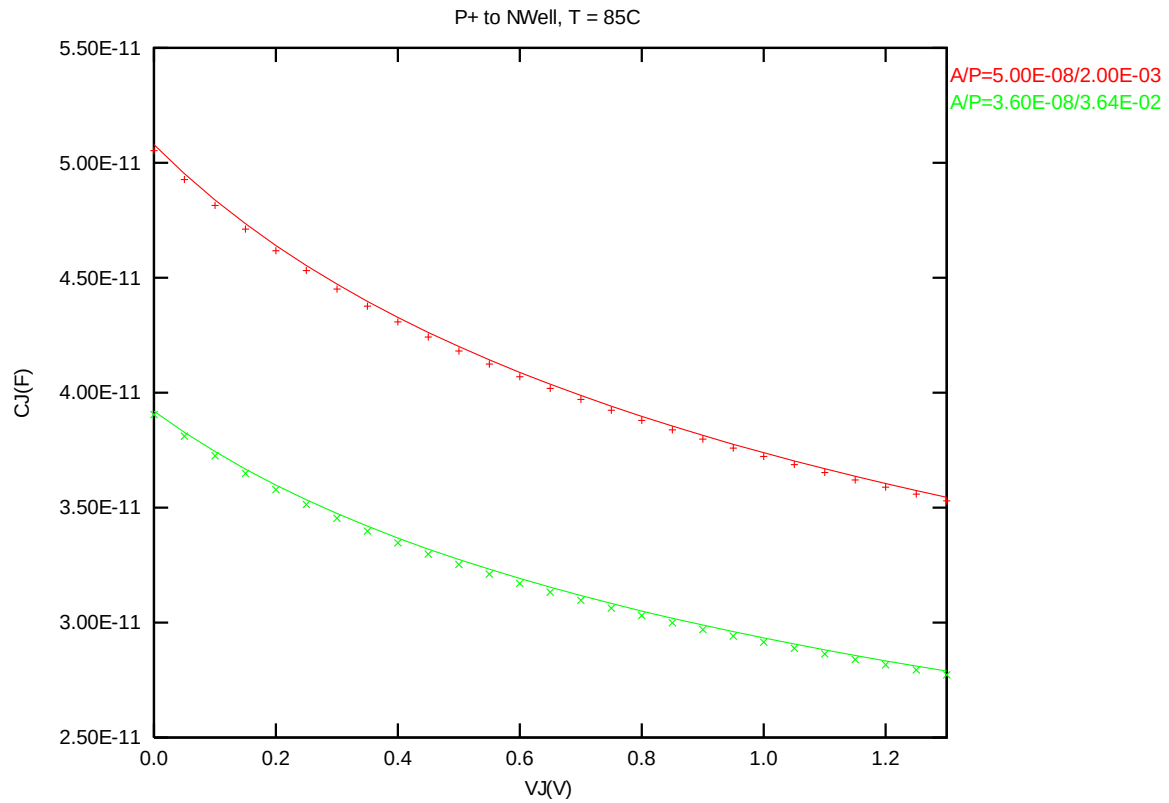


Figure 3-7 Diode junction CV curve at 85 °C

3.3.4 1.2V DIOP model verification at 0 °C

➤ Diode junction CV curve at 0 °C.

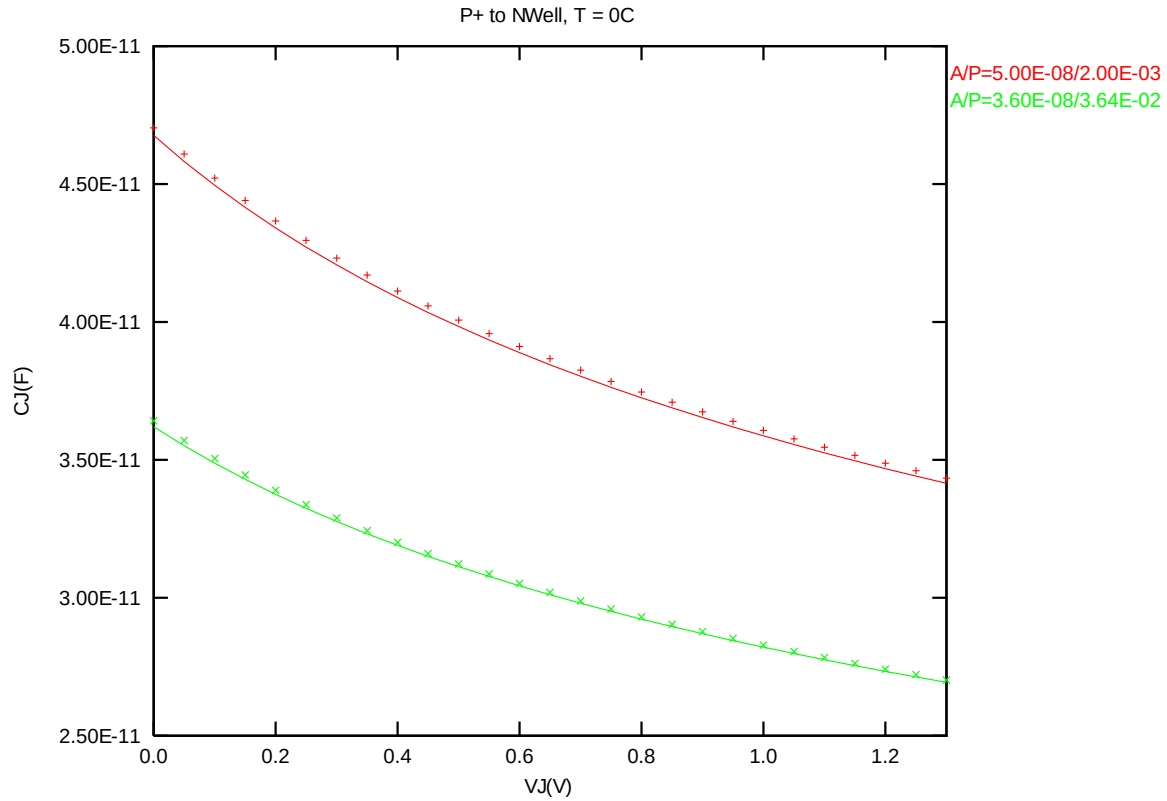


Figure 3-8 Diode junction CV curve at 0 °C

3.3.5 1.2V DIOP model verification at -50 °C

➤ Diode junction CV curve at -50 °C.

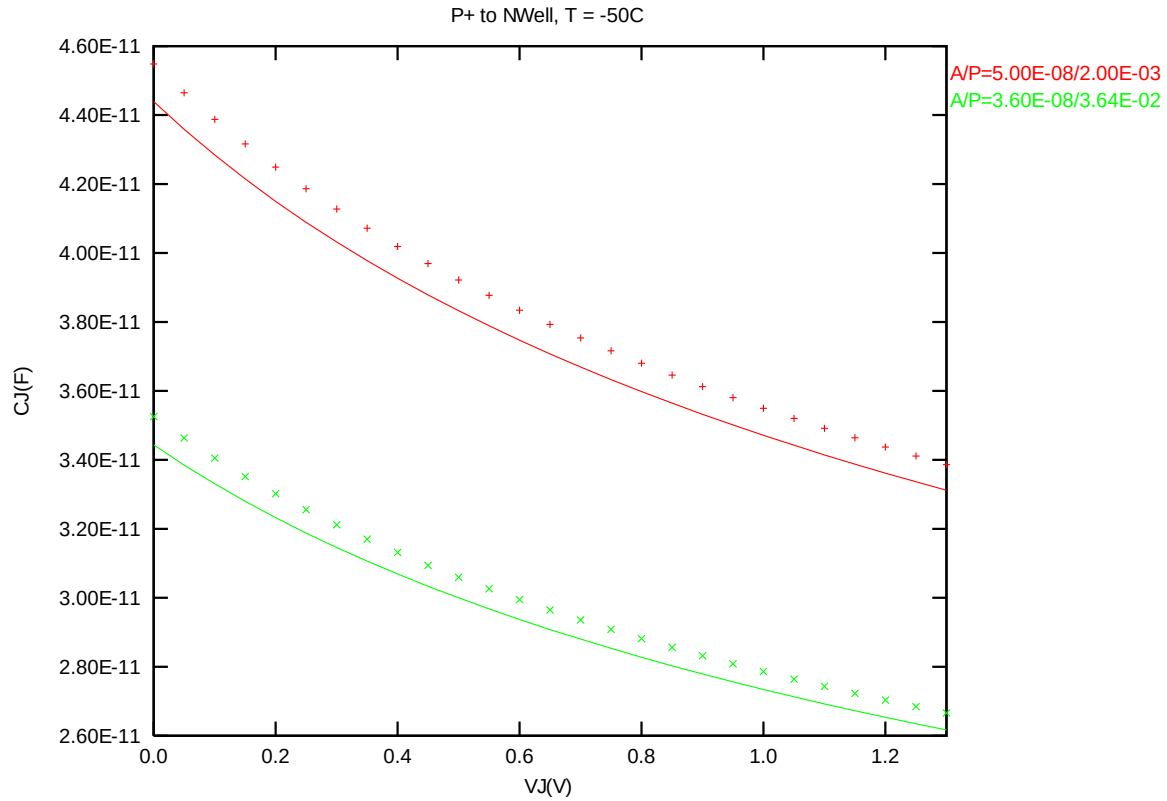


Figure 3-9 Diode junction CV curve at -50 °C

➤ **Forward IV curve at -50 °C.**

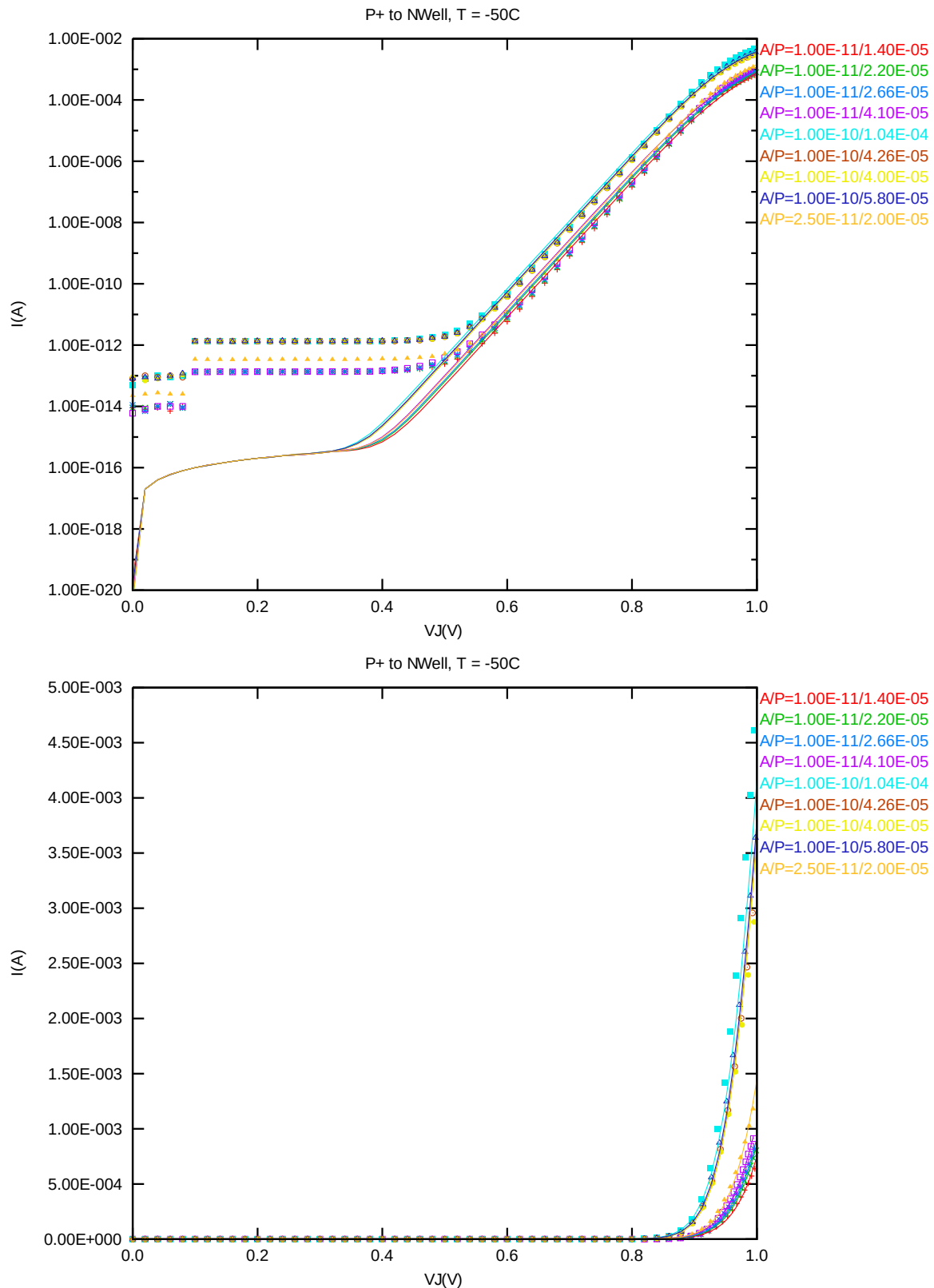


Figure 3-10 Forward IV curve at -50 °C

➤ **Reverse IV curve at -50 °C.**

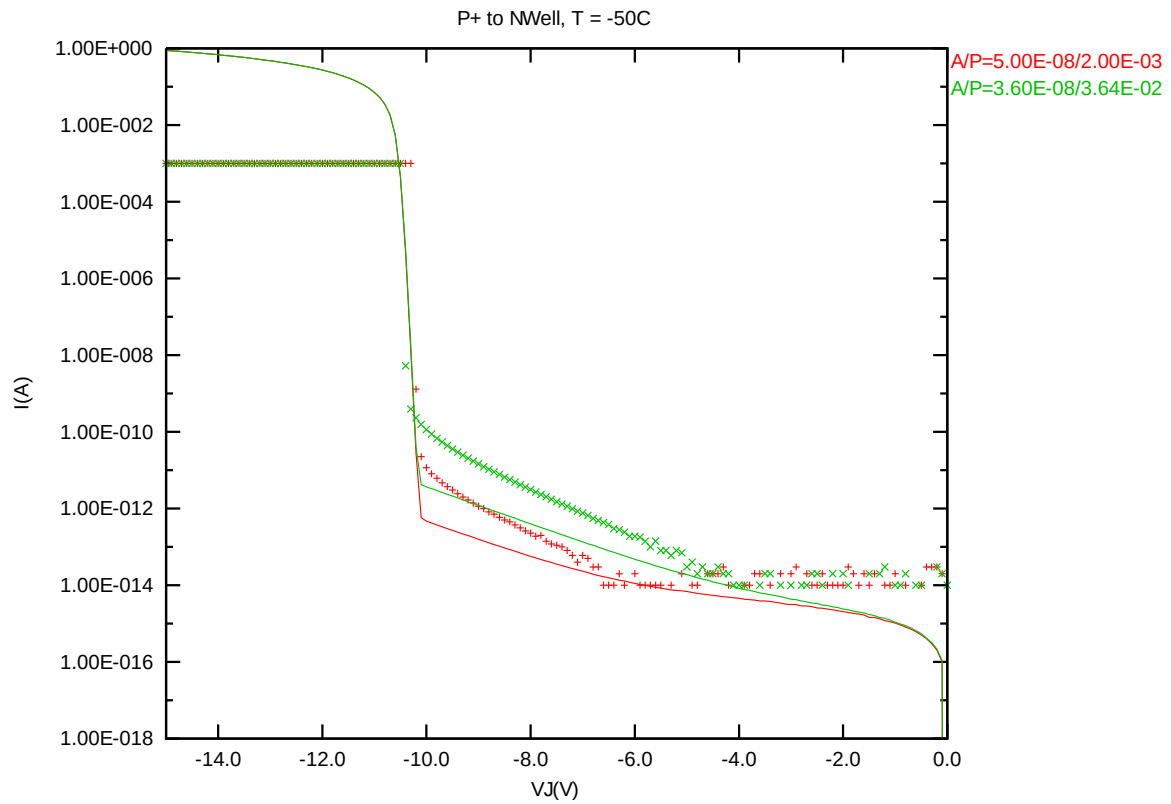


Figure 3-11 Reverse IV curve at -50 °C

3.3.6 1.2V DIOP model verification for temperature effect

➤ *Diode junction CV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C.*

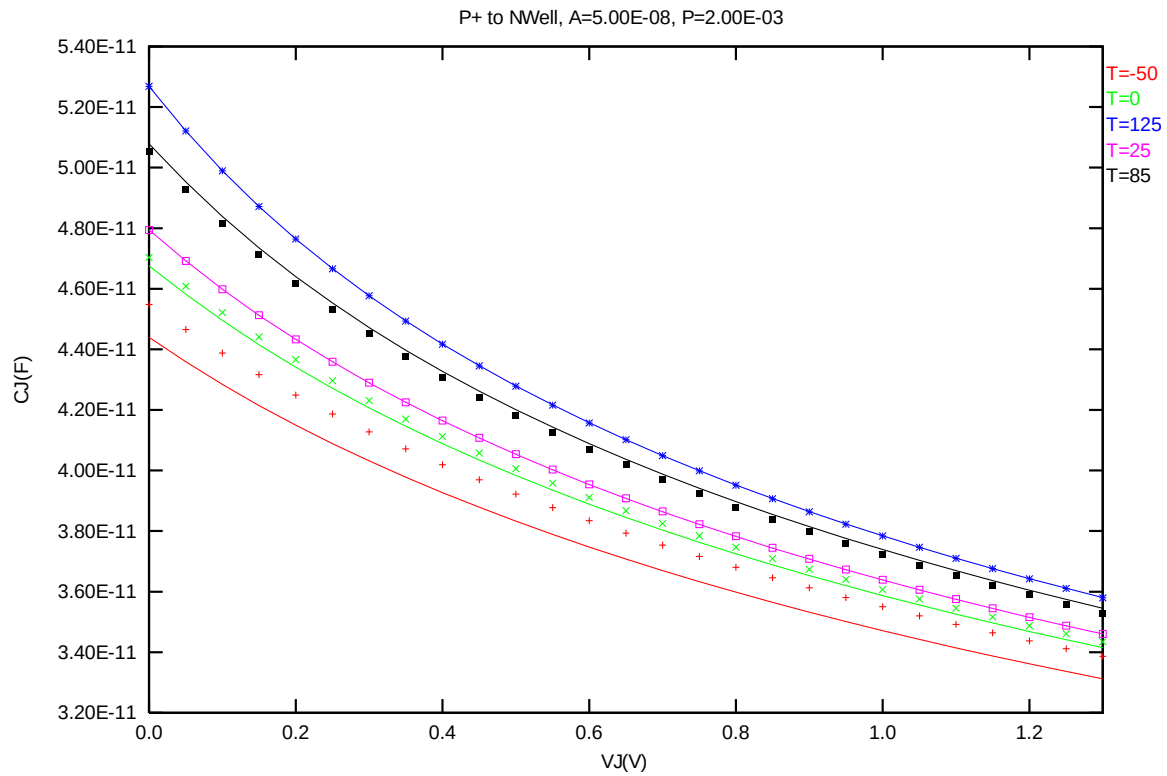


Figure 3-12 Diode junction CV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C.

➤ **Forward IV curve at 25 °C, 125 °C, -50 °C.**

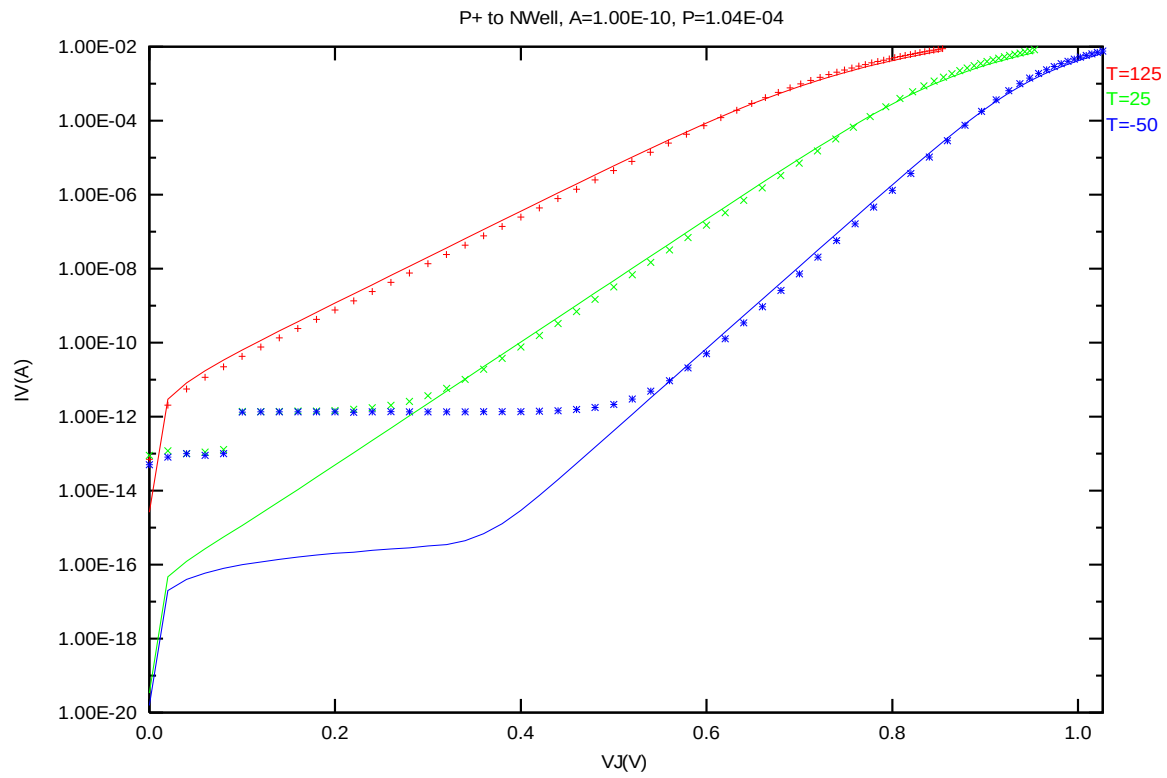


Figure 3-13 Forward IV curve at 25 °C, 125 °C, -50 °C

➤ **Reverse IV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C**

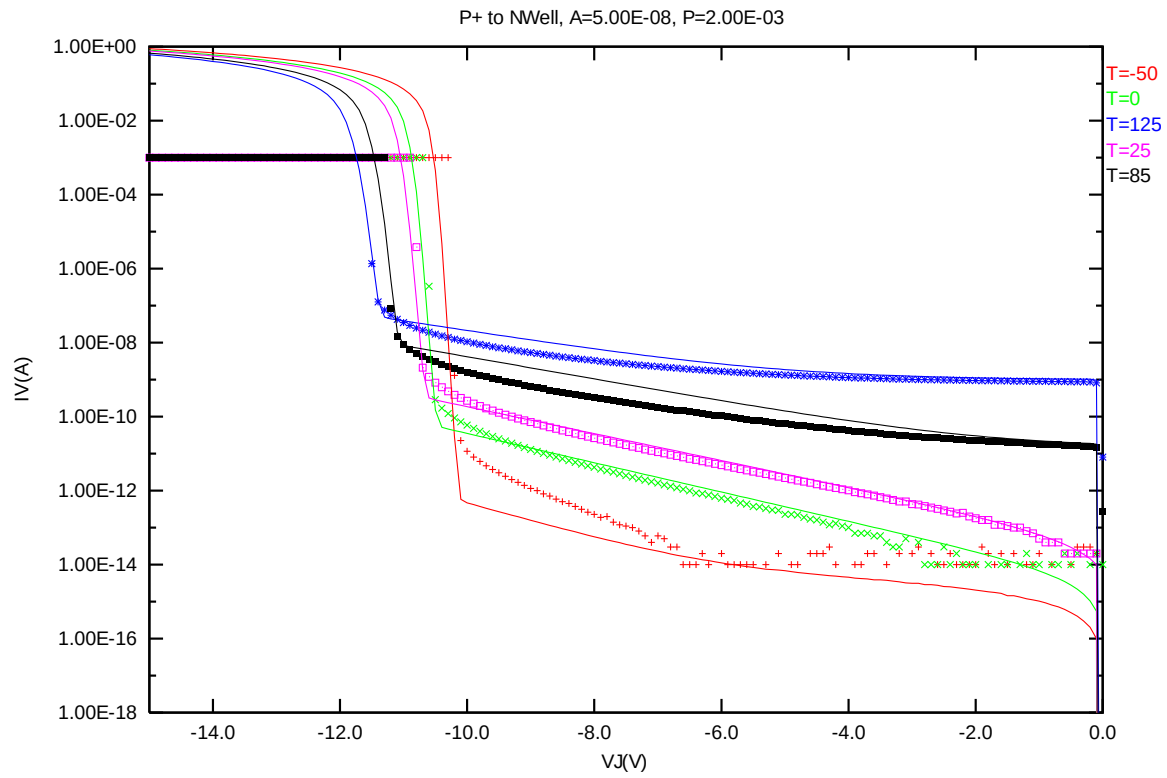


Figure 3-14 Reverse IV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C

4 3.3V DION Diode Model Verification

4.1 Silicon Wafer for Modeling

The information of the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 4-1 Test wafer information for DION_33

	Dion_33 diode model
Mask ID	BSW0FA
Lot ID	D566R.1
Wafer Number	#01

This model is characterized as the following geometry and temperature ranges depicted. Device behavior beyond these ranges may not be well described by the model. This model are extracted by devices with these two dimensions :

Junction Area Devices :

Area = $5.0\text{e-}8 \text{ m}^2$

Perimeter = $2.0\text{e-}3 \text{ m}$

Junction Edge Devices :

Area = $3.6\text{e-}8 \text{ m}^2$

Perimeter = $3.64\text{e-}2 \text{ m}$

4.2 Testing Conditions

The model is verified to measurement data under following bias conditions

Forward and reverse diode IV curve:

Table 4-2 Forward and reverse diode IV measurement conditions

Forward Diode

	Start	Stop	Step
$V_J(V)$	0	1	0.05

Reverse Diode

	Start	Stop	Step
$V_J(V)$	0	-15	-0.1

Diode junction CV:

Table 4-3 Diode junction CV measurement conditions

$V_{BIAS} [V]$	VH: pwell & VL: n+ From 0 to 3.63 V in 0.1 V step
Frequency [kHz]	100
$V_{OSC} [mV]$	50
Temp [°C]	-50, 0, 25, 85, 125

4.3 3.3V DION model verification

4.3.1 3.3V DION model verification at room temperature (25 °C)

➤ *Diode junction CV curve at 25 °C.*

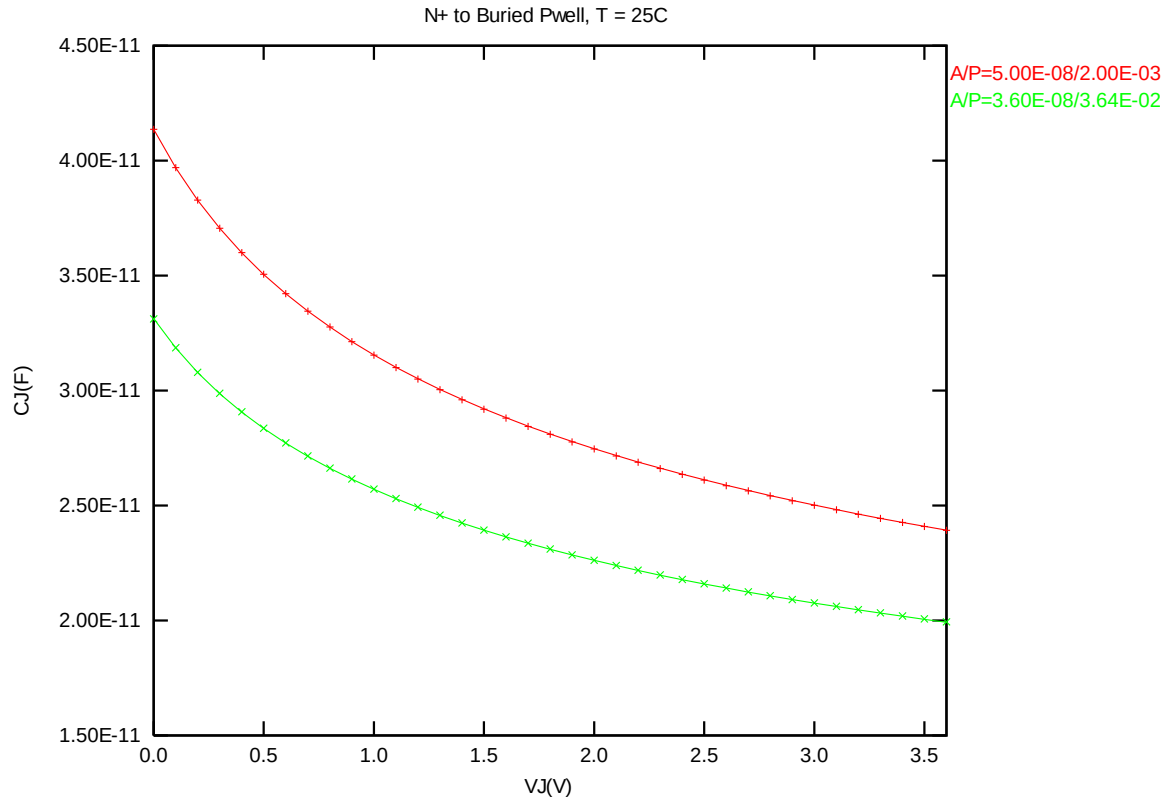


Figure 4-1 Diode junction CV curve at 25 °C

➤ **Forward IV curve at 25 °C.**

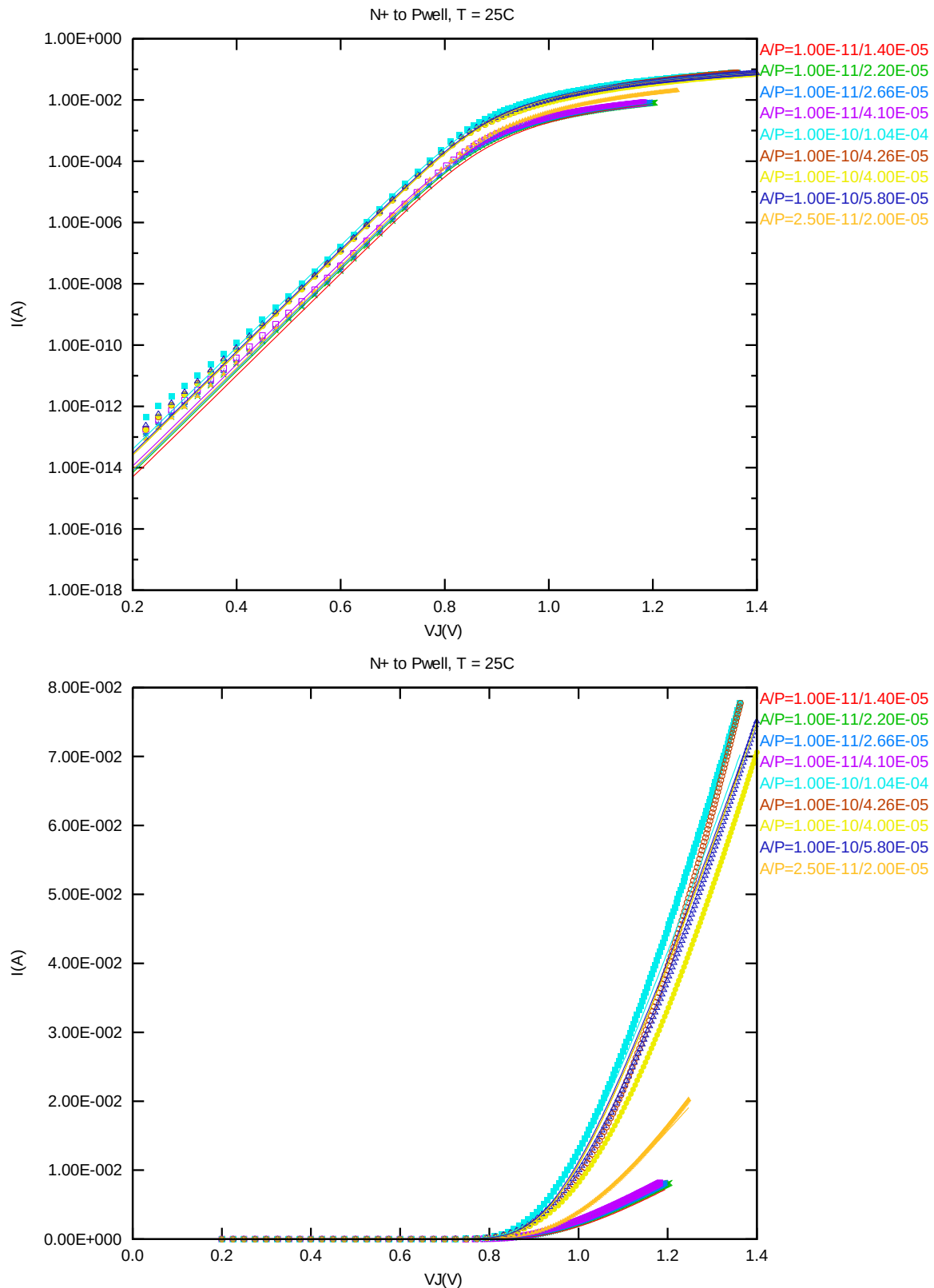


Figure 4-2 Forward IV curve at 25 °C

➤ **Reverse IV curve at 25 °C.**

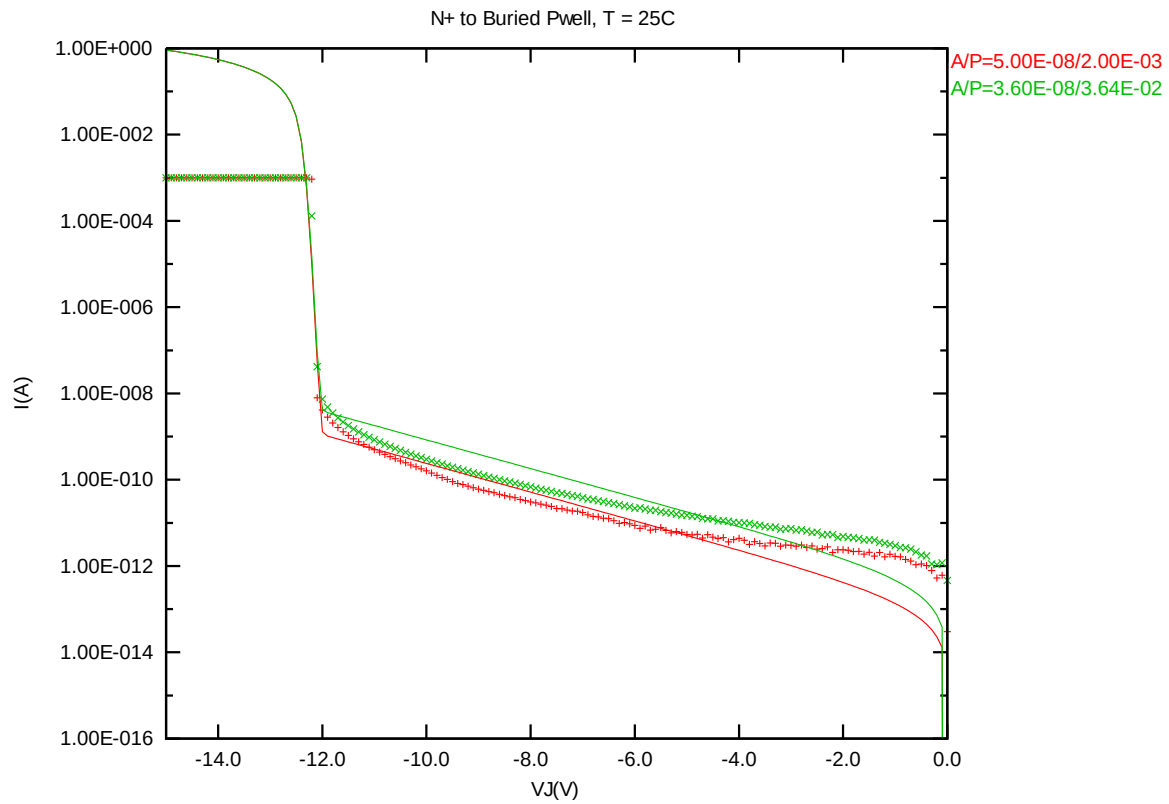


Figure 4-3 Reverse IV curve at 25 °C

4.3.2 3.3V DION model verification at 125 °C

➤ *Diode junction CV curve at 125 °C.*

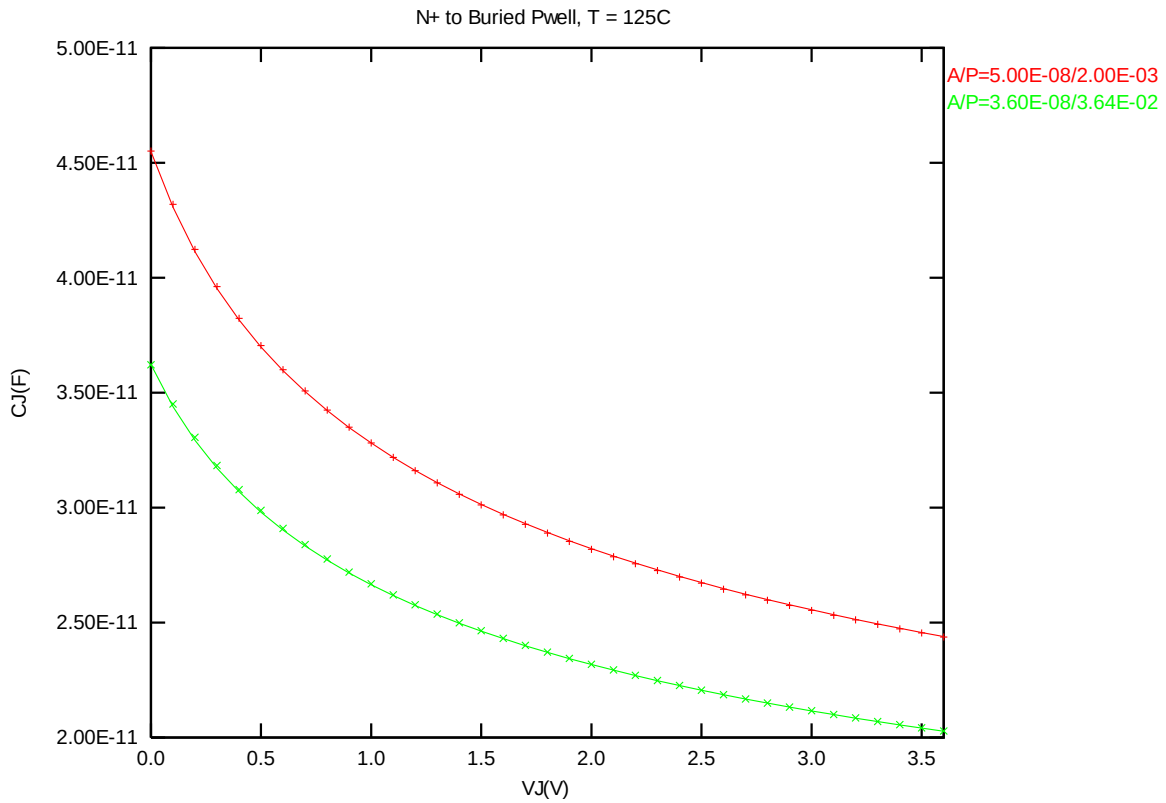


Figure 4-4 Diode junction CV curve at 125 °C

➤ **Forward IV curve at 125 °C.**

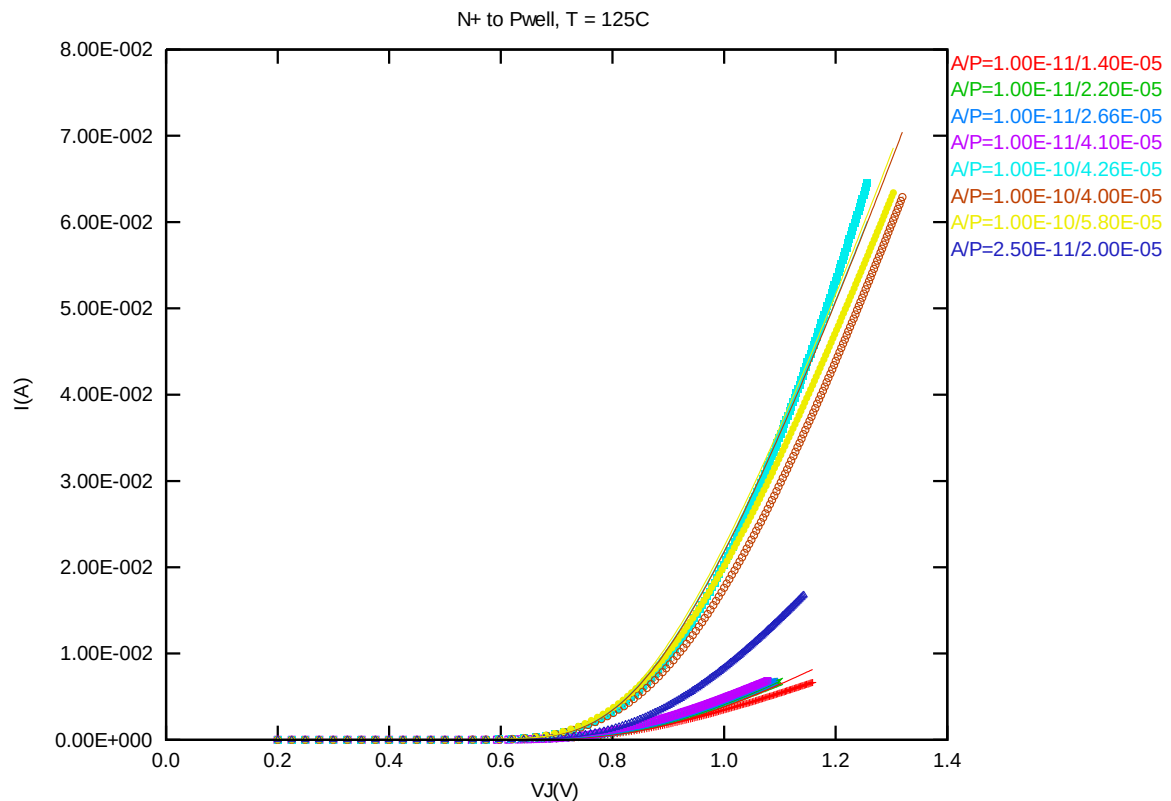
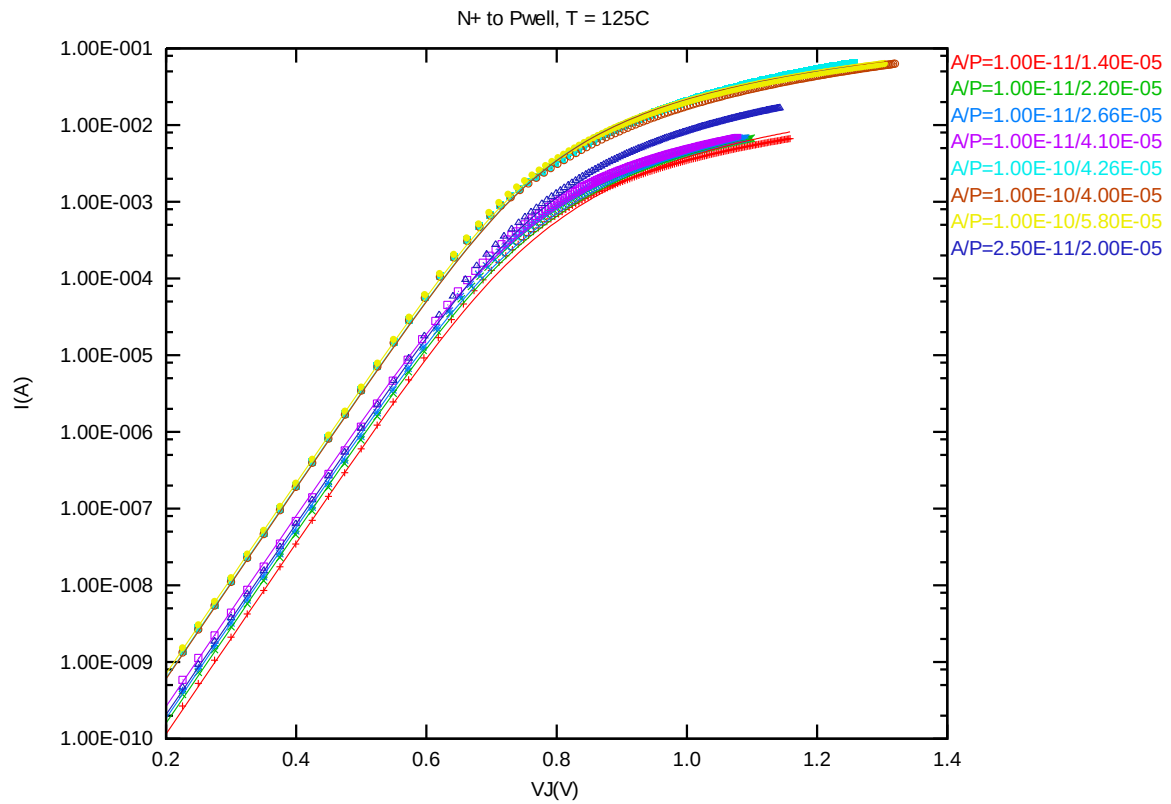


Figure 4-5 Forward IV curve at 125 °C

➤ **Reverse IV curve at 125 °C.**

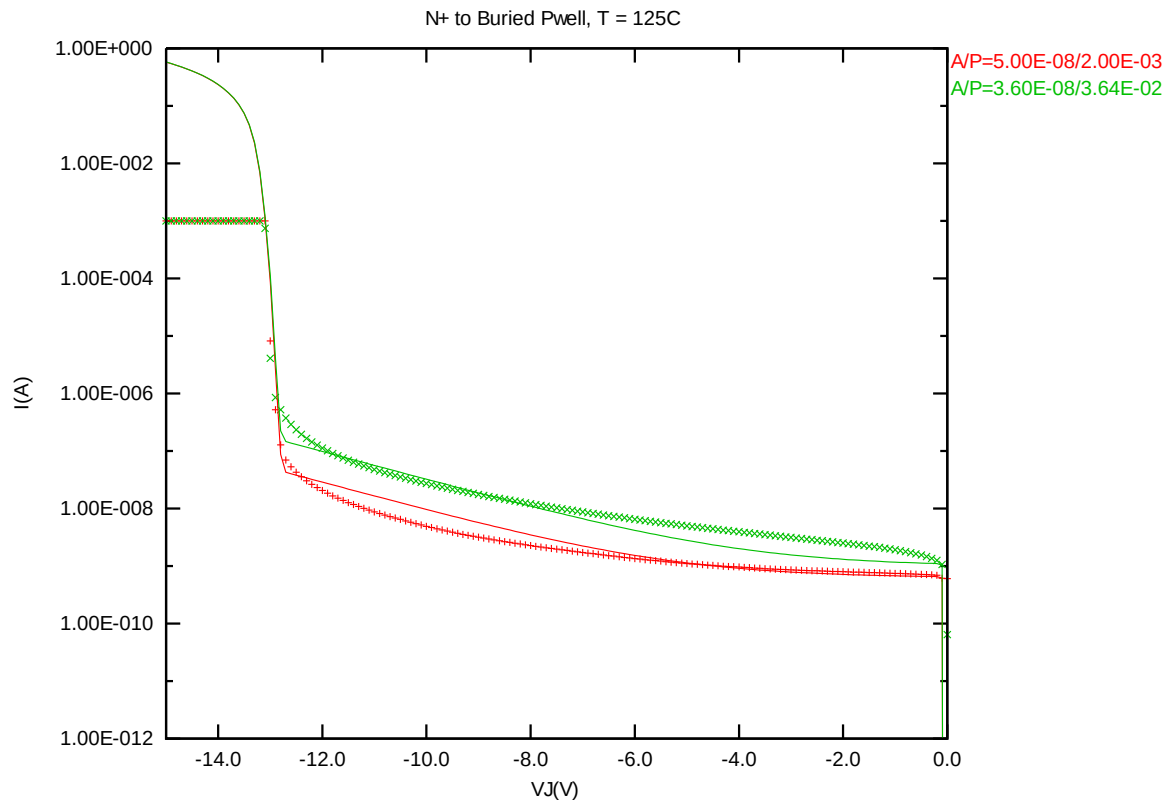


Figure 4-6 Reverse IV curve at 125 °C

4.3.3 3.3V DION model verification at 85 °C

➤ *Diode junction CV curve at 85 °C.*

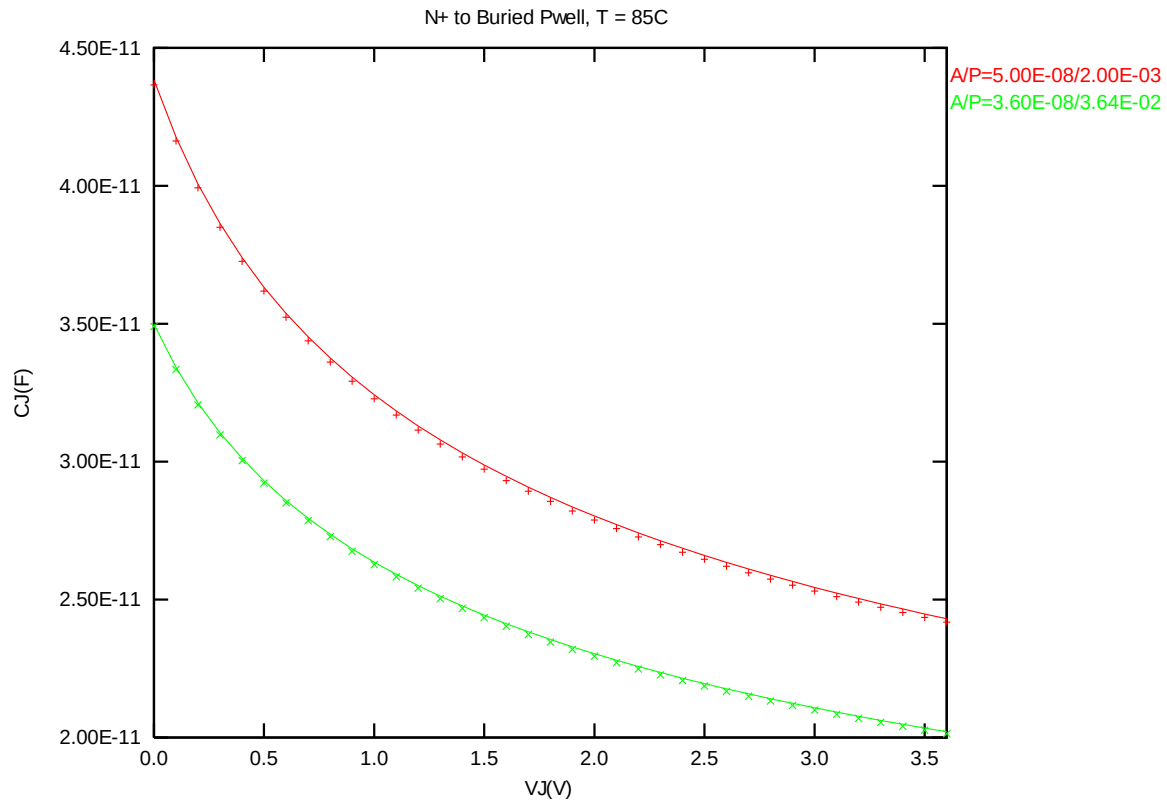


Figure 4-7 Diode junction CV curve at 85 °C

4.3.4 3.3V DION model verification at 0 °C

➤ *Diode junction CV curve at 0 °C.*

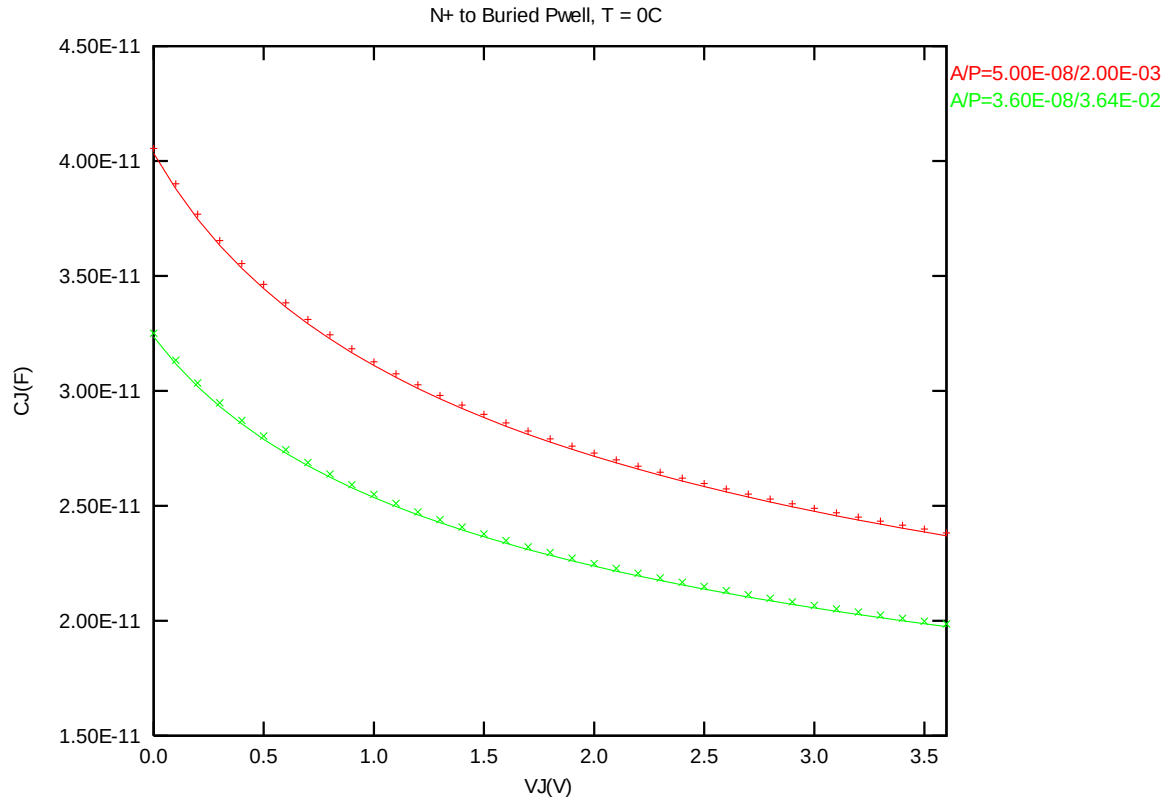


Figure 4-8 Diode junction CV curve at 0 °C

4.3.5 3.3V DION model verification at -50 °C

➤ *Diode junction CV curve at -50 °C.*

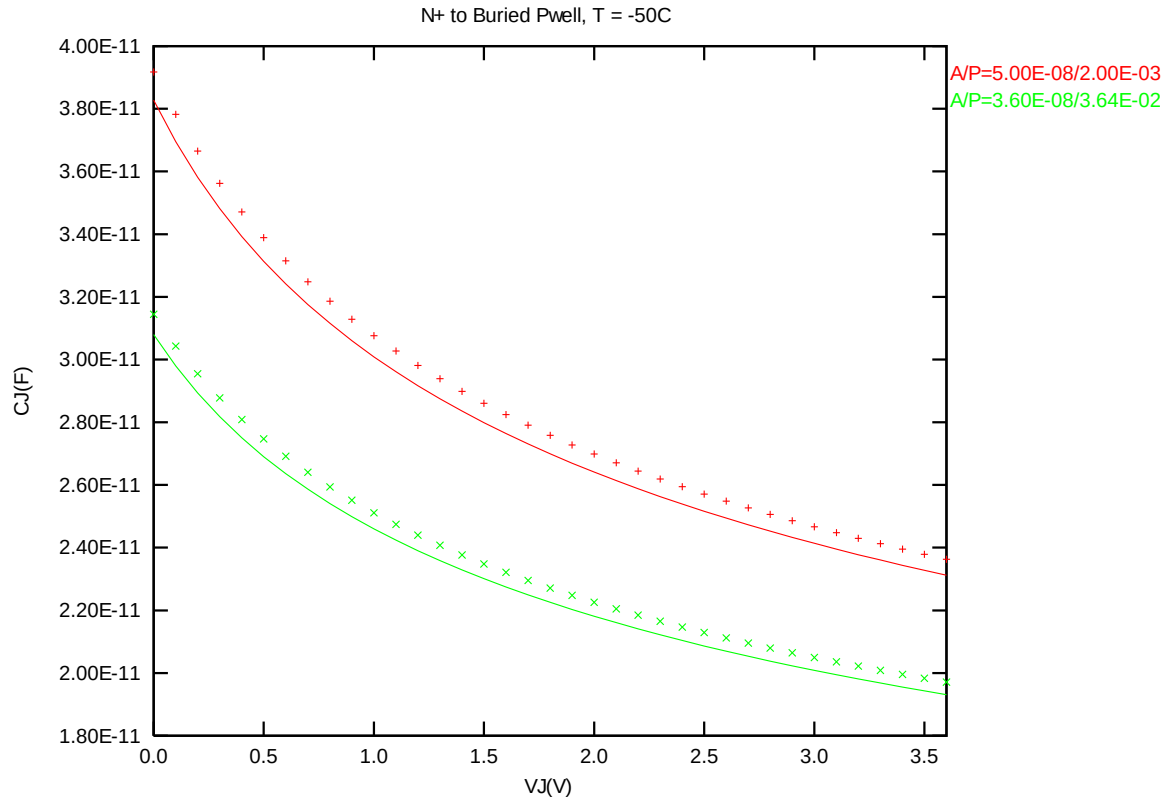


Figure 4-9 Diode junction CV curve at -50 °C

➤ **Forward IV curve at -50 °C.**

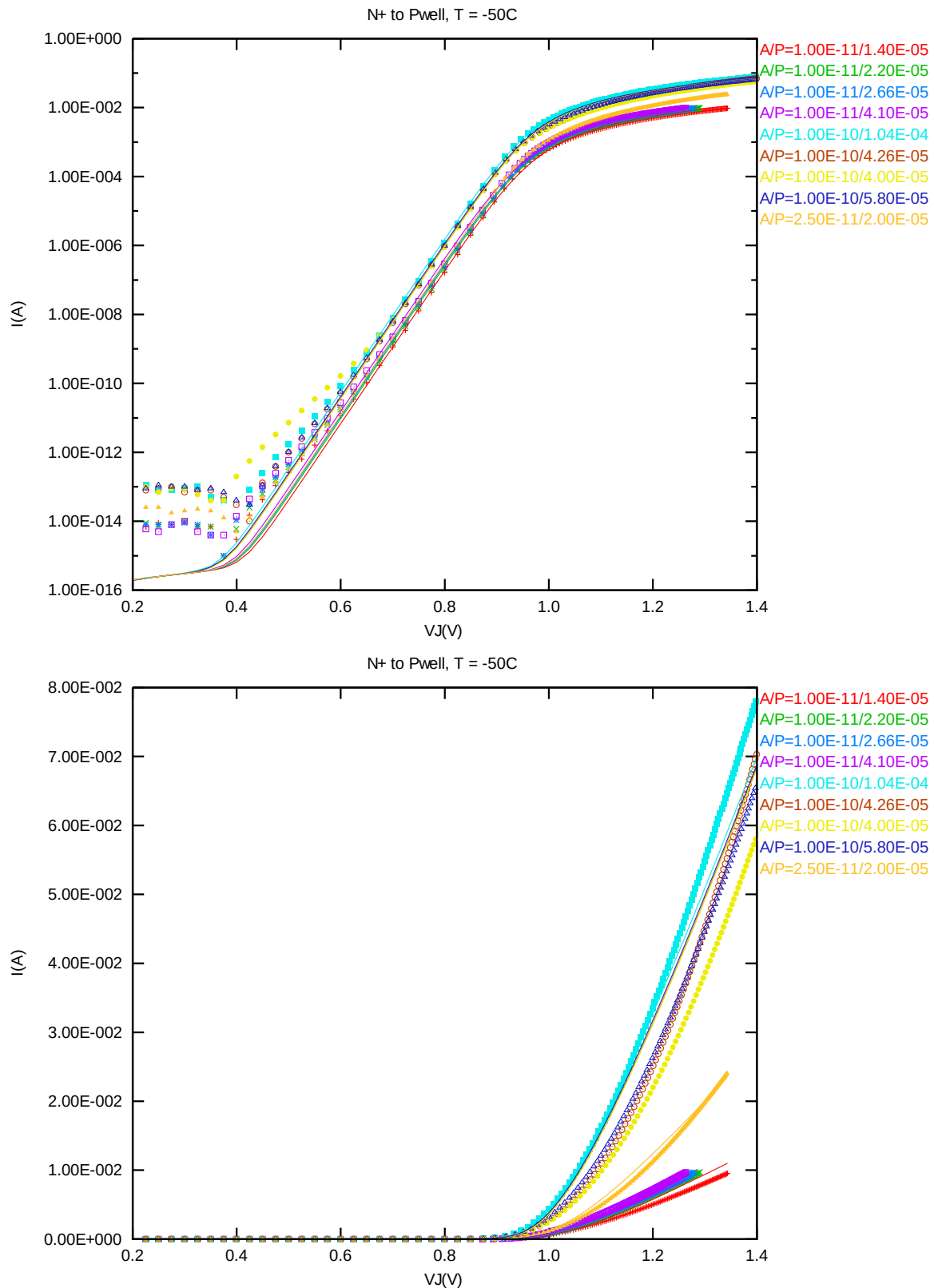


Figure 4-10 Forward IV curve at -50 °C

➤ **Reverse IV curve at -50 °C.**

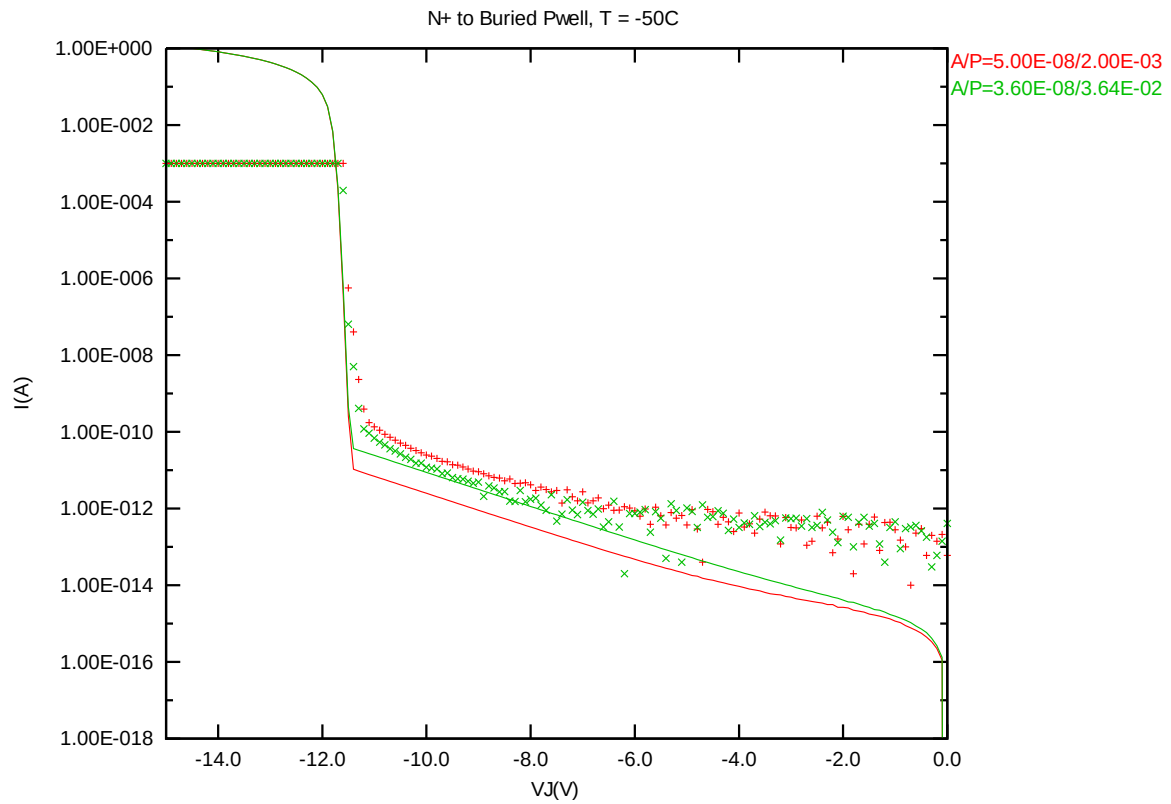


Figure 4-11 Reverse IV curve at -50 °C

4.3.6 3.3V DION model verification for temperature effect

➤ *Diode junction CV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C.*

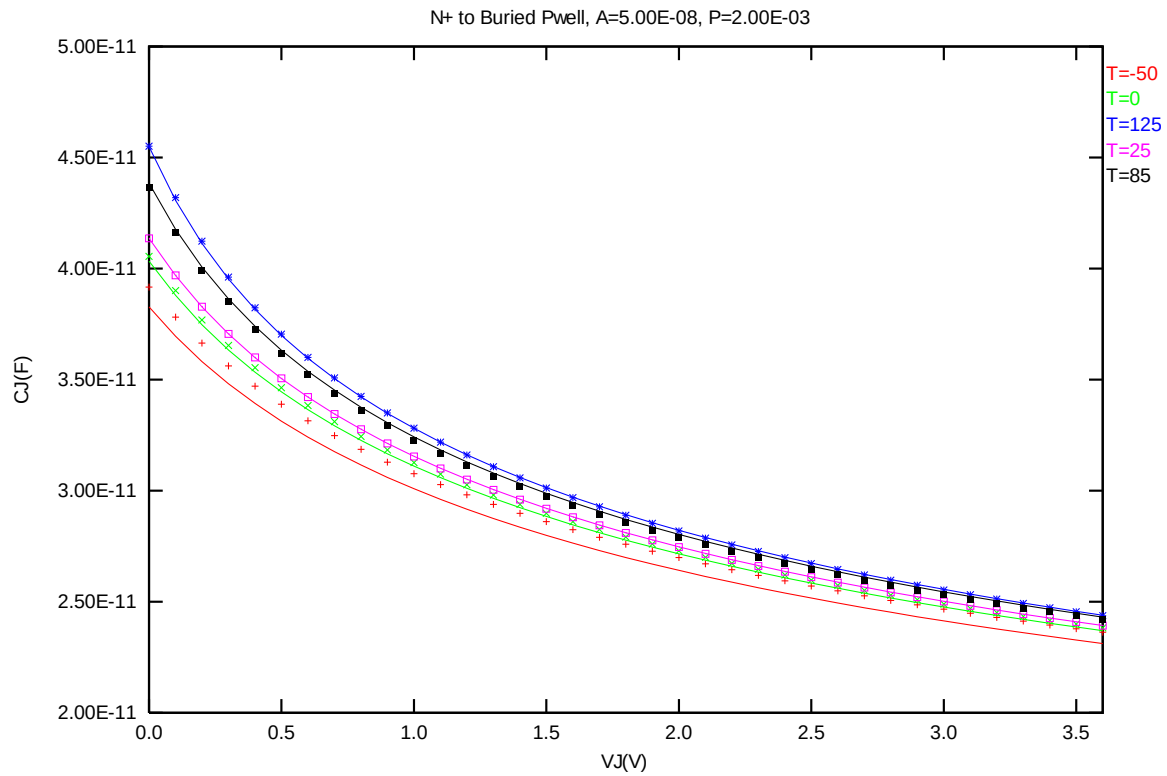


Figure 4-12 Diode junction CV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C.

➤ **Forward IV curve at 25 °C, 125 °C, -50 °C.**

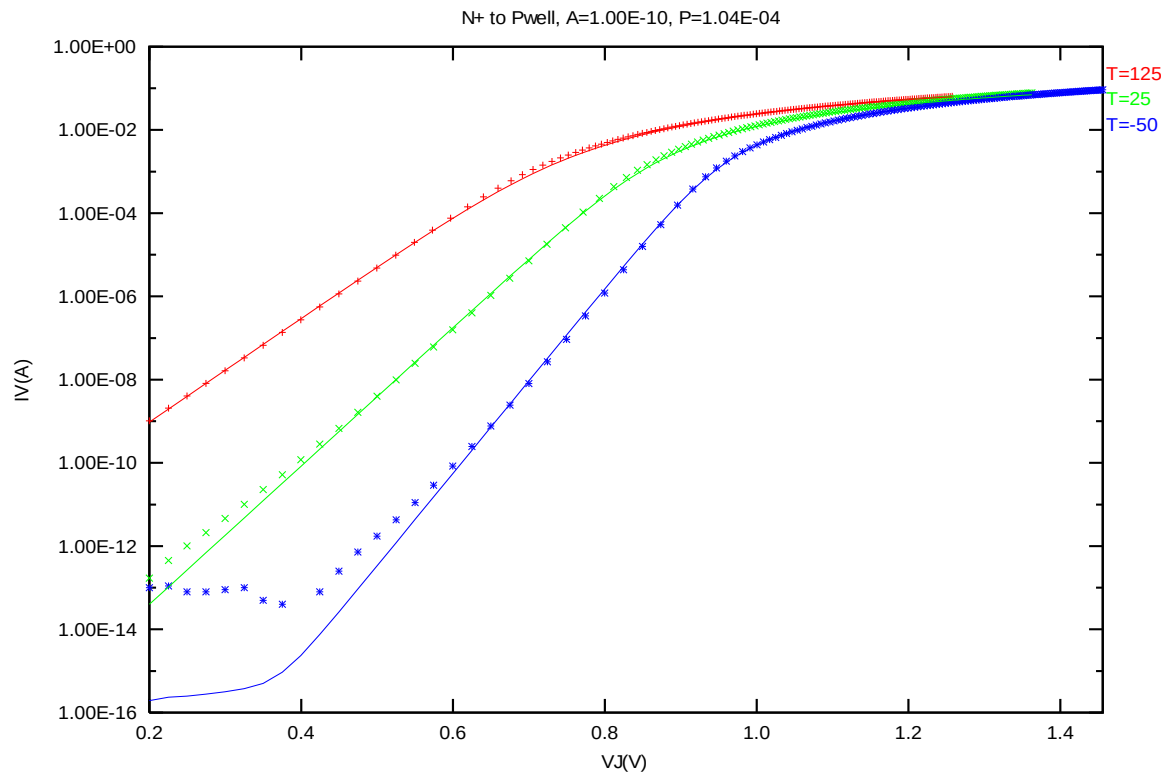


Figure 4-13 Forward IV curve at 25 °C, 125 °C, -50 °C

➤ **Reverse IV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C**

N+ to Buried Pwell, A=5.00E-08, P=2.00E-03

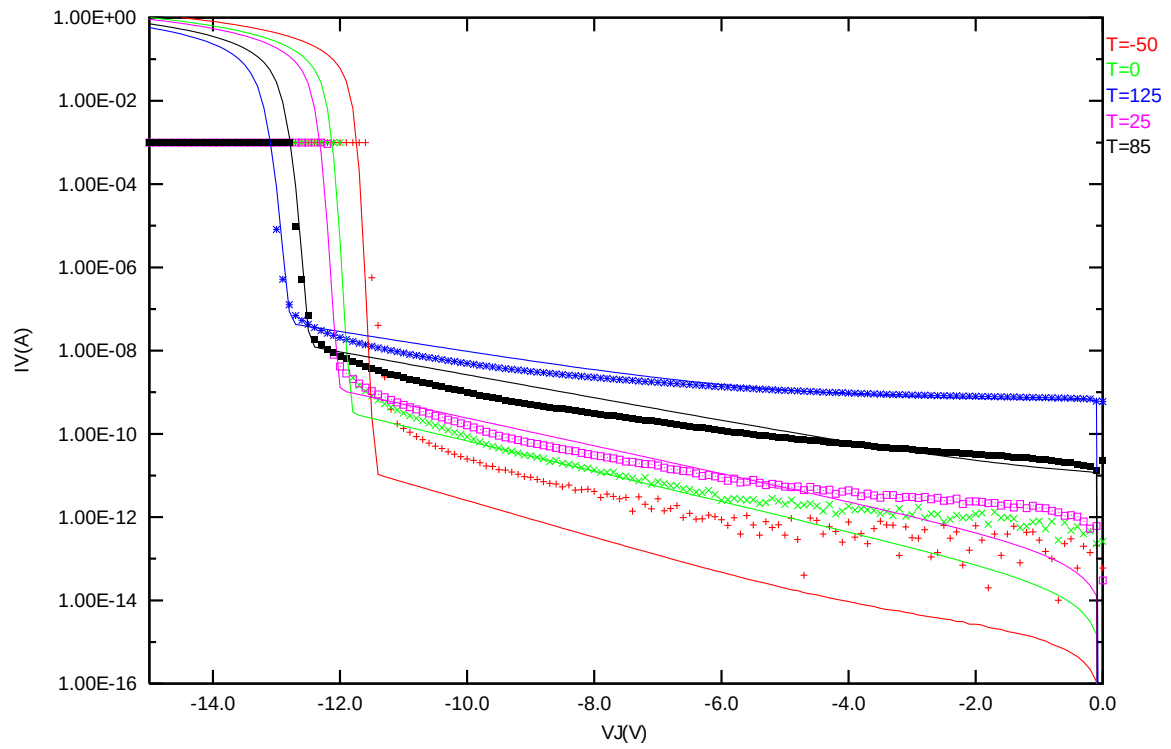


Figure 4-14 Reverse IV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C

5 3.3V DIOP Diode Model Verification

5.1 Silicon Wafer for Modeling

The information of the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 5-1 Test wafer information for DIOP_33

	Diop_33 diode model
Mask ID	BSW0FA
Lot ID	D566R.1
Wafer Number	#01

This model is characterized as the following geometry and temperature ranges depicted. Device behavior beyond these ranges may not be well described by the model. This model are extracted by devices with these two dimensions :

Junction Area Devices :

Area = $5.0\text{e-}8 \text{ m}^2$

Perimeter = $2.0\text{e-}3 \text{ m}$

Junction Edge Devices :

Area = $3.6\text{e-}8 \text{ m}^2$

Perimeter = $3.64\text{e-}2 \text{ m}$

5.2 Testing Conditions

The model is verified to measurement data under following bias conditions

Forward and reverse diode IV curve:

Table 5-2 Forward and reverse diode IV measurement conditions

Forward Diode

	Start	Stop	Step
$V_J(V)$	0	1	0.05

Reverse Diode

	Start	Stop	Step
$V_J(V)$	0	-15	-0.1

Diode junction CV:

Table 5-3 Diode junction CV measurement conditions

$V_{BIAS} [V]$	VH: nwell & VL: p+ From 0 to 3.63 V in 0.1 V step
Frequency [kHz]	100
$V_{OSC} [mV]$	50
Temp [°C]	-50, 0, 25, 85, 125

5.3 3.3V DIOP model verification

5.3.1 3.3V DIOP model verification at room temperature (25 °C)

➤ *Diode junction CV curve at 25 °C.*

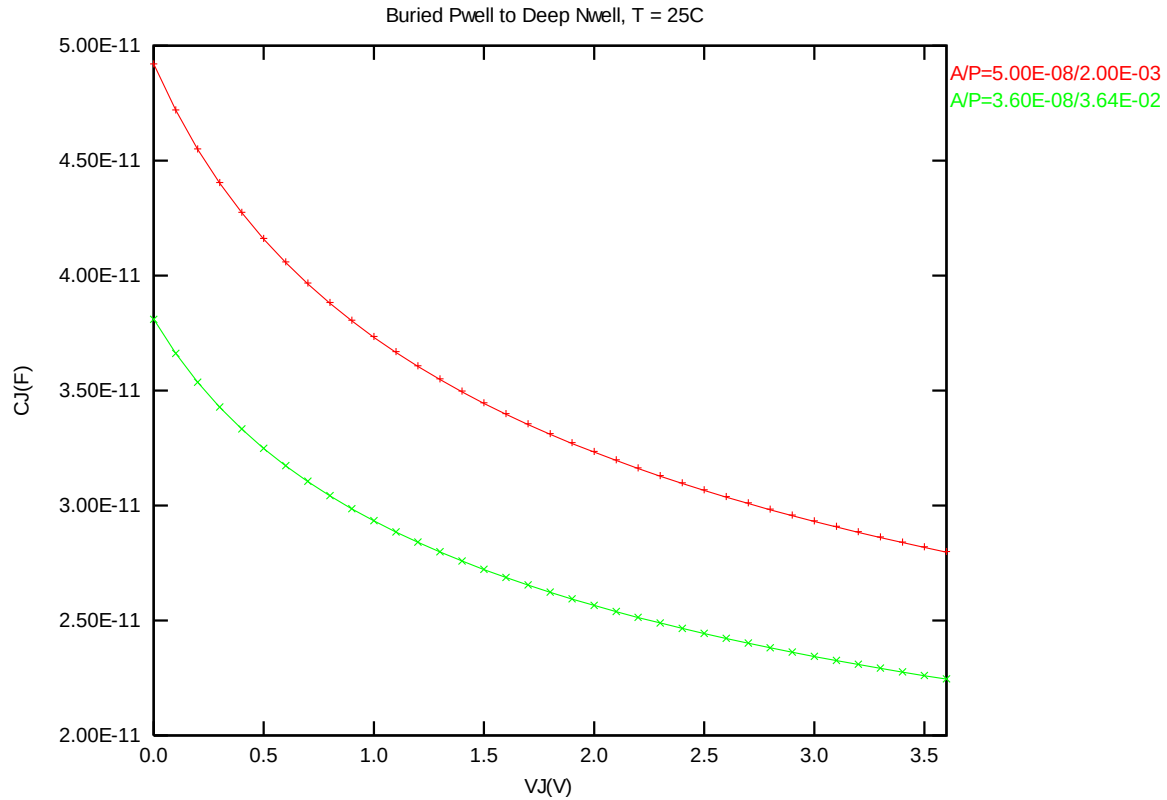


Figure 5-1 Diode junction CV curve at 25 °C

➤ **Forward IV curve at 25 °C.**

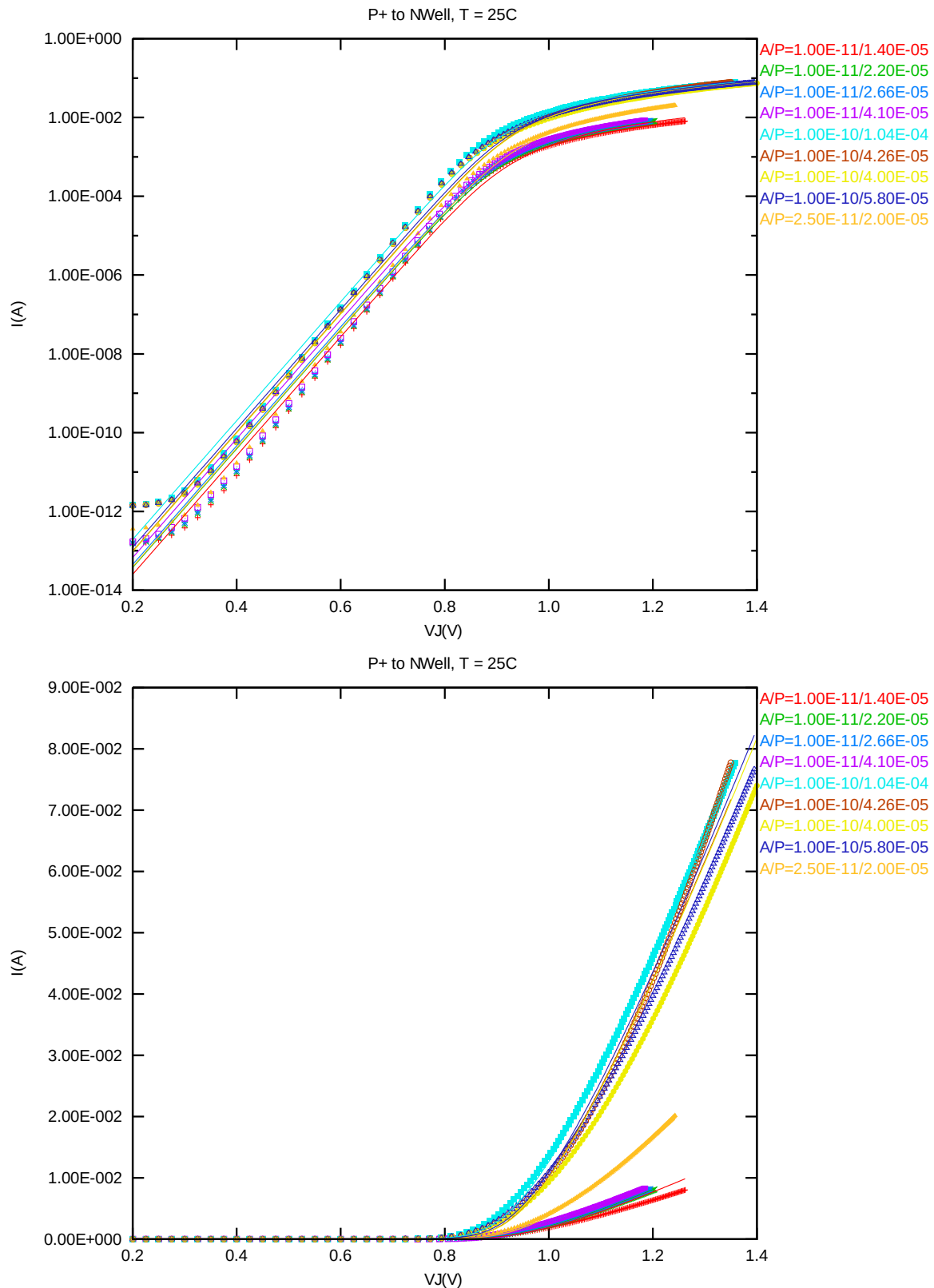


Figure 5-2 Forward IV curve at 25 °C

➤ **Reverse IV curve at 25 °C.**

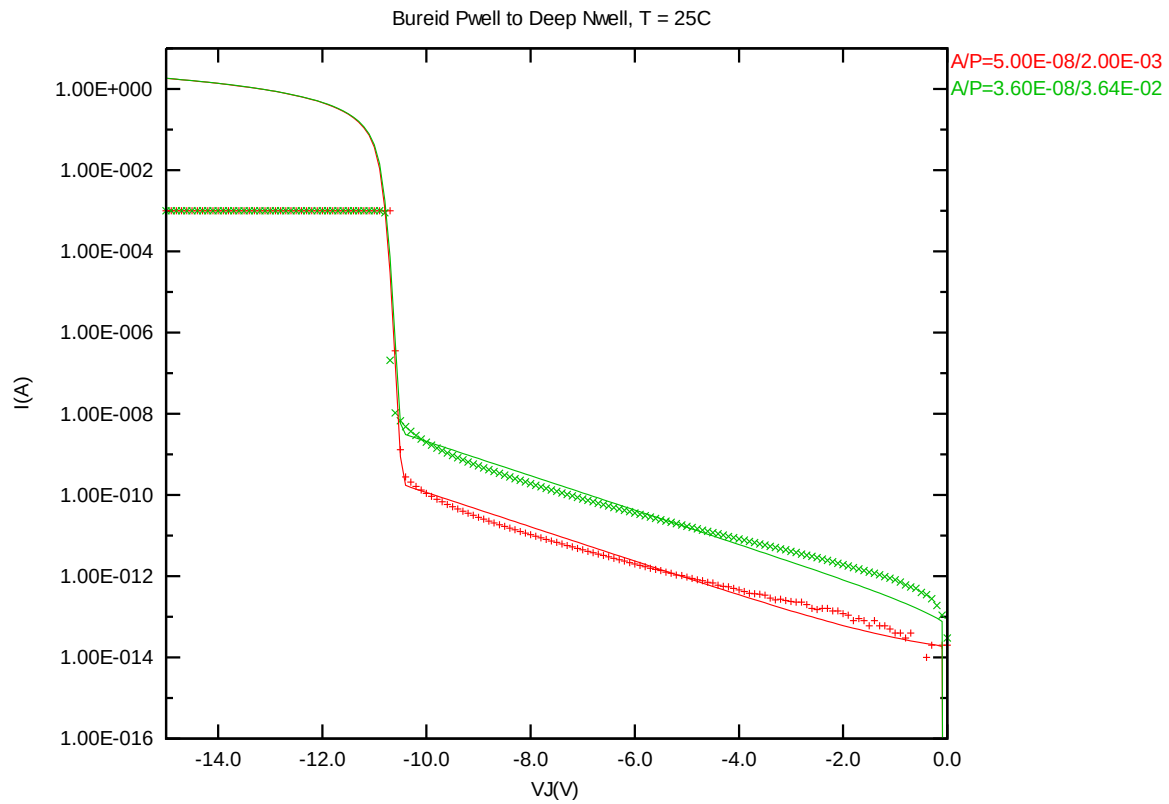


Figure 5-3 Reverse IV curve at 25 °C

5.3.2 3.3V DIOP model verification at 125 °C

➤ Diode junction CV curve at 125 °C.

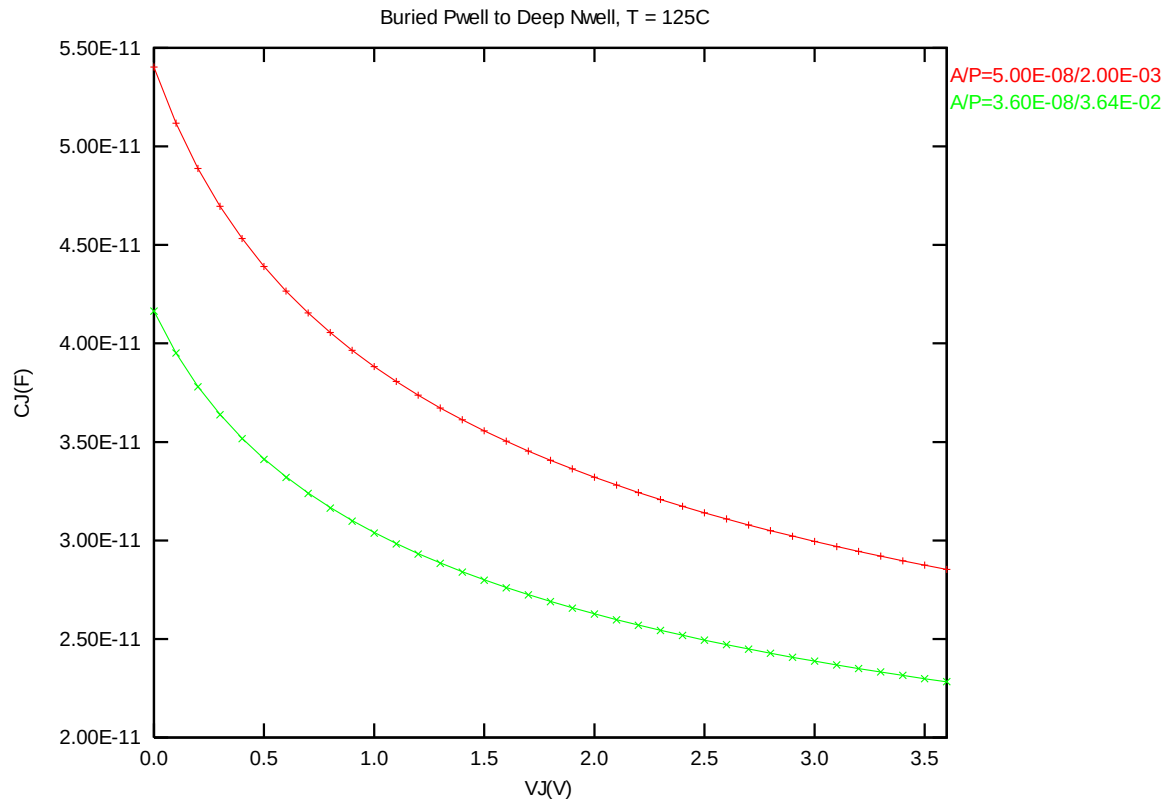
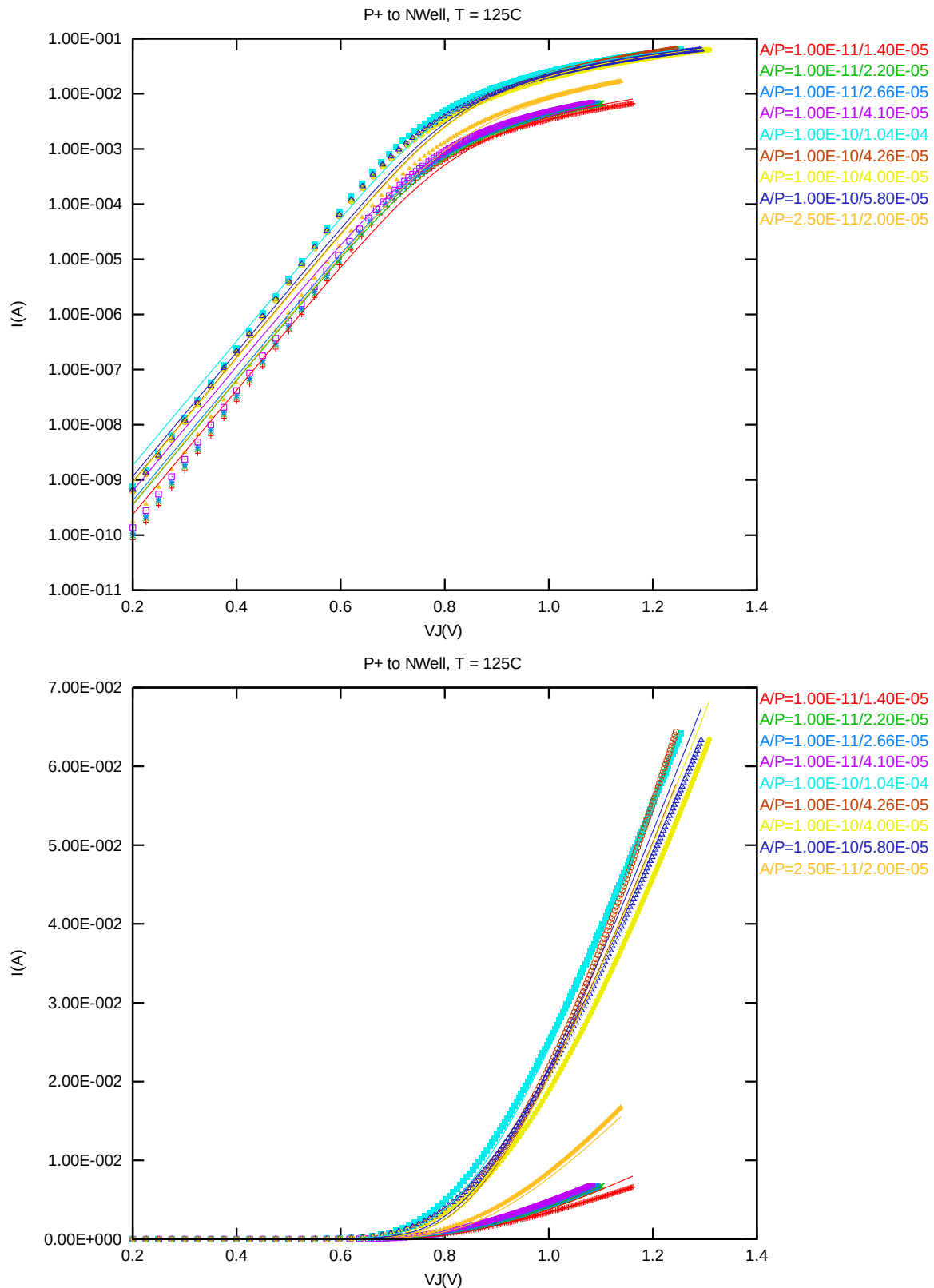


Figure 5-4 Diode junction CV curve at 125 °C

➤ **Forward IV curve at 125 °C.**



➤ **Reverse IV curve at 125 °C.**

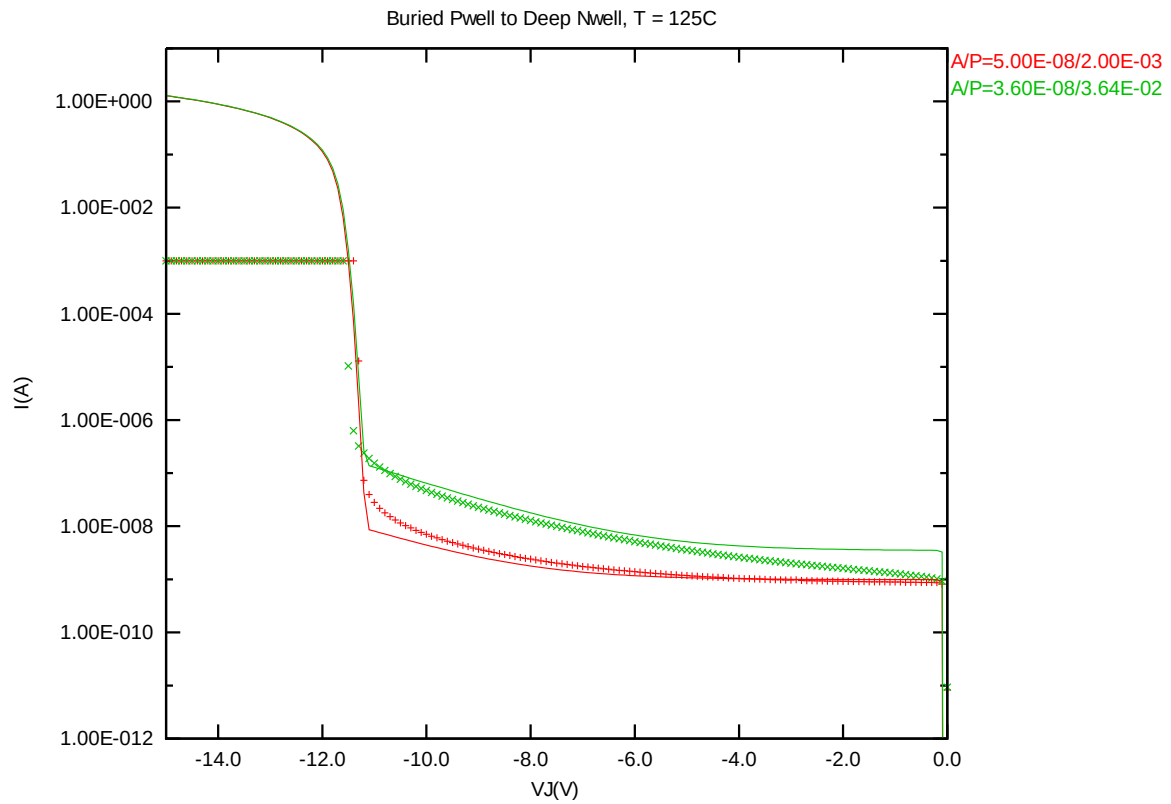


Figure 5-6 Reverse IV curve at 125 °C

5.3.3 3.3V DIOP model verification at 85 °C

➤ *Diode junction CV curve at 85 °C.*

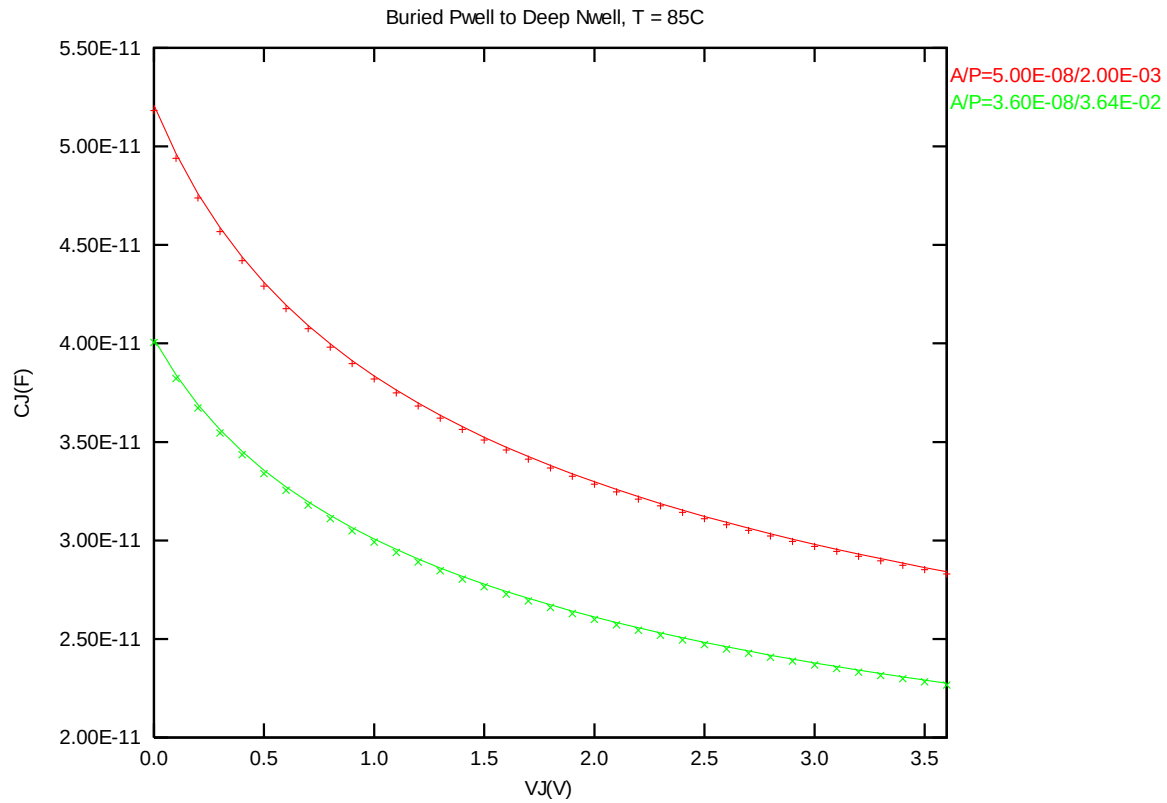


Figure 5-7 Diode junction CV curve at 85 °C

5.3.4 3.3V DIOP model verification at 0 °C

➤ Diode junction CV curve at 0 °C.

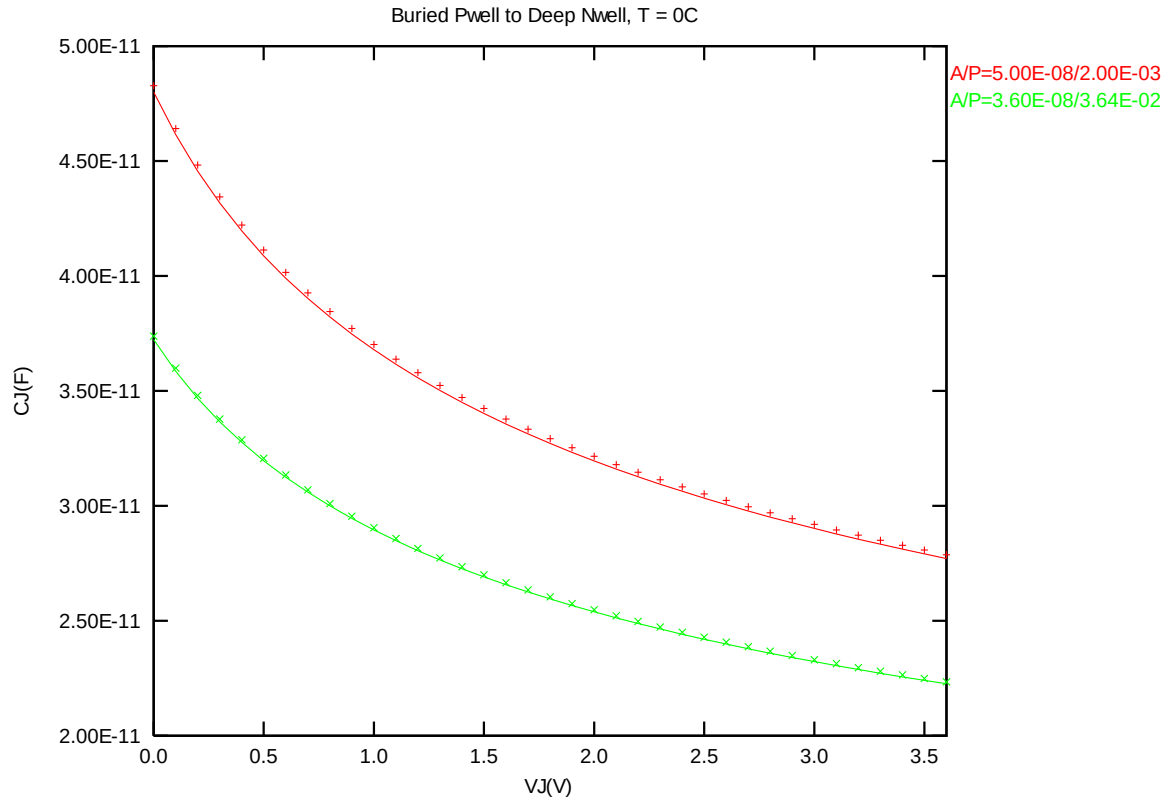


Figure 5-8 Diode junction CV curve at 0 °C

5.3.5 3.3V DIOP model verification at -50 °C

➤ Diode junction CV curve at -50 °C.

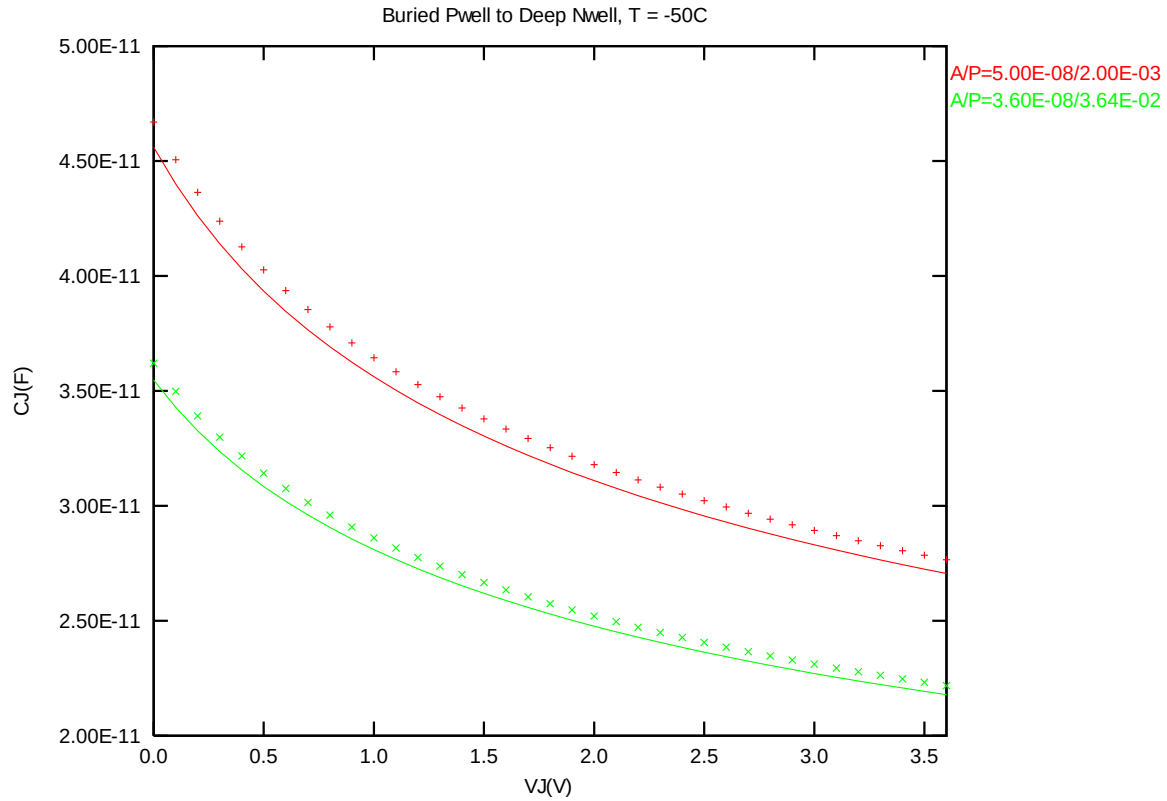
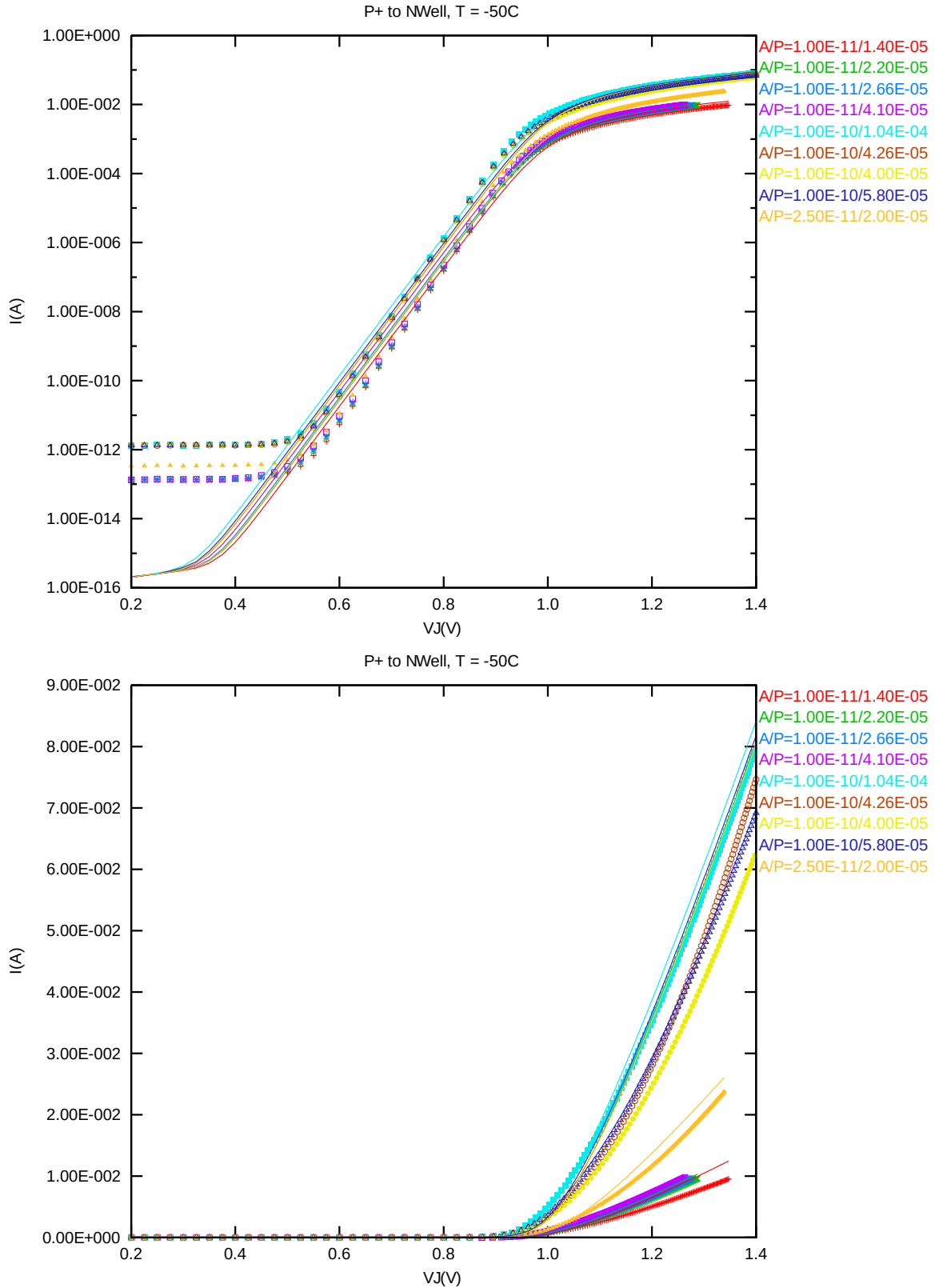


Figure 5-9 Diode junction CV curve at -50 °C

➤ **Forward IV curve at -50 °C.**



➤ **Reverse IV curve at -50 °C.**

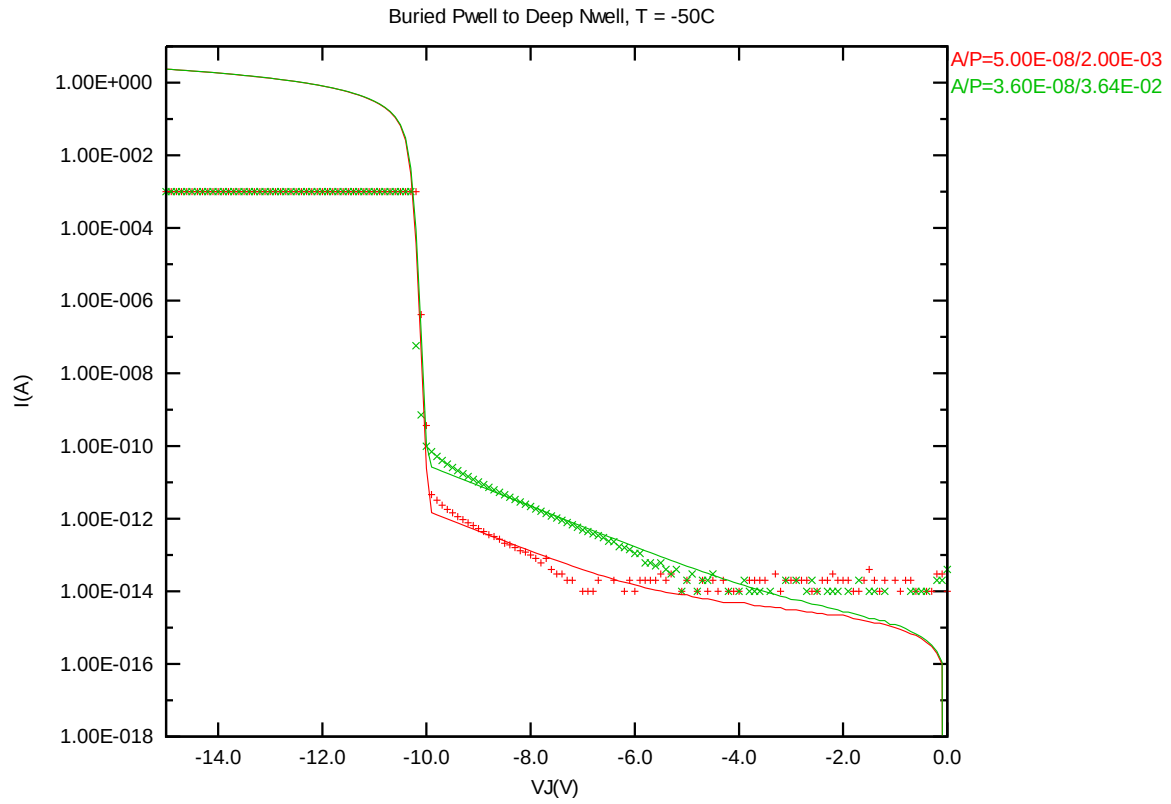


Figure 5-11 Reverse IV curve at -50 °C

5.3.6 3.3V DIOP model verification for temperature effect

➤ *Diode junction CV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C.*

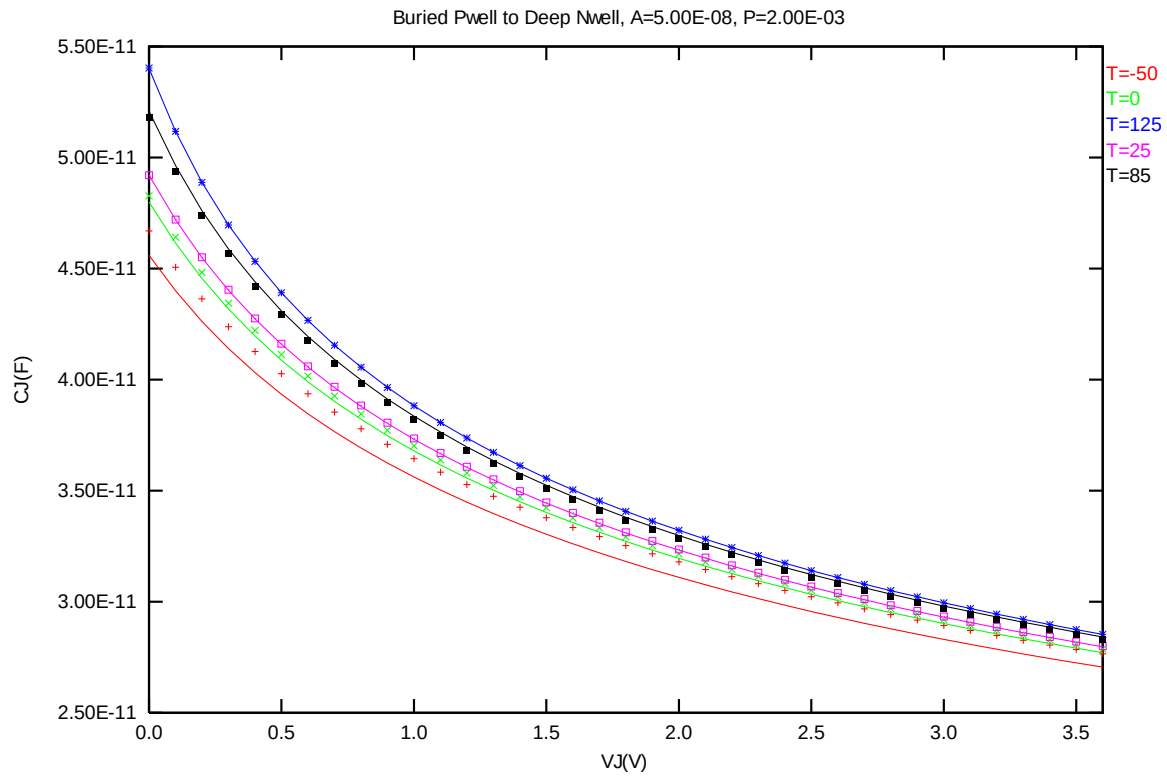


Figure 5-12 Diode junction CV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C.

➤ **Forward IV curve at 25 °C, 125 °C, -50 °C.**

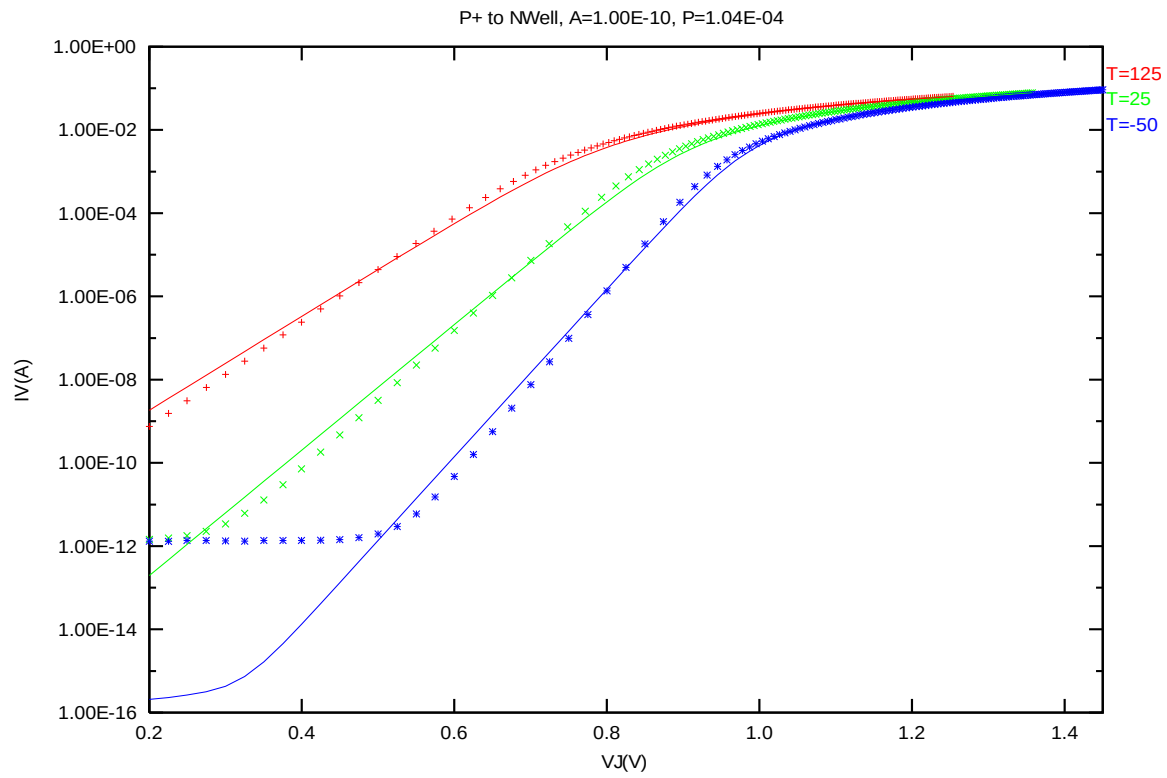


Figure 5-13 Forward IV curve at 25 °C, 125 °C, -50 °C

➤ **Reverse IV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C**

Buried Pwell to Deep Nwell, A=5.00E-08, P=2.00E-03

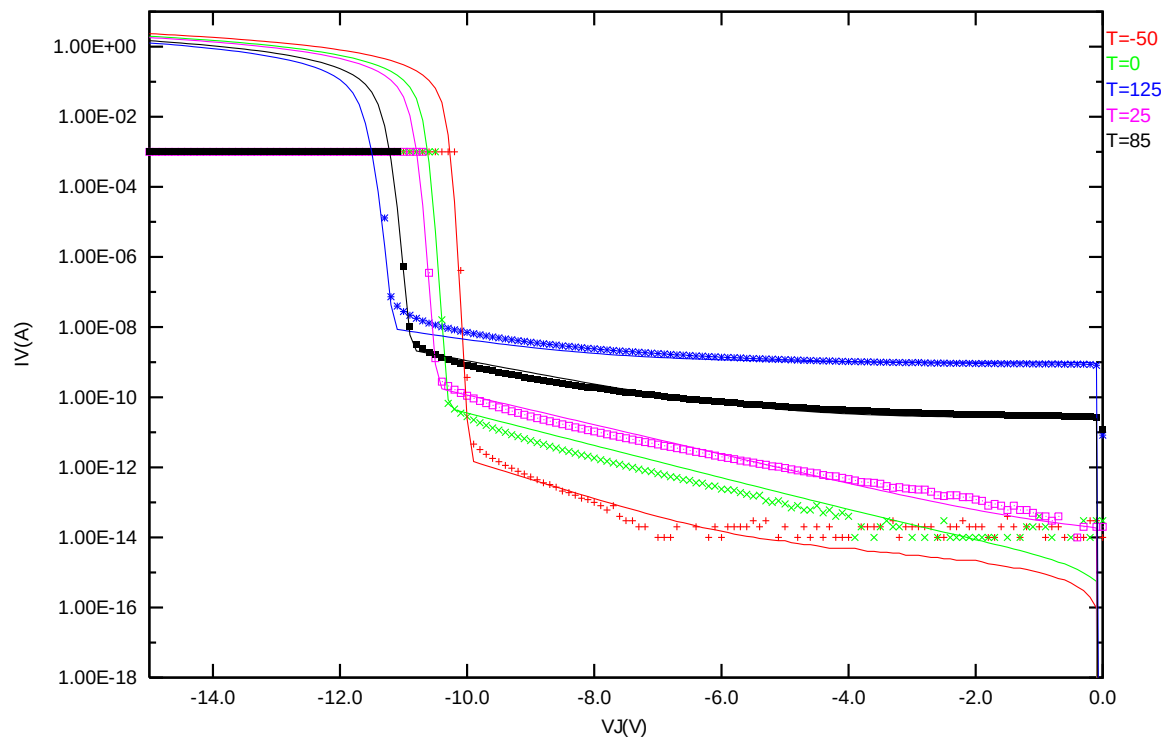


Figure 5-14 Reverse IV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C

6 1.2V DIONW Diode Model Verification

6.1 Silicon Wafer for Modeling

The information of the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 6-1 Test wafer information for DIONW

	Dionw diode model
Mask ID	BSW07A
Lot ID	D4MLP
Wafer Number	#02

This model is characterized as the following geometry and temperature ranges depicted. Device behavior beyond these ranges may not be well described by the model. This model are extracted by devices with these two dimensions :

Junction Area Devices :

Area = $1.77\text{e-}7 \text{ m}^2$

Perimeter = $4.93\text{e-}3 \text{ m}$

6.2 Testing Conditions

The model is verified to measurement data under following bias conditions

Forward and reverse diode IV curve:

Table 6-2 Forward and reverse diode IV measurement conditions

Forward Diode

	Start	Stop	Step
$V_J(V)$	0	1	0.05

Reverse Diode

	Start	Stop	Step
$V_J(V)$	0	-15	-0.1

Diode junction CV:

Table 6-3 Diode junction CV measurement conditions

V_{BIAS} [V]	VH: psub & VL: nwell From 0 to 1.32 V in 0.05 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50, 0, 25, 85, 125

6.3 1.2V DIONW model verification

6.3.1 1.2V DIONW model verification at room temperature (25 °C)

➤ *Diode junction CV curve at 25 °C.*

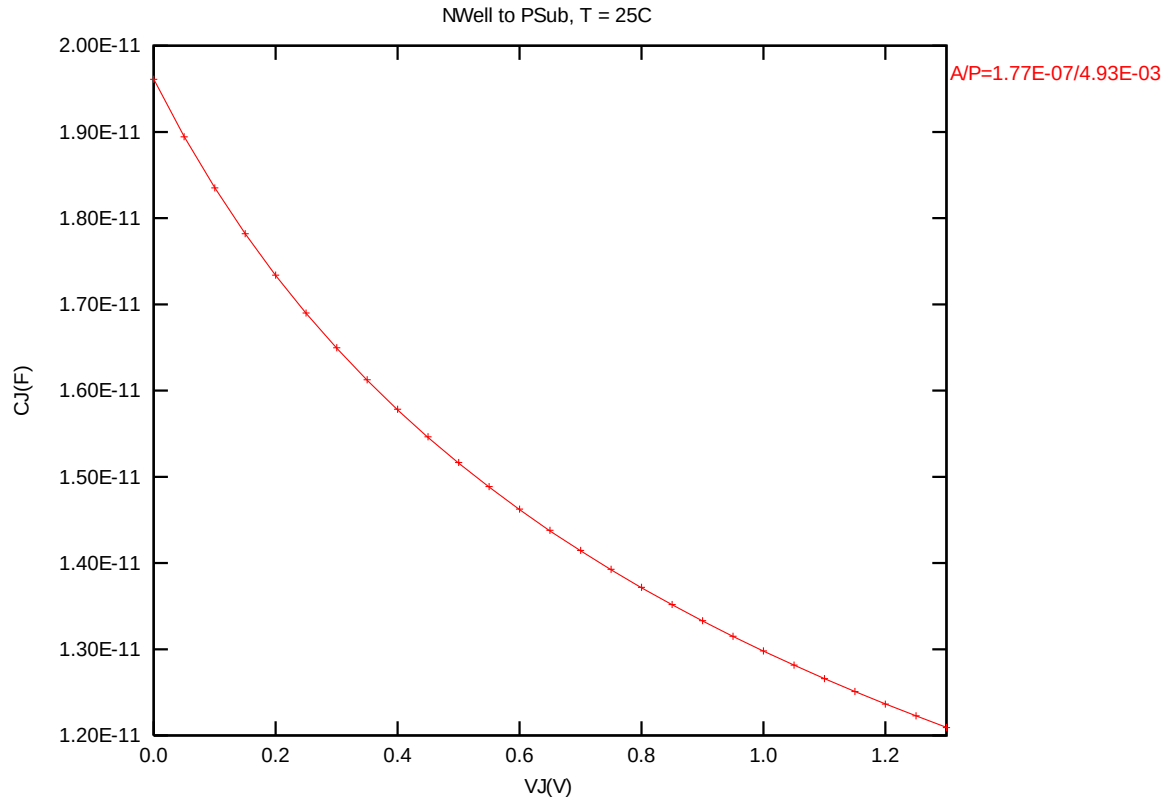


Figure 6-1 Diode junction CV curve at 25 °C

➤ **Forward IV curve at 25 °C.**

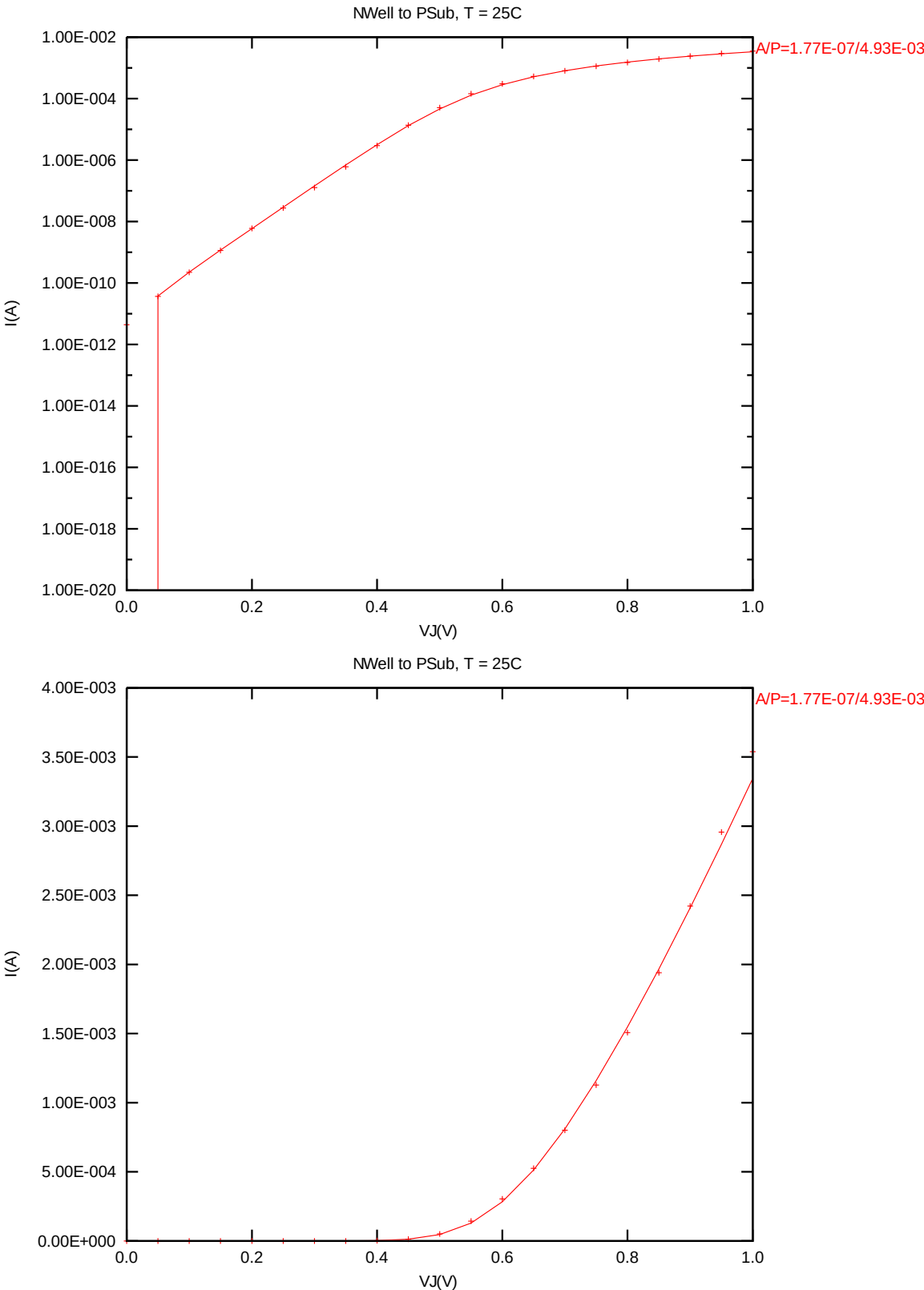


Figure 6-2 Forward IV curve at 25 °C

➤ **Reverse IV curve at 25 °C.**

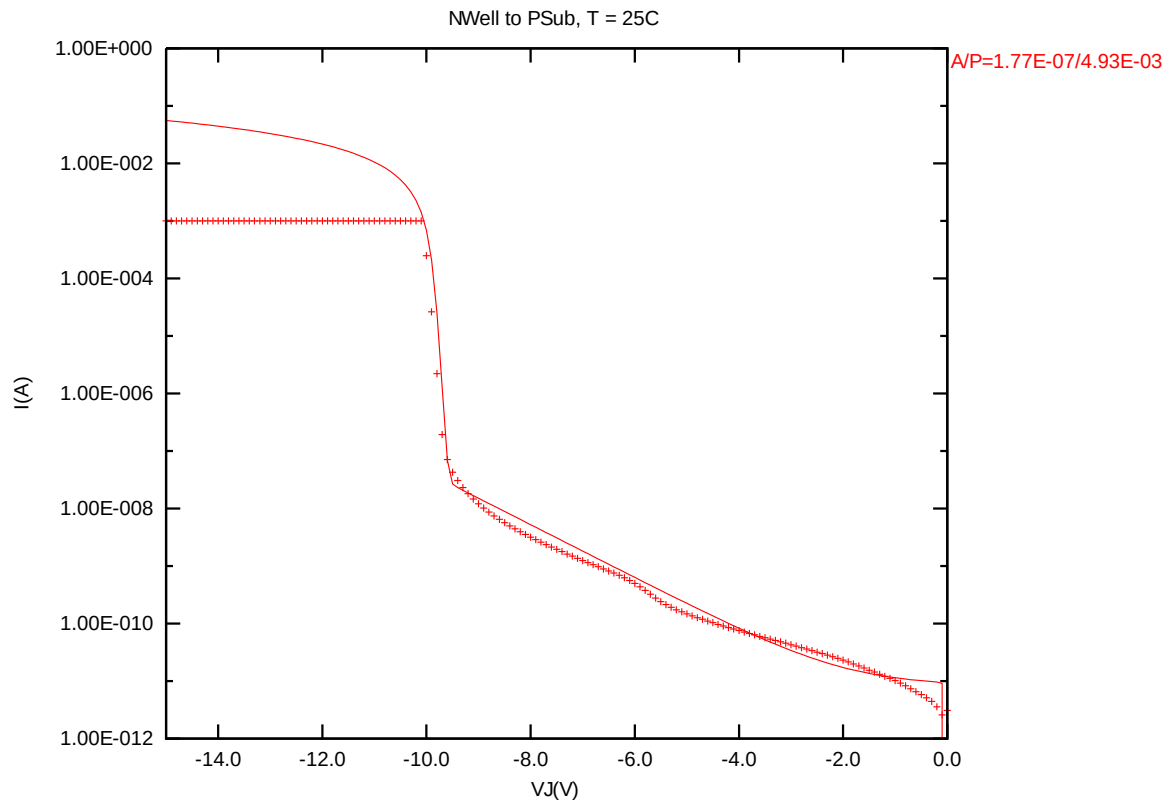


Figure 6-3 Reverse IV curve at 25 °C

6.3.2 1.2V DIONW model verification at 125 °C

➤ *Diode junction CV curve at 125 °C.*

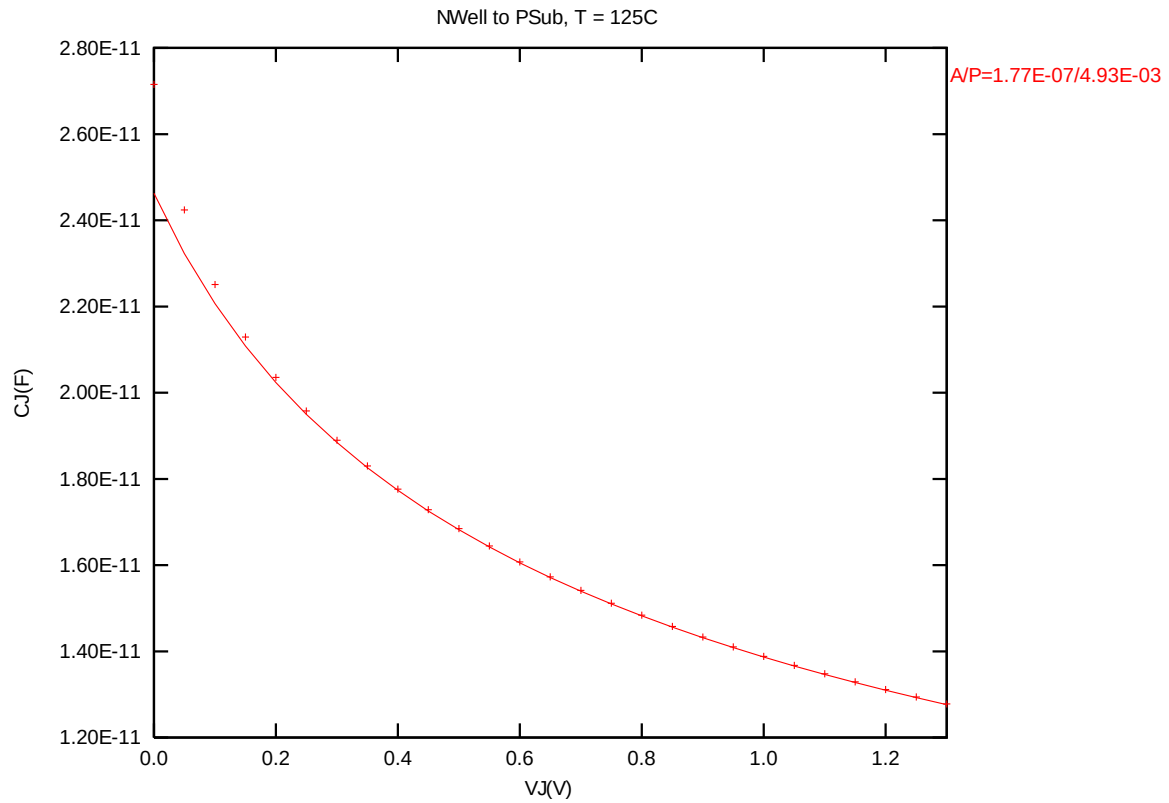


Figure 6-4 Diode junction CV curve at 125 °C

➤ **Forward IV curve at 125 °C.**

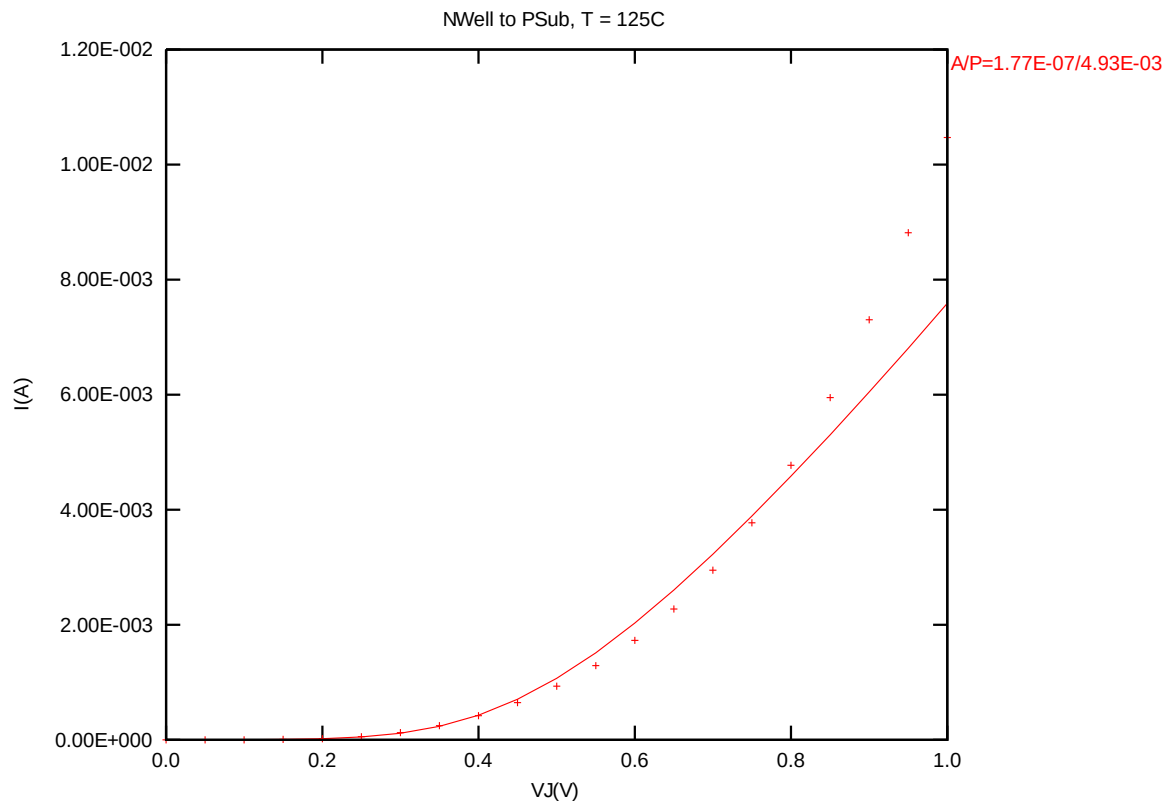
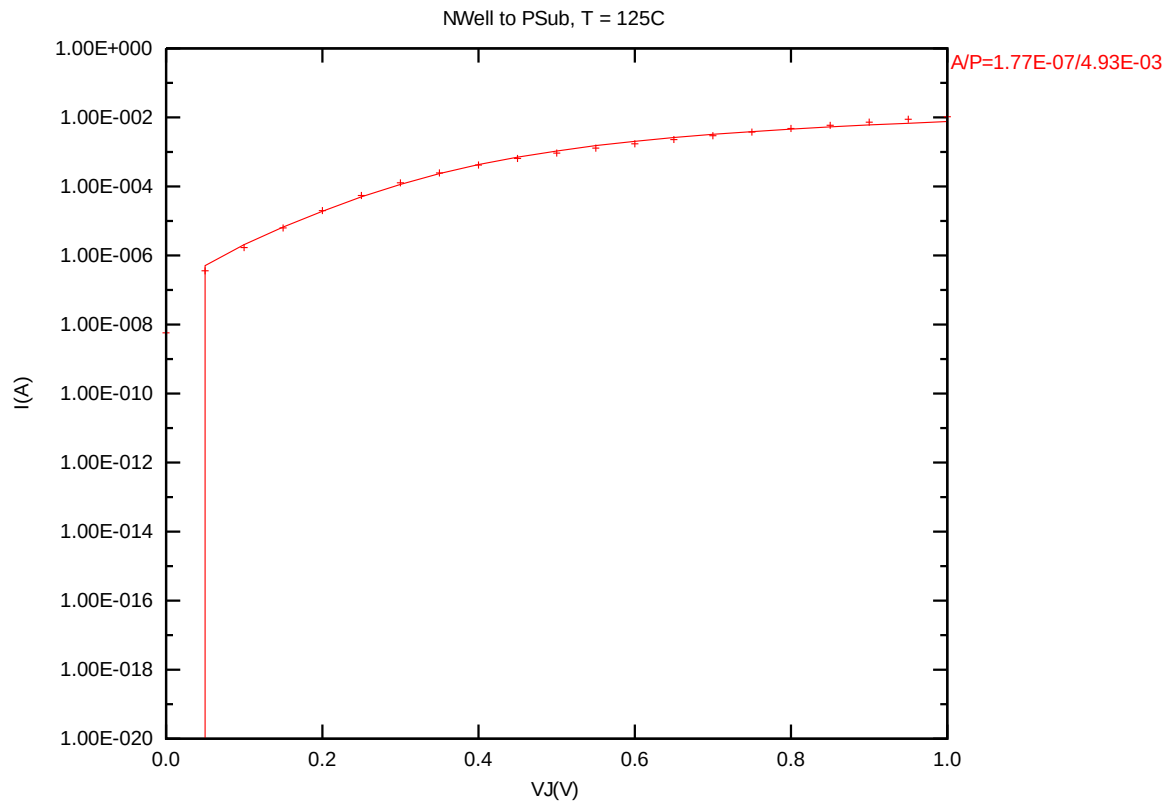


Figure 6-5 Forward IV curve at 125 °C

➤ **Reverse IV curve at 125 °C.**

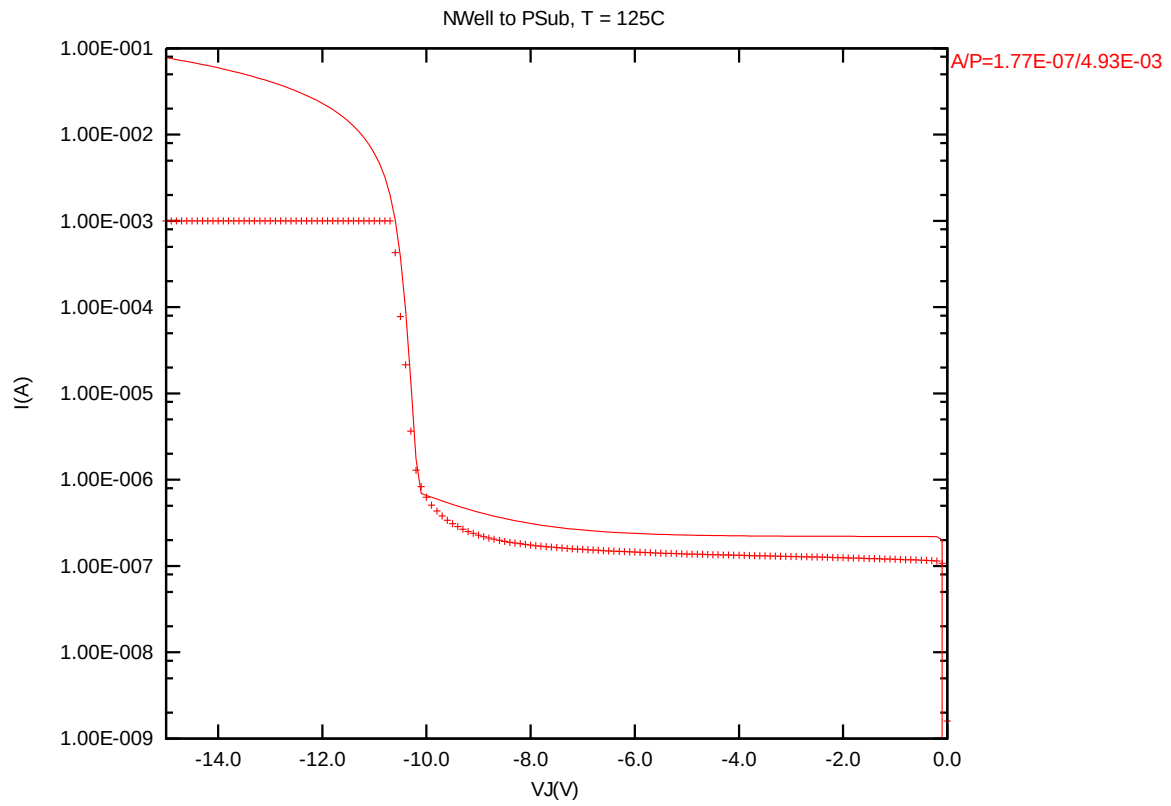


Figure 6-6 Reverse IV curve at 125 °C

6.3.3 1.2V DIONW model verification at 85 °C

➤ *Diode junction CV curve at 85 °C.*

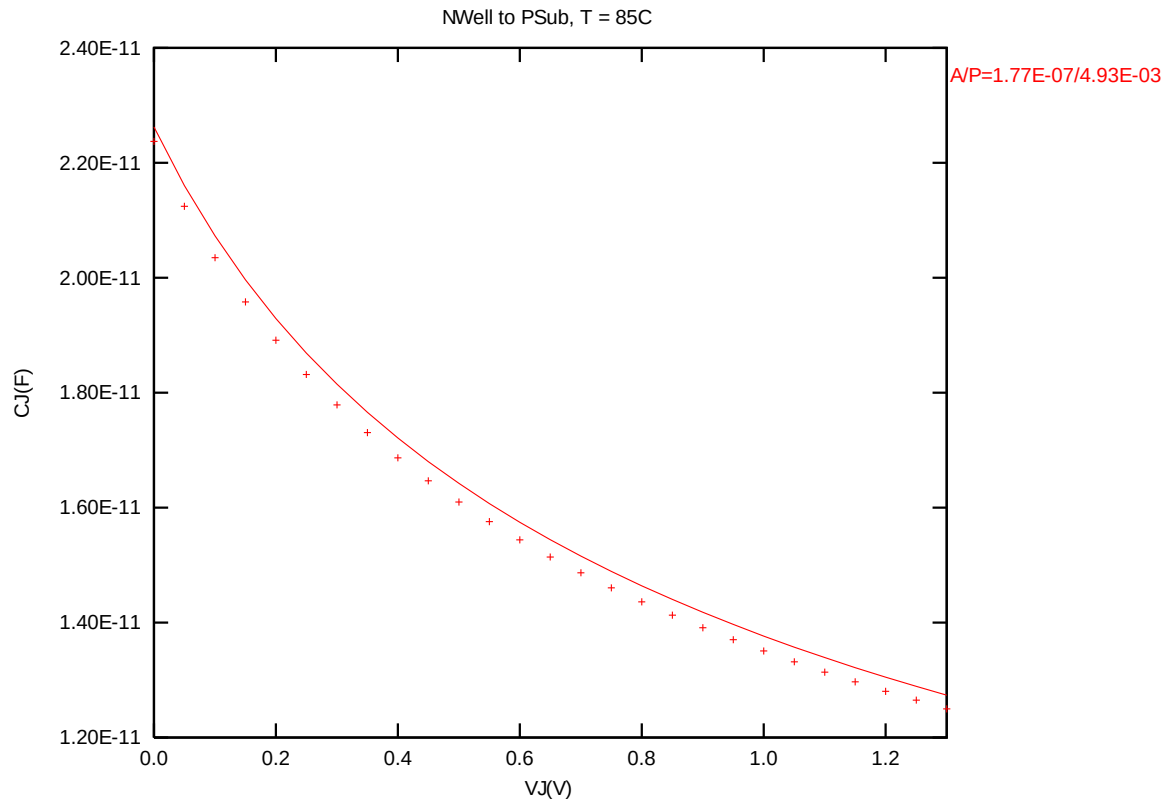


Figure 6-7 Diode junction CV curve at 85 °C

➤ **Forward IV curve at 85 °C.**

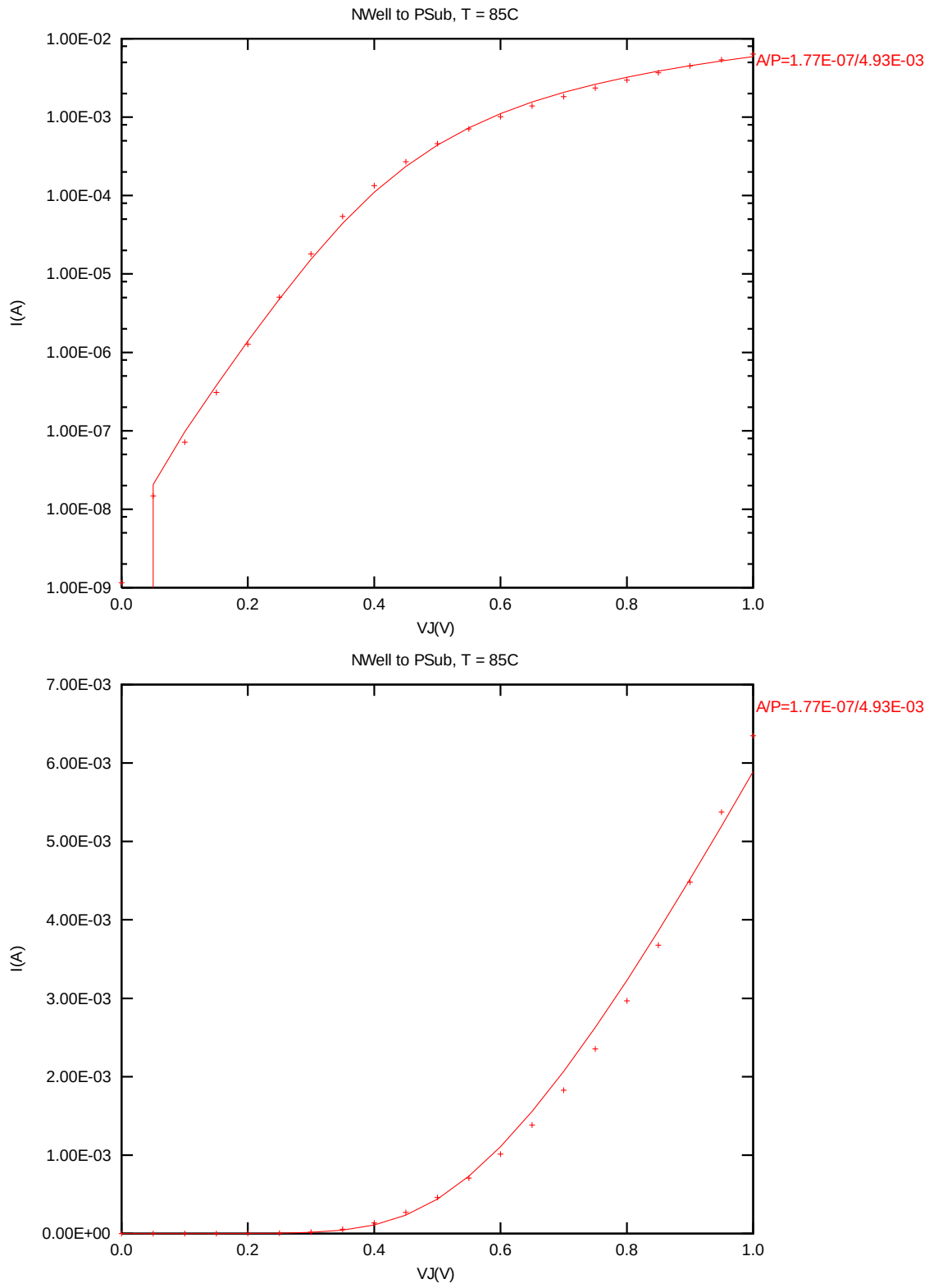


Figure 6-8 Forward IV curve at 85 °C

➤ **Reverse IV curve at 85 °C.**

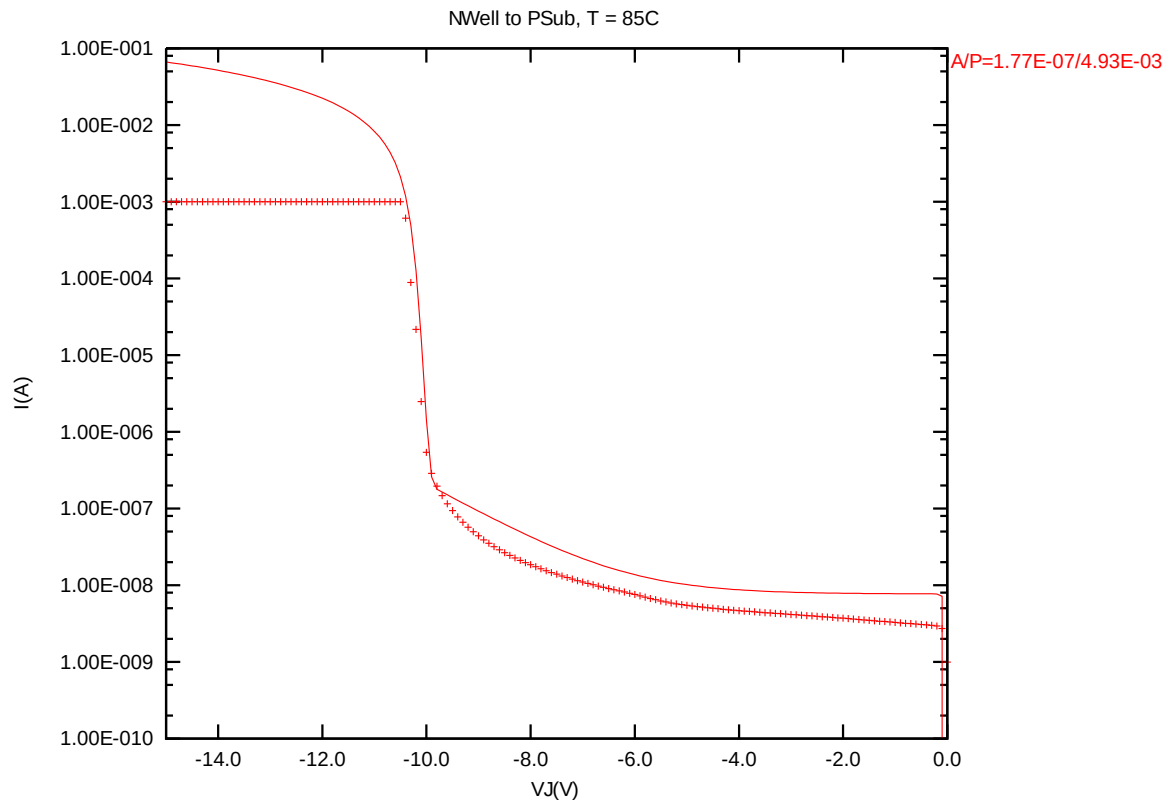


Figure 6-9 Reverse IV curve at 85 °C

6.3.4 1.2V DIONW model verification at 0 °C

➤ *Diode junction CV curve at 0 °C.*

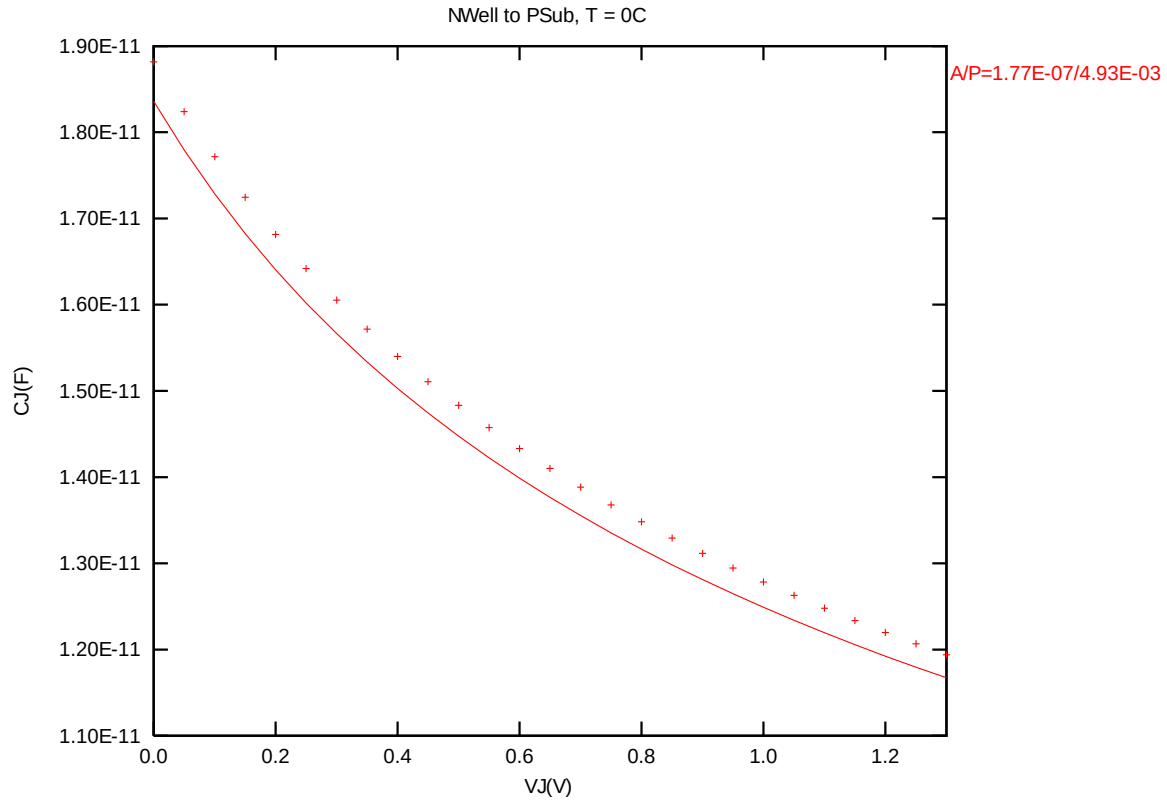


Figure 6-10 Diode junction CV curve at 0 °C

➤ **Forward IV curve at 0 °C.**

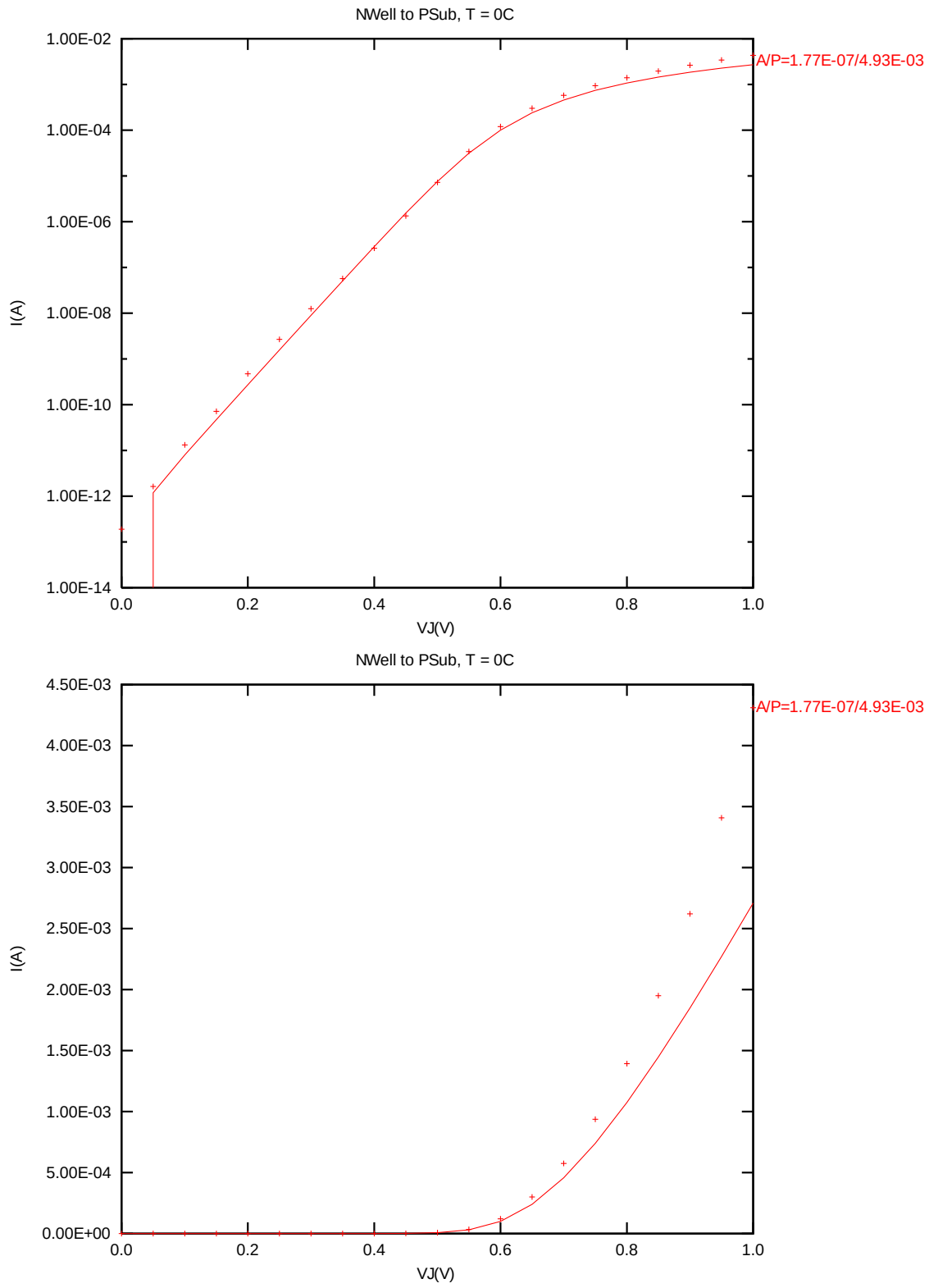


Figure 6-11 Forward IV curve at 0 °C

➤ **Reverse IV curve at 0 °C.**

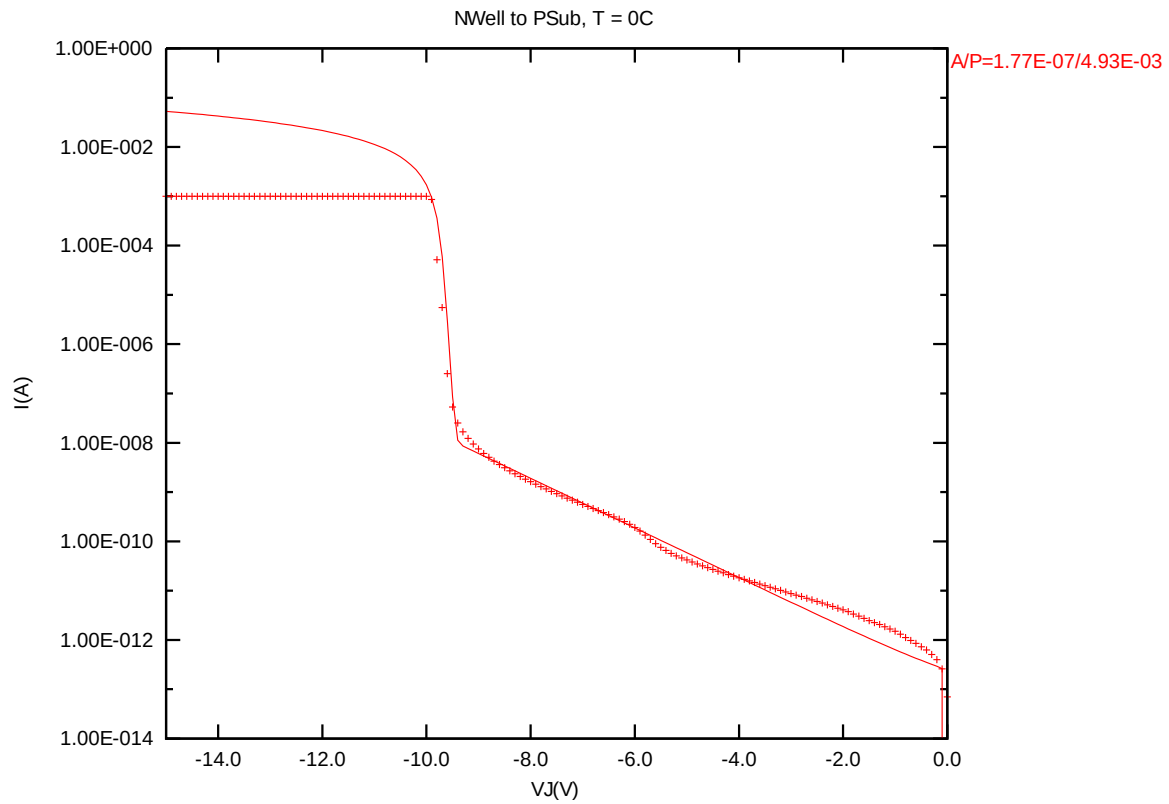


Figure 6-12 Reverse IV curve at 0 °C

6.3.5 1.2V DIONW model verification at -50 °C

➤ *Diode junction CV curve at -50 °C.*

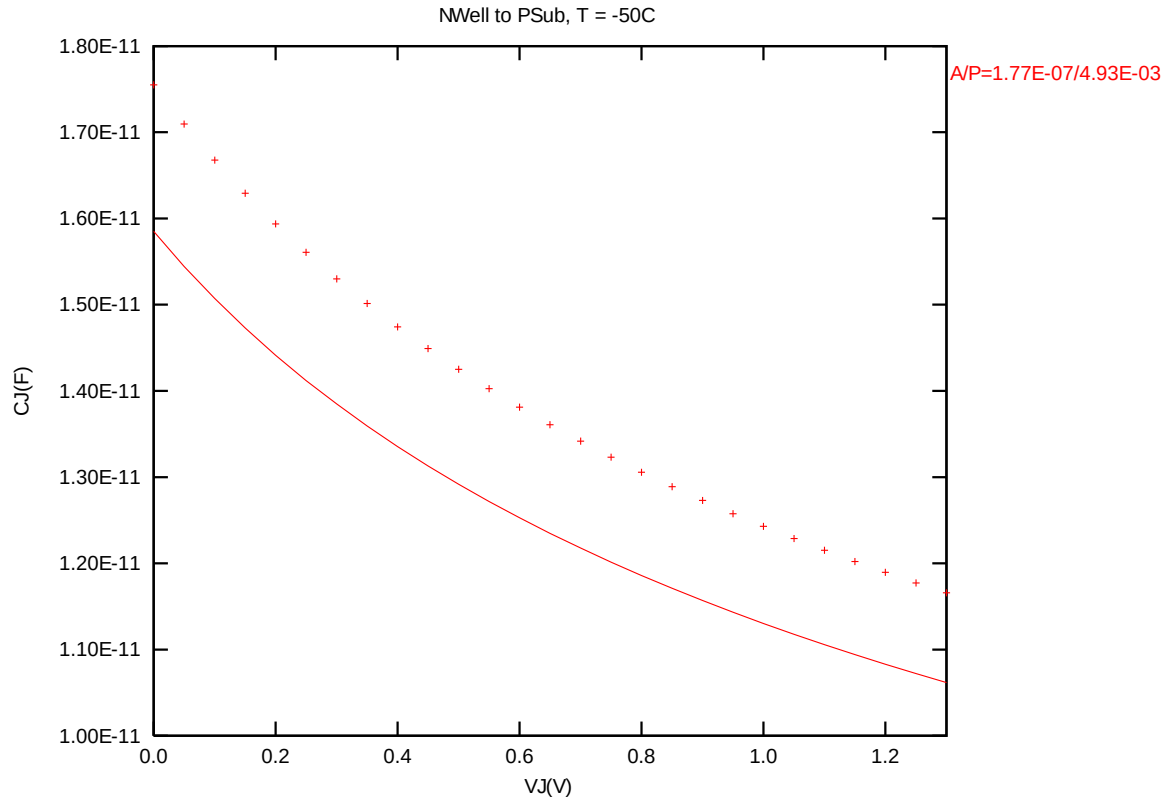


Figure 6-13 Diode junction CV curve at -50 °C

➤ **Forward IV curve at -50 °C.**

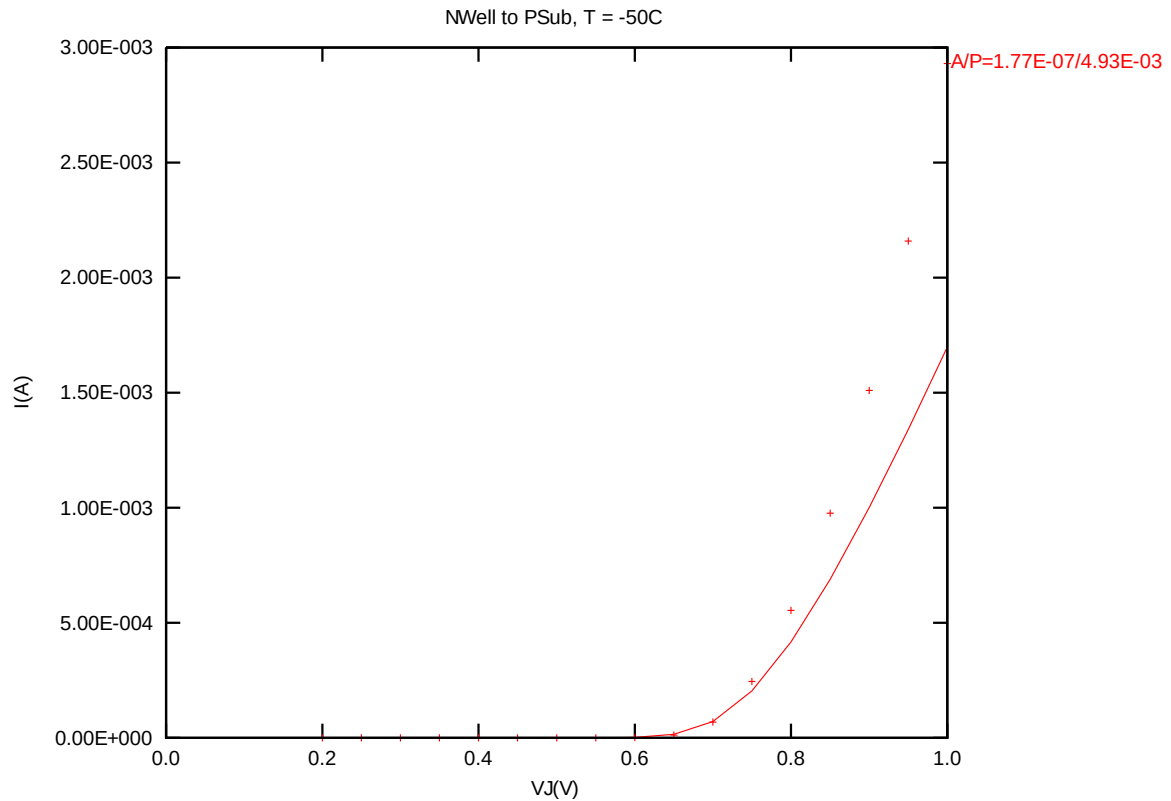
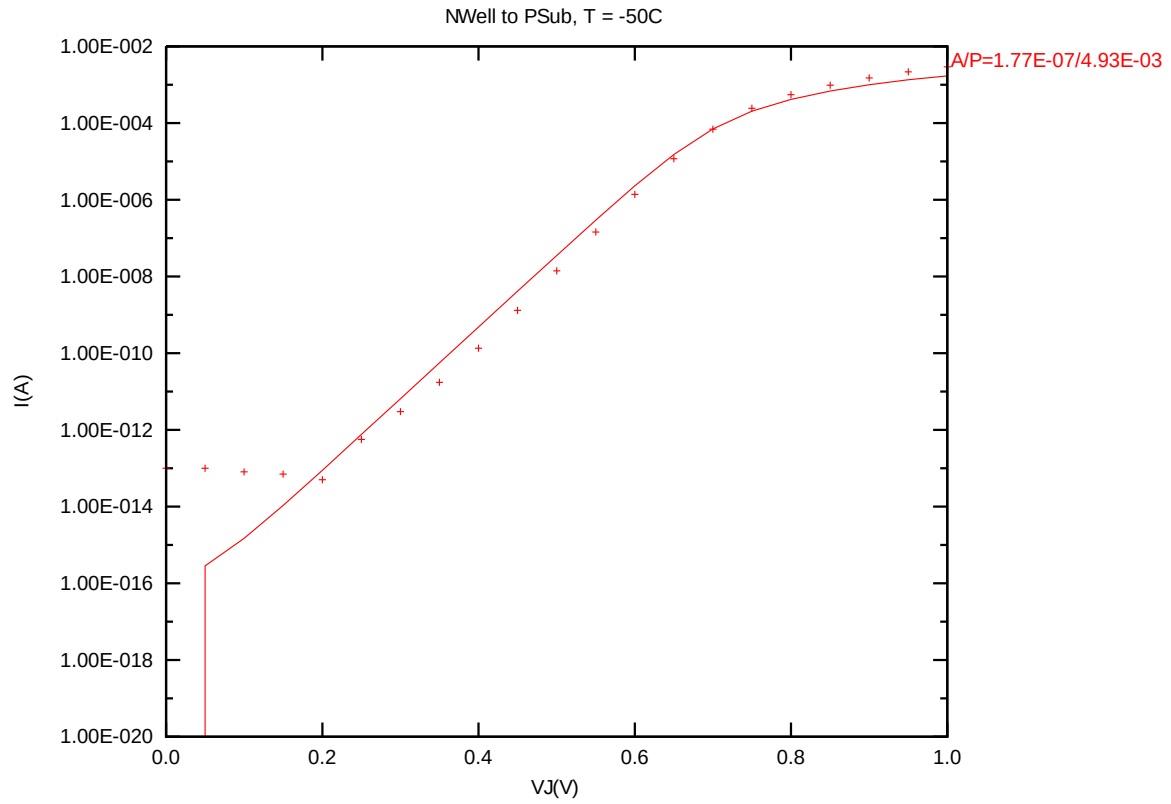


Figure 6-14 Forward IV curve at -50 °C

➤ **Reverse IV curve at -50 °C.**

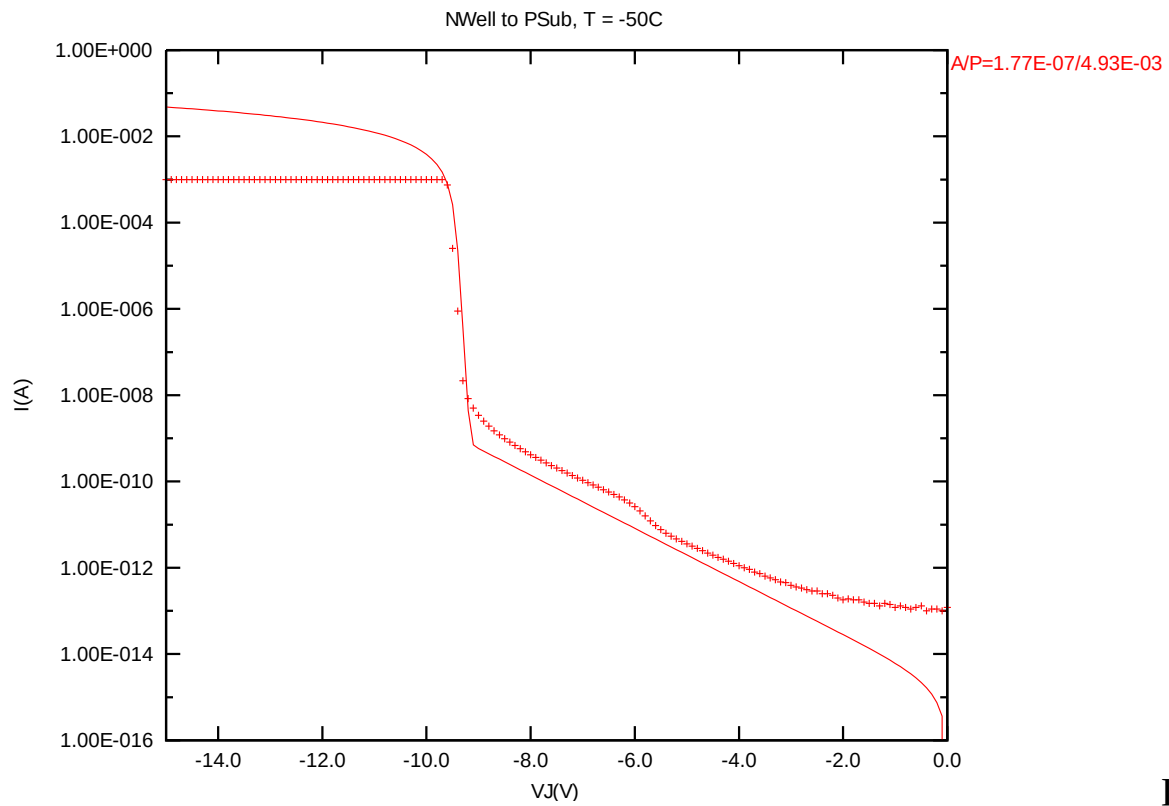


Figure 6-15 Reverse IV curve at -50 °C

6.3.6 1.2V DIONW model verification for temperature effect

➤ *Diode junction CV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C.*

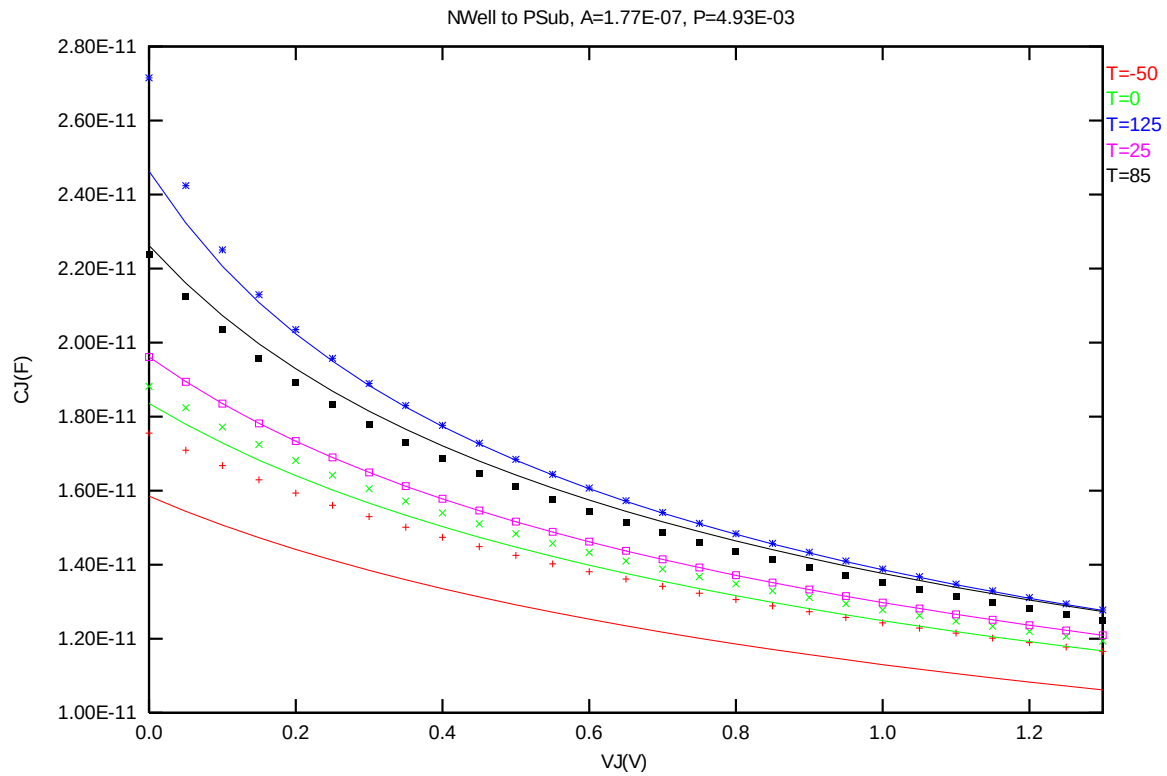


Figure 6-16 Diode junction CV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C.

➤ **Forward IV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C.**

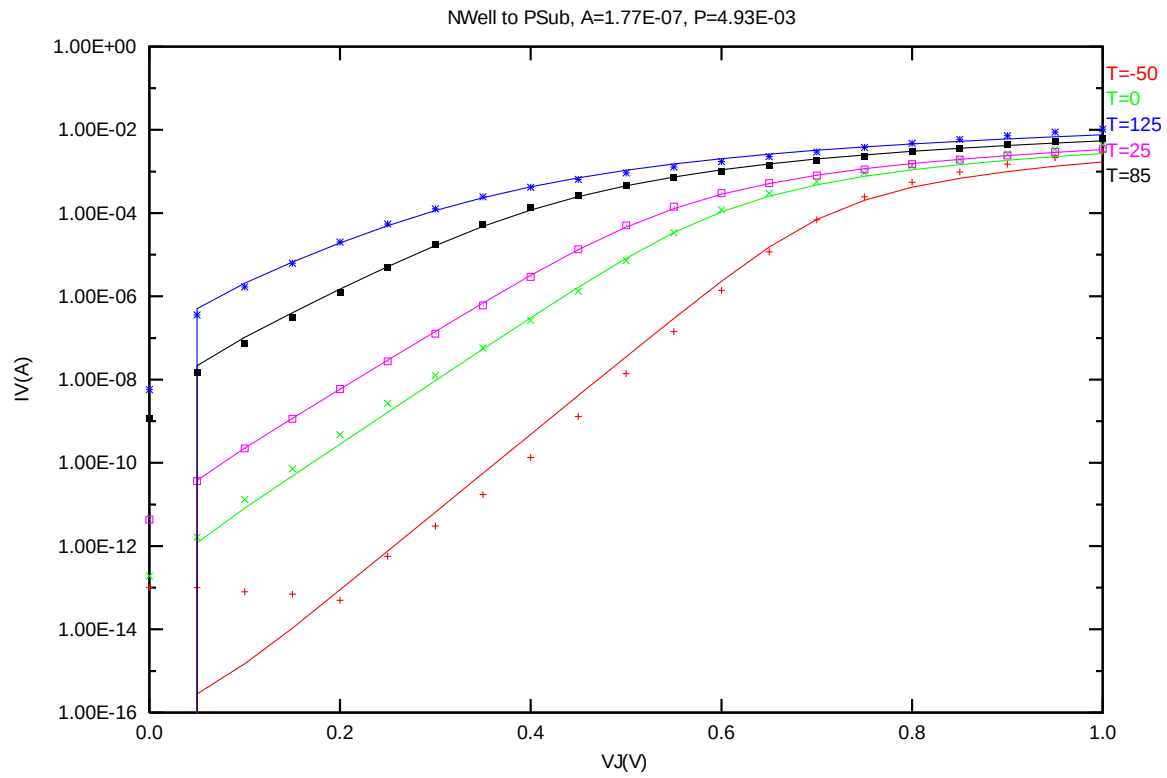


Figure 6-17 Forward IV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C

➤ **Reverse IV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C**

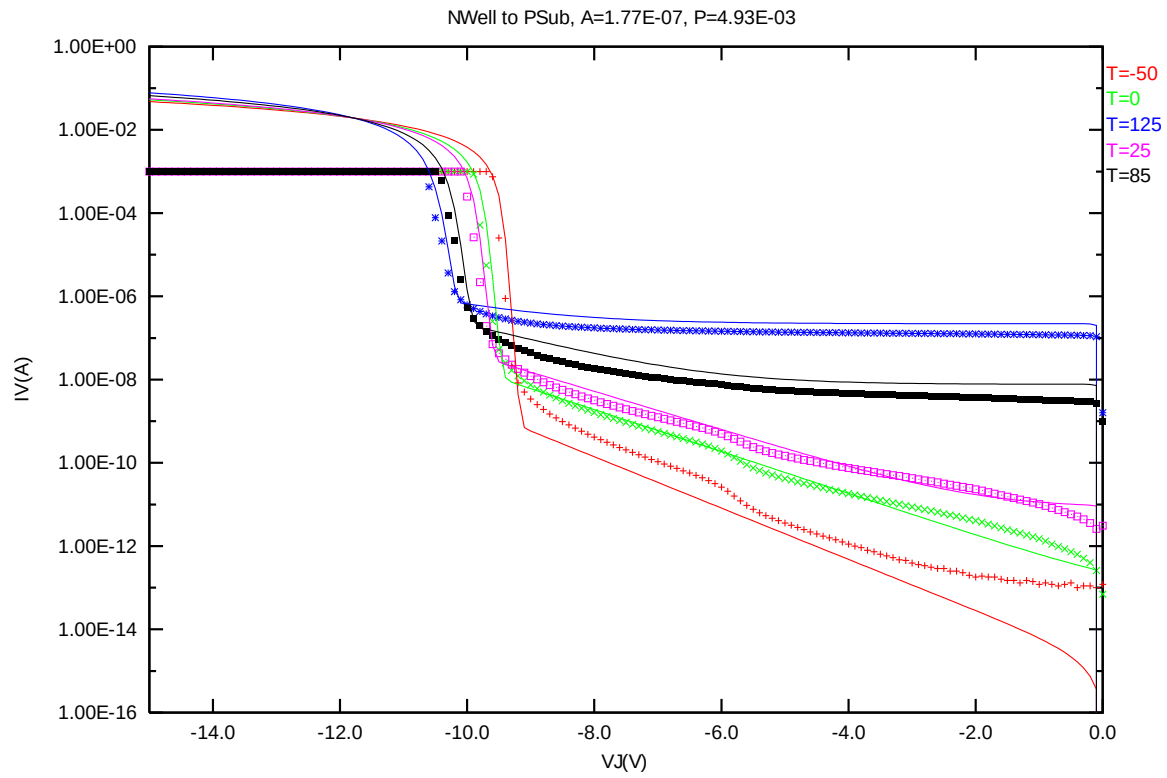


Figure 6-18 Reverse IV curve at 25 °C, 125 °C, 85 °C, 0 °C -50 °C

7 DIODNW Diode Model Verification

7.1 Silicon Wafer for Modeling

The information of the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 7-1 Test wafer information for DIODNW

	Diodnw diode model
Mask ID	BSW0FA
Lot ID	D566R.1
Wafer Number	#01

This model is characterized as the following geometry and temperature ranges depicted. Device behavior beyond these ranges may not be well described by the model. This model are extracted by devices with these two dimensions :

Junction Area Devices :

$$\text{Area} = 3.2\text{e-}8 \text{ m}^2$$

$$\text{Perimeter} = 3.2\text{e-}4 \text{ m}$$

Junction Edge Devices :

$$\text{Area} = 3.2\text{e-}8 \text{ m}^2$$

$$\text{Perimeter} = 6.4\text{e-}3 \text{ m}$$

7.2 Testing Conditions

The model is verified to measurement data under following bias conditions

Forward and reverse diode IV curve:

Table 7-2 Forward and reverse diode IV measurement conditions

Forward Diode

	Start	Stop	Step
$V_J(V)$	0	1	0.05

Reverse Diode

	Start	Stop	Step
$V_J(V)$	0	-15	-0.1

Diode junction CV:

Table 7-3 Diode junction CV measurement conditions

$V_{BIAS} [V]$	VH: psub & VL: nwell From 0 to 3.63 V in 0.033 V step
Frequency [kHz]	100
$V_{OSC} [mV]$	50
Temp [°C]	-50, 25, 125

7.3 DIODNW model verification

7.3.1 DIODNW model verification at room temperature (25 °C)

➤ *Diode junction CV curve at 25 °C.*

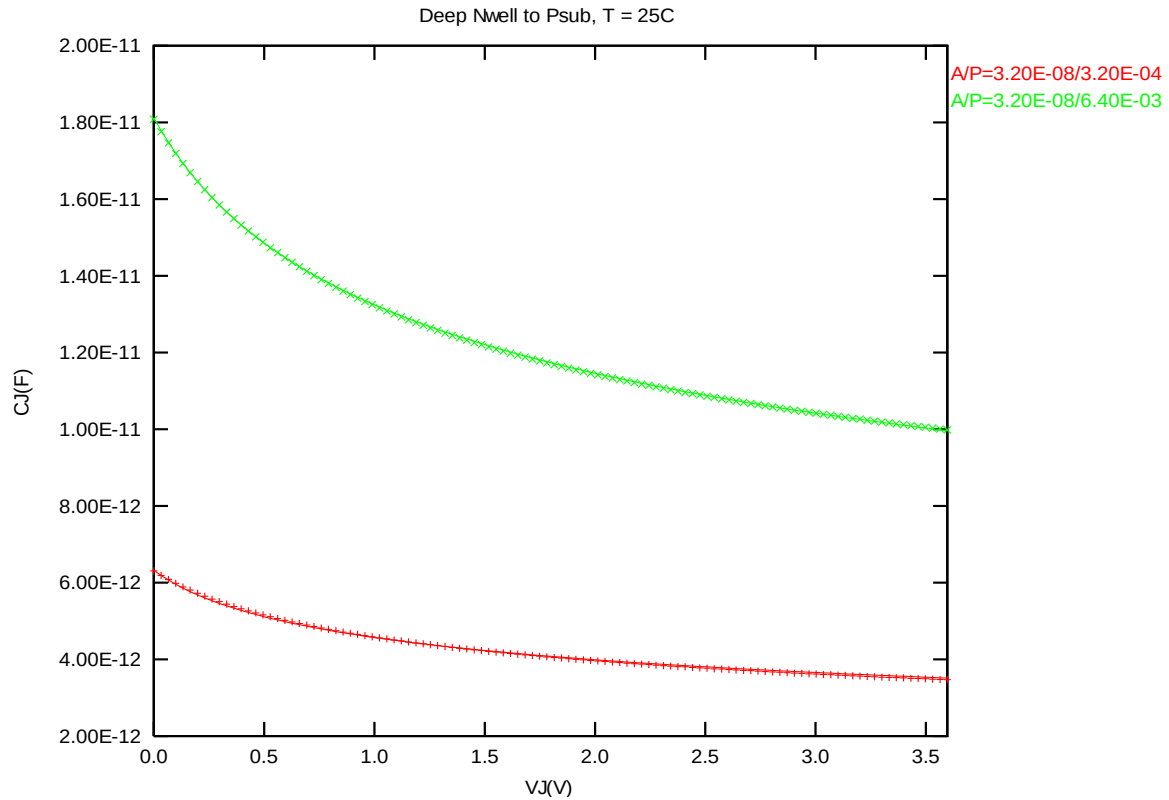


Figure 7-1 Diode junction CV curve at 25 °C

➤ **Forward IV curve at 25 °C.**

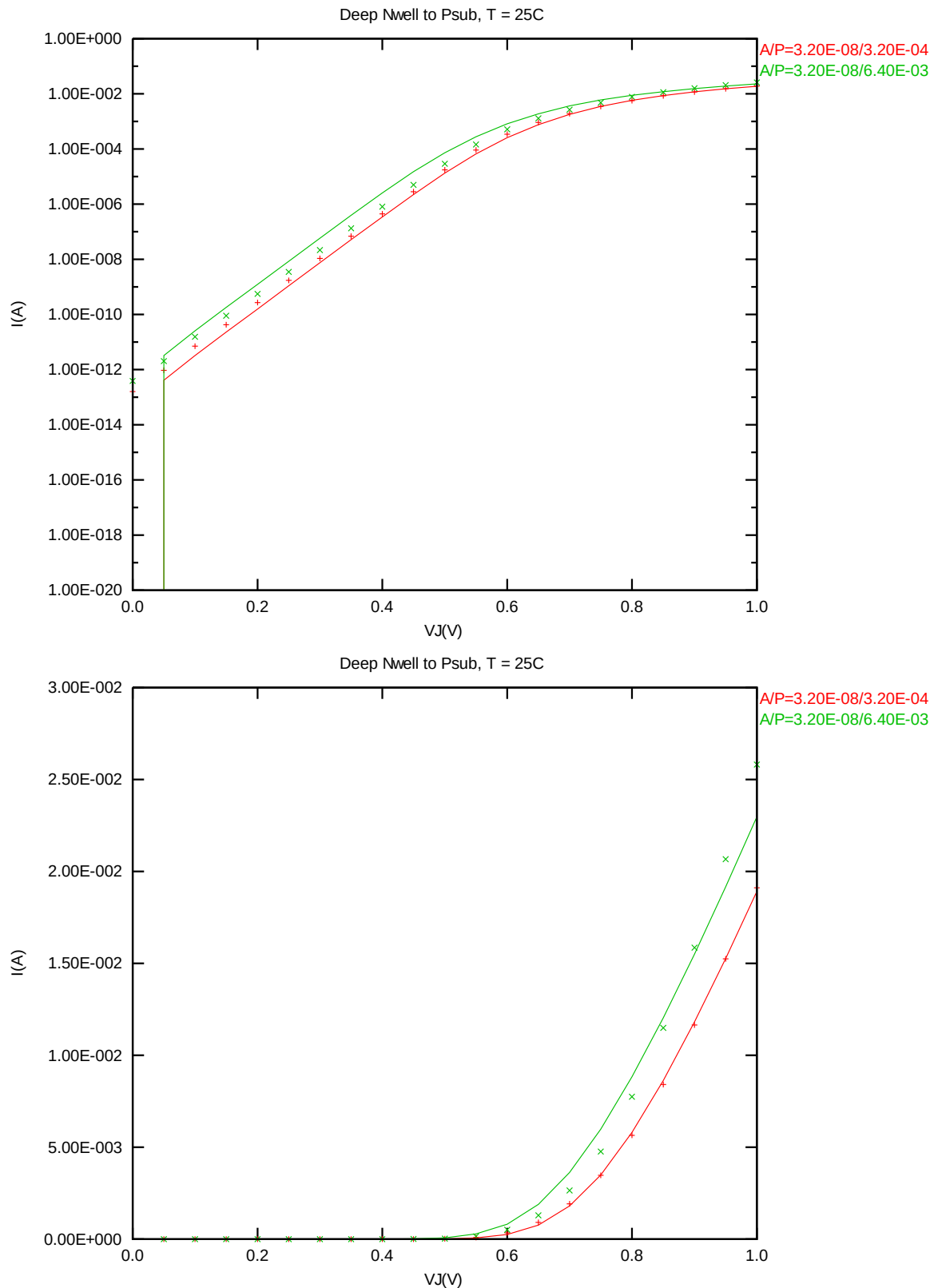


Figure 7-2 Forward IV curve at 25 °C

➤ **Reverse IV curve at 25 °C.**

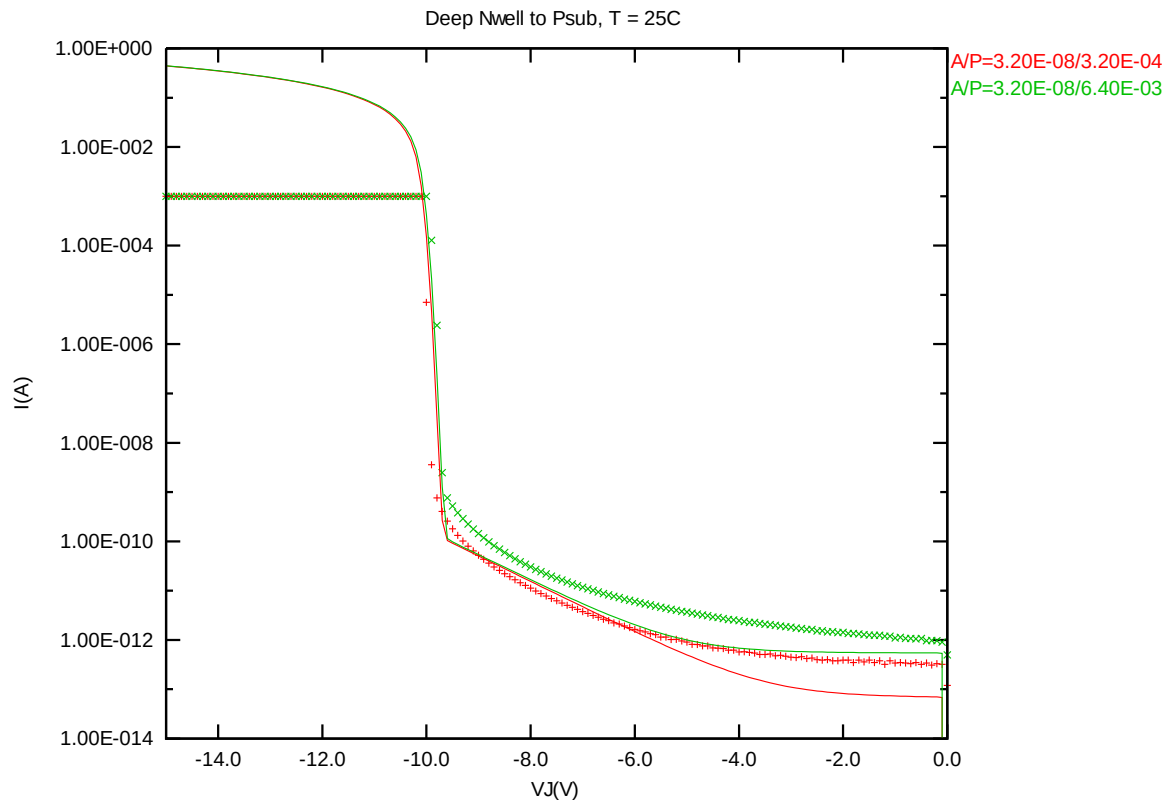


Figure 7-3 Reverse IV curve at 25 °C

7.3.2 DIODNW model verification at 125 °C

➤ *Diode junction CV curve at 125 °C.*

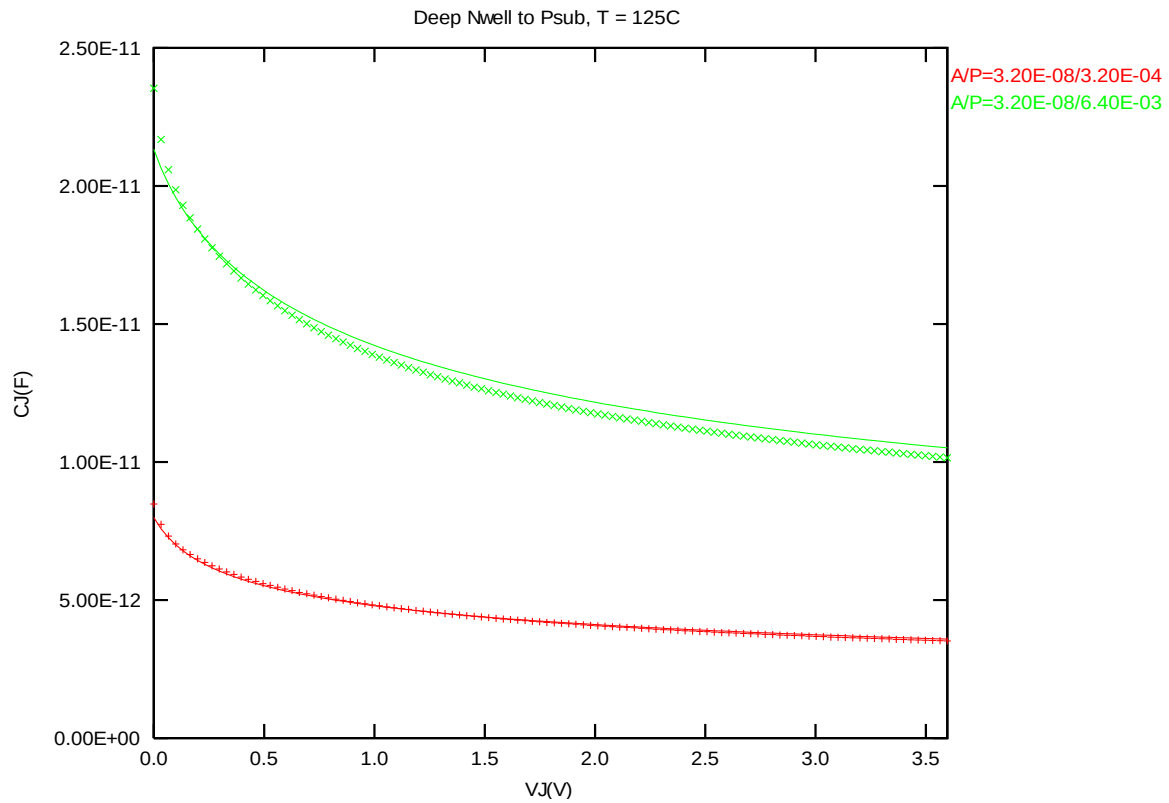


Figure 7-4 Diode junction CV curve at 125 °C

➤ **Forward IV curve at 125 °C.**

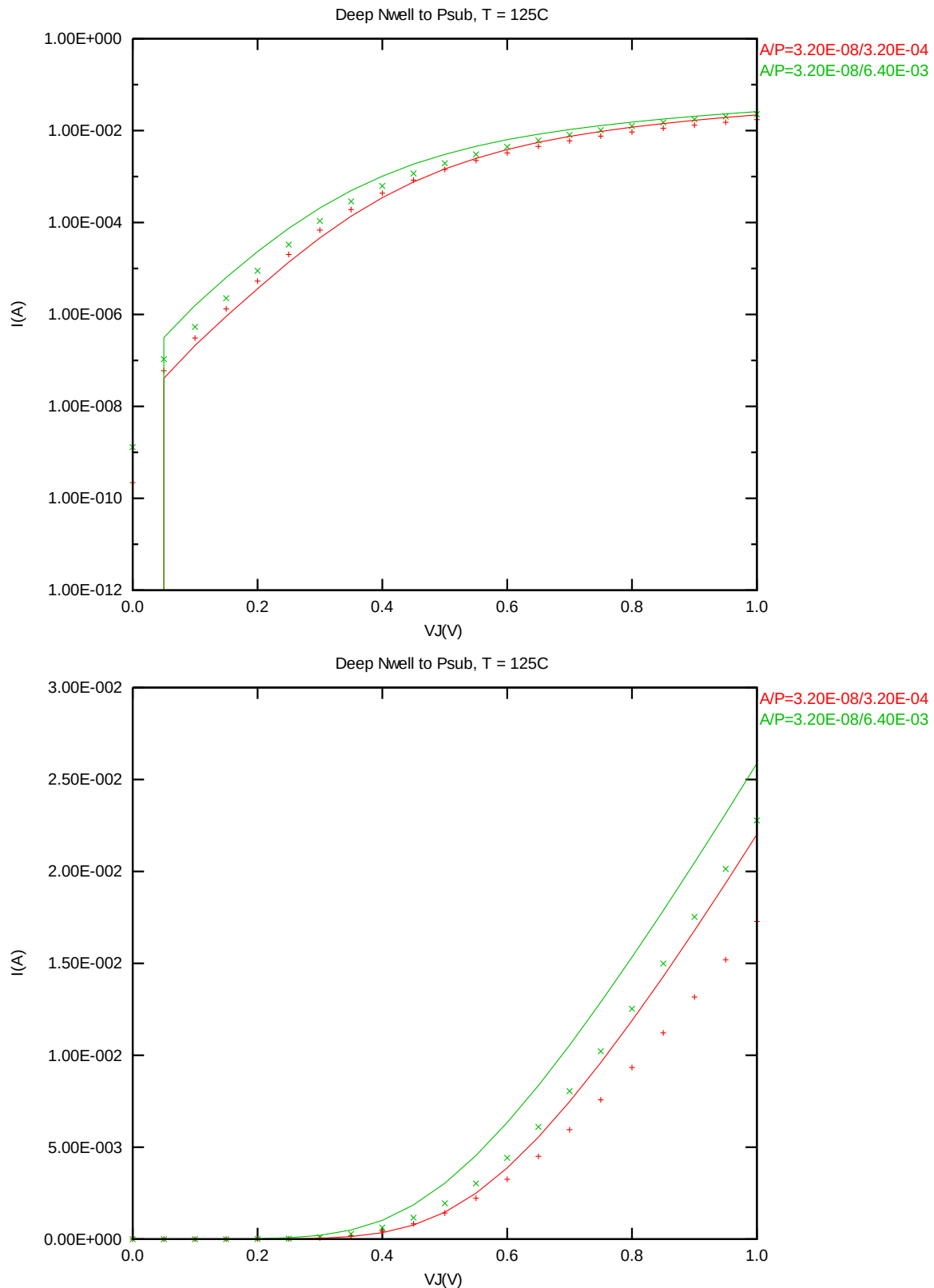


Figure 7-5 Forward IV curve at 125 °C

➤ **Reverse IV curve at 125 °C.**

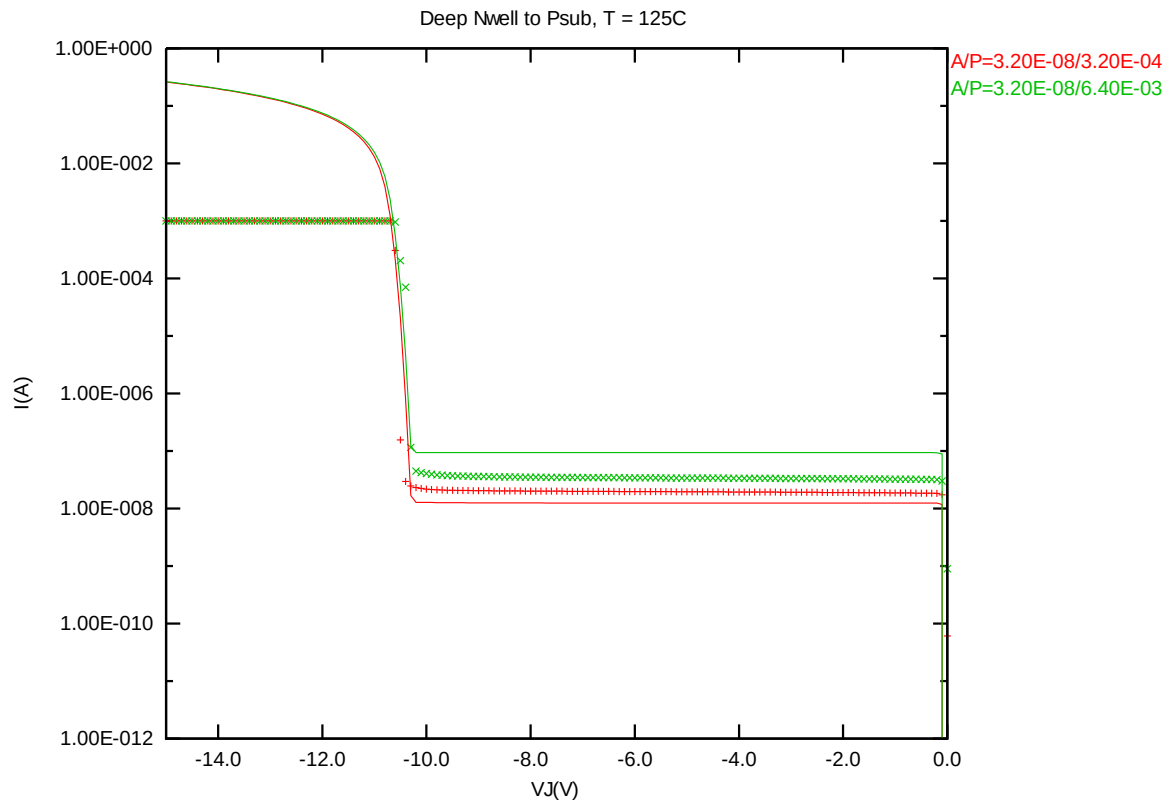


Figure 7-6 Reverse IV curve at 125 °C

7.3.3 DIODNW model verification at -50 °C

➤ *Diode junction CV curve at -50 °C.*

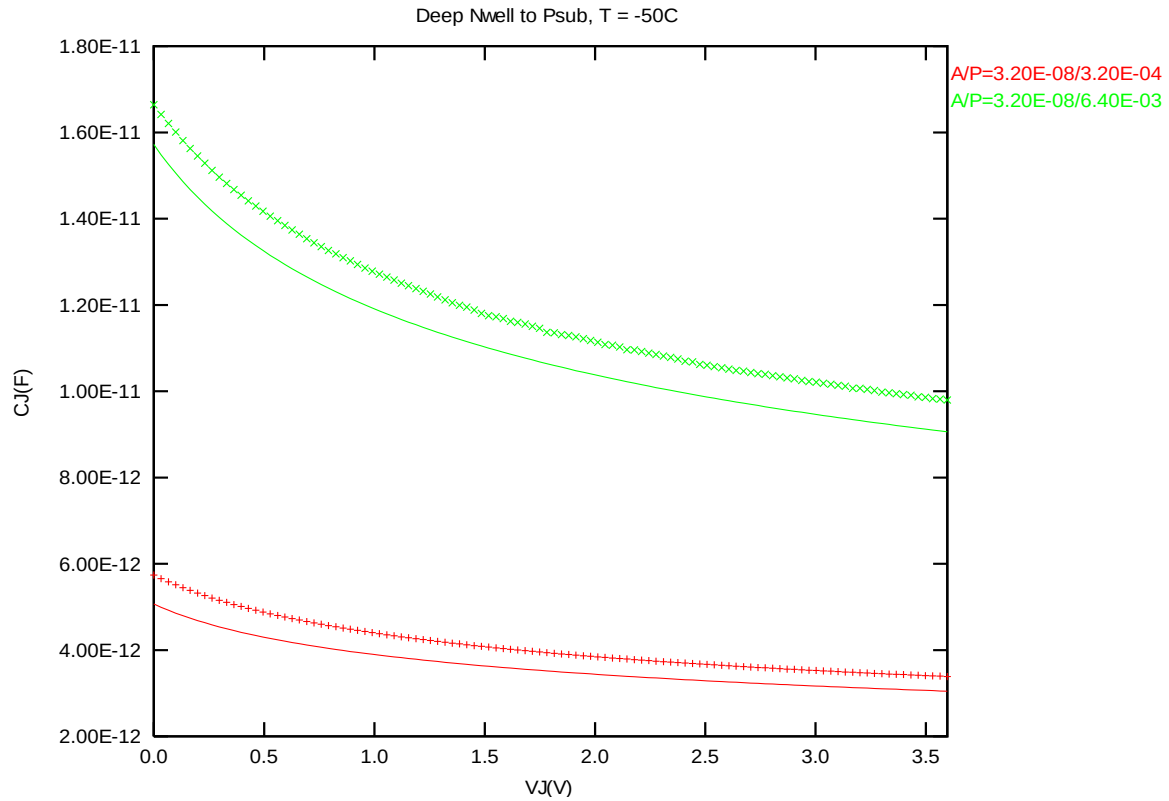


Figure 7-7 Diode junction CV curve at -50 °C

➤ **Forward IV curve at -50 °C.**

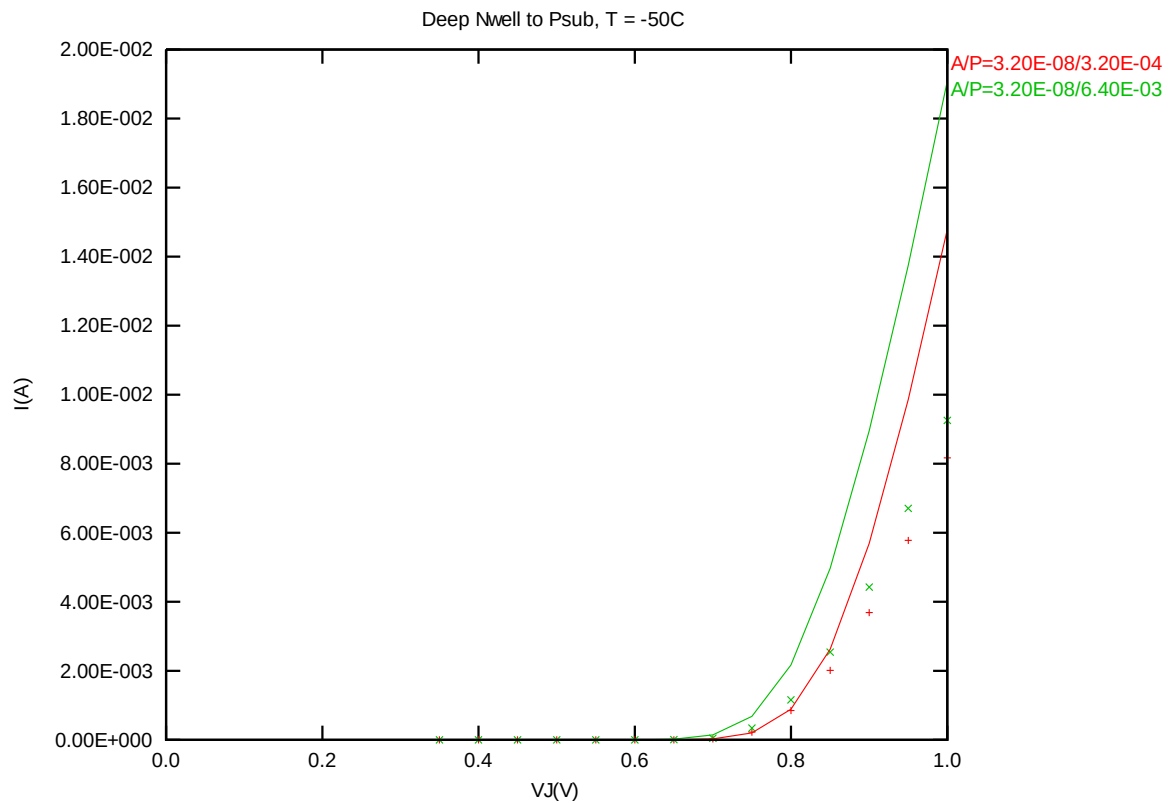
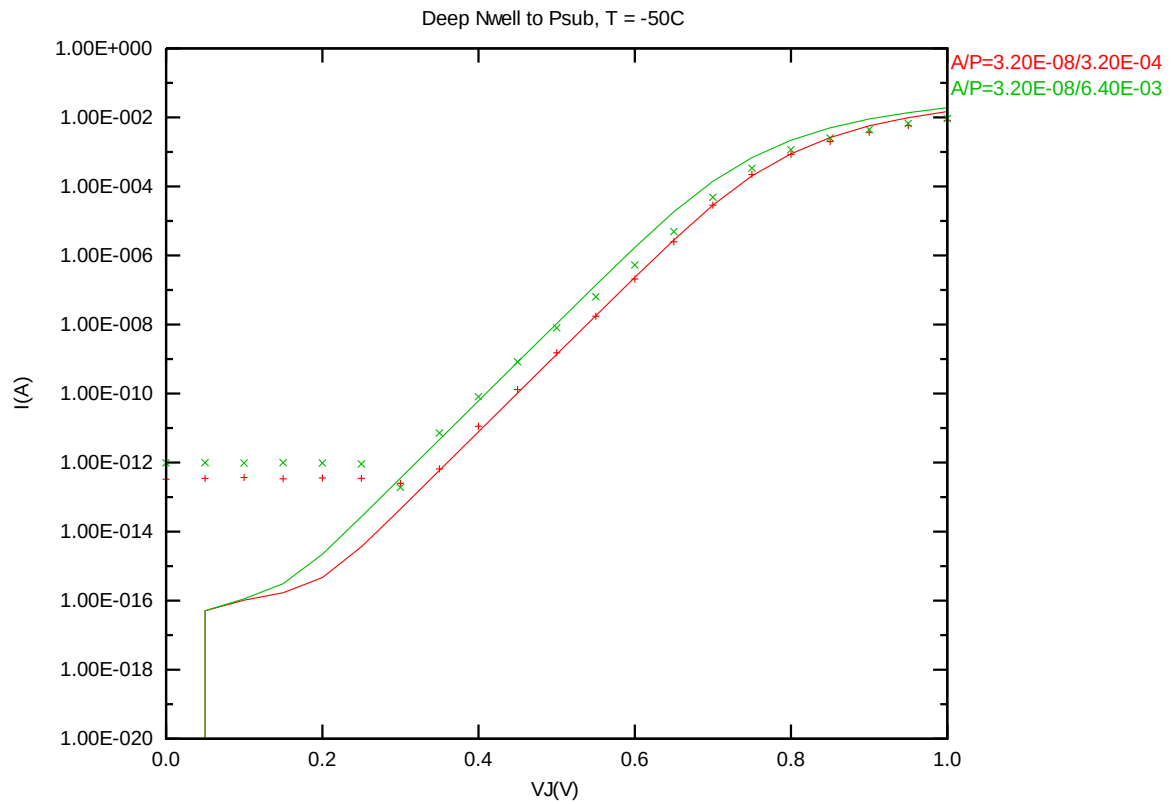


Figure 7-8 Forward IV curve at -50 °C

➤ **Reverse IV curve at -50 °C.**

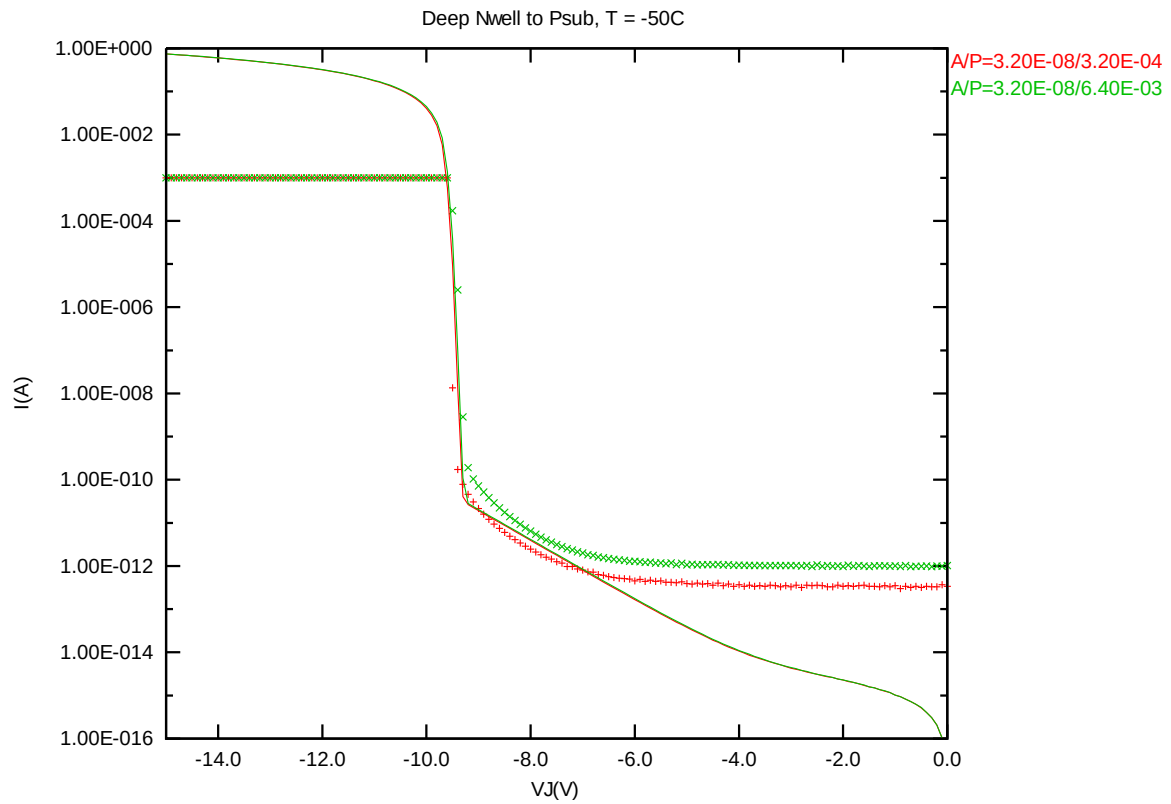


Figure 7-9 Reverse IV curve at -50 °C

7.3.4 DIODNW model verification for temperature effect

➤ *Diode junction CV curve at 25 °C, 125 °C, -50 °C.*

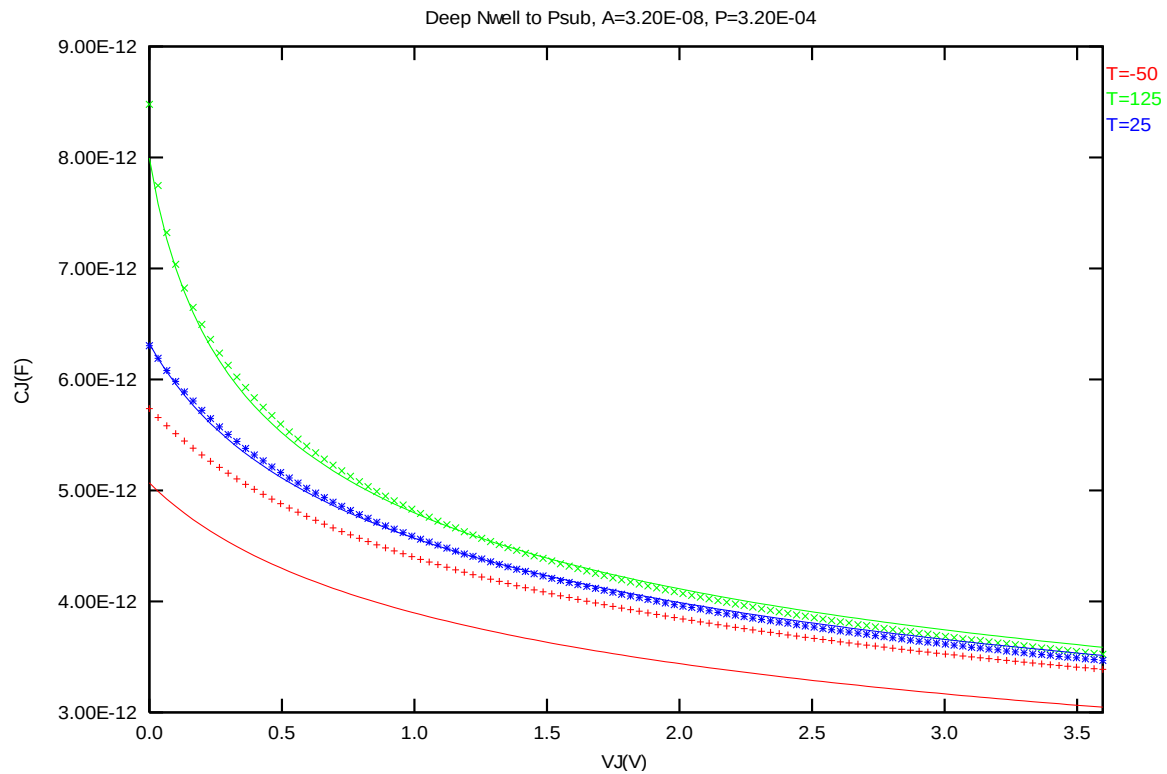


Figure 7-10 Diode junction CV curve at 25 °C, 125 °C, -50 °C.

➤ **Forward IV curve at 25 °C, 125 °C, -50 °C.**

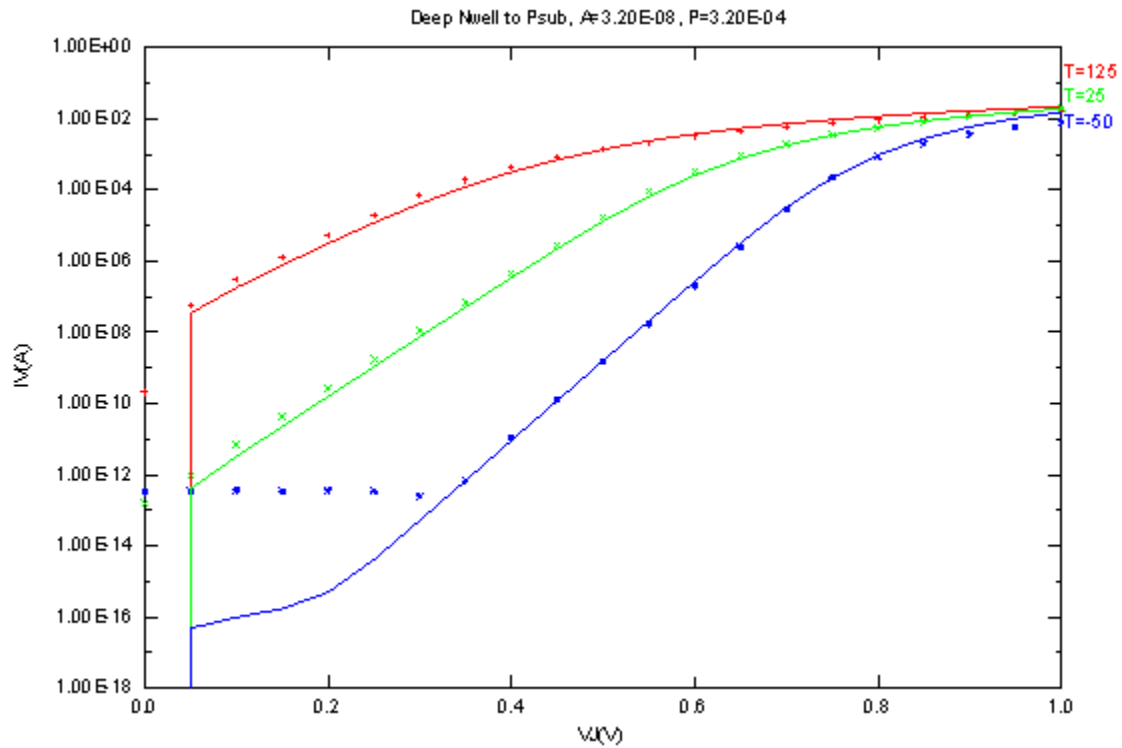


Figure 7-11 Forward IV curve at 25 °C, 125 °C, -50 °C

➤ **Reverse IV curve at 25 °C, 125 °C, -50 °C**

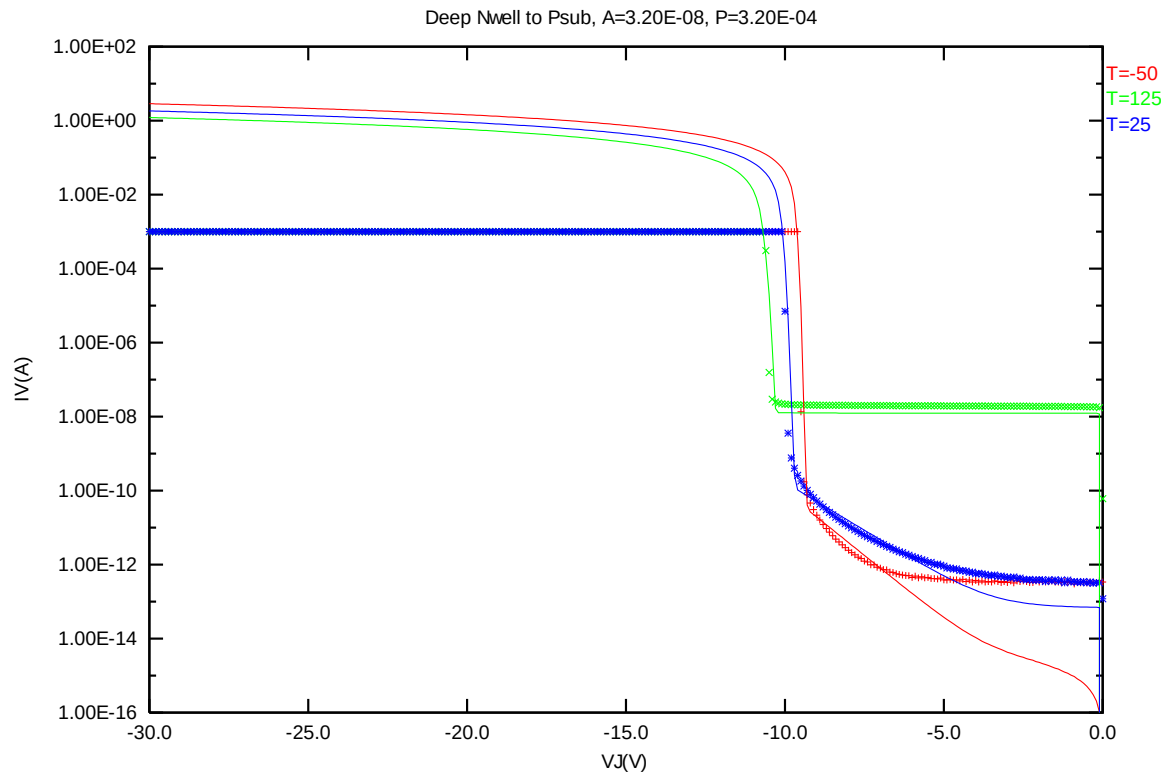


Figure 7-12 Reverse IV curve at 25 °C, 125 °C, -50 °C

8 DIOPW Diode Model Verification

8.1 Silicon Wafer for Modeling

The information of the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 8-1 Test wafer information for DIOPW

	Diopw diode model
Mask ID	BSW0FA
Lot ID	D566R.1
Wafer Number	#01

This model is characterized as the following geometry and temperature ranges depicted. Device behavior beyond these ranges may not be well described by the model. This model are extracted by devices with these two dimensions :

Junction Area Devices :

Area = $3.2 \times 10^{-8} \text{ m}^2$

Perimeter = $3.2 \times 10^{-4} \text{ m}$

8.2 Testing Conditions

The model is verified to measurement data under following bias conditions

Forward and reverse diode IV curve:

Table 8-2 Forward and reverse diode IV measurement conditions

Forward Diode

	Start	Stop	Step
$V_J(V)$	0	1	0.05

Reverse Diode

	Start	Stop	Step
$V_J(V)$	0	-15	-0.1

Diode junction CV:

Table 8-3 Diode junction CV measurement conditions

V_{BIAS} [V]	VH: psub & VL: nwell From 0 to 3.63 V in 0.033 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50, 25, 125

8.3 DIOPW model verification

8.3.1 DIOPW model verification at room temperature (25 °C)

➤ *Diode junction CV curve at 25 °C.*

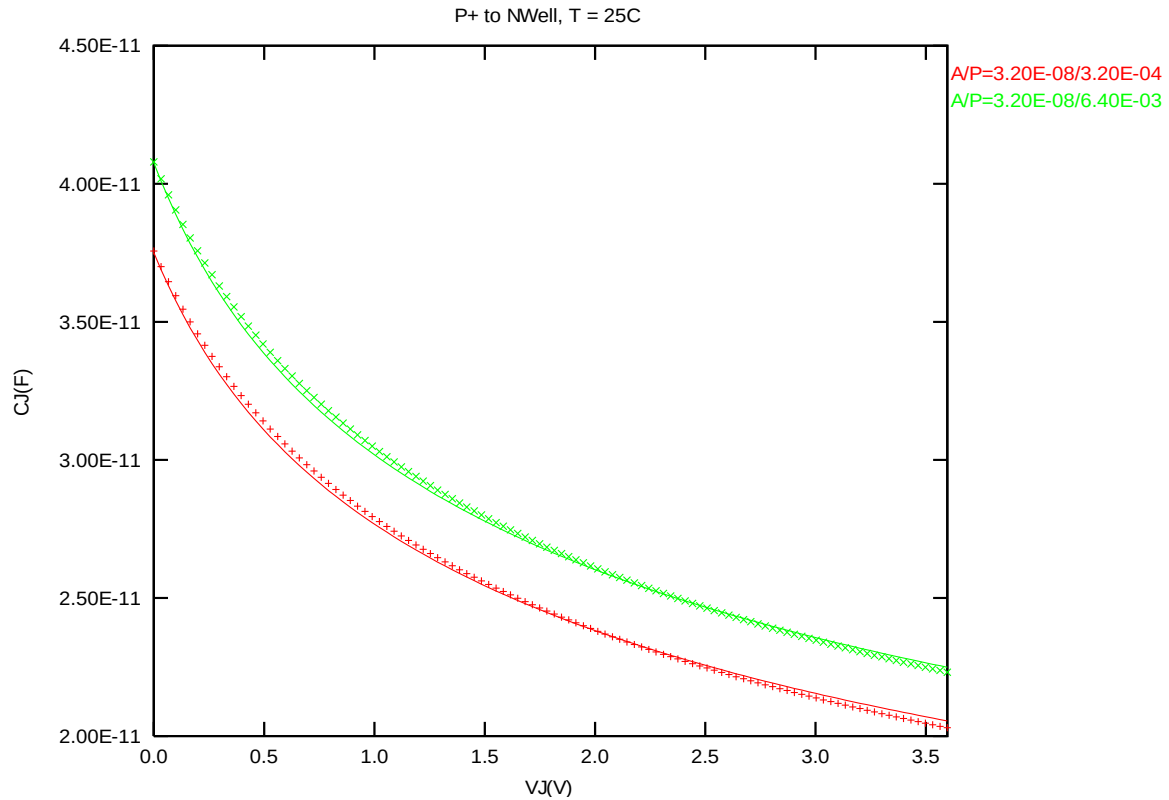


Figure 8-1 Diode junction CV curve at 25 °C

➤ **Forward IV curve at 25 °C.**

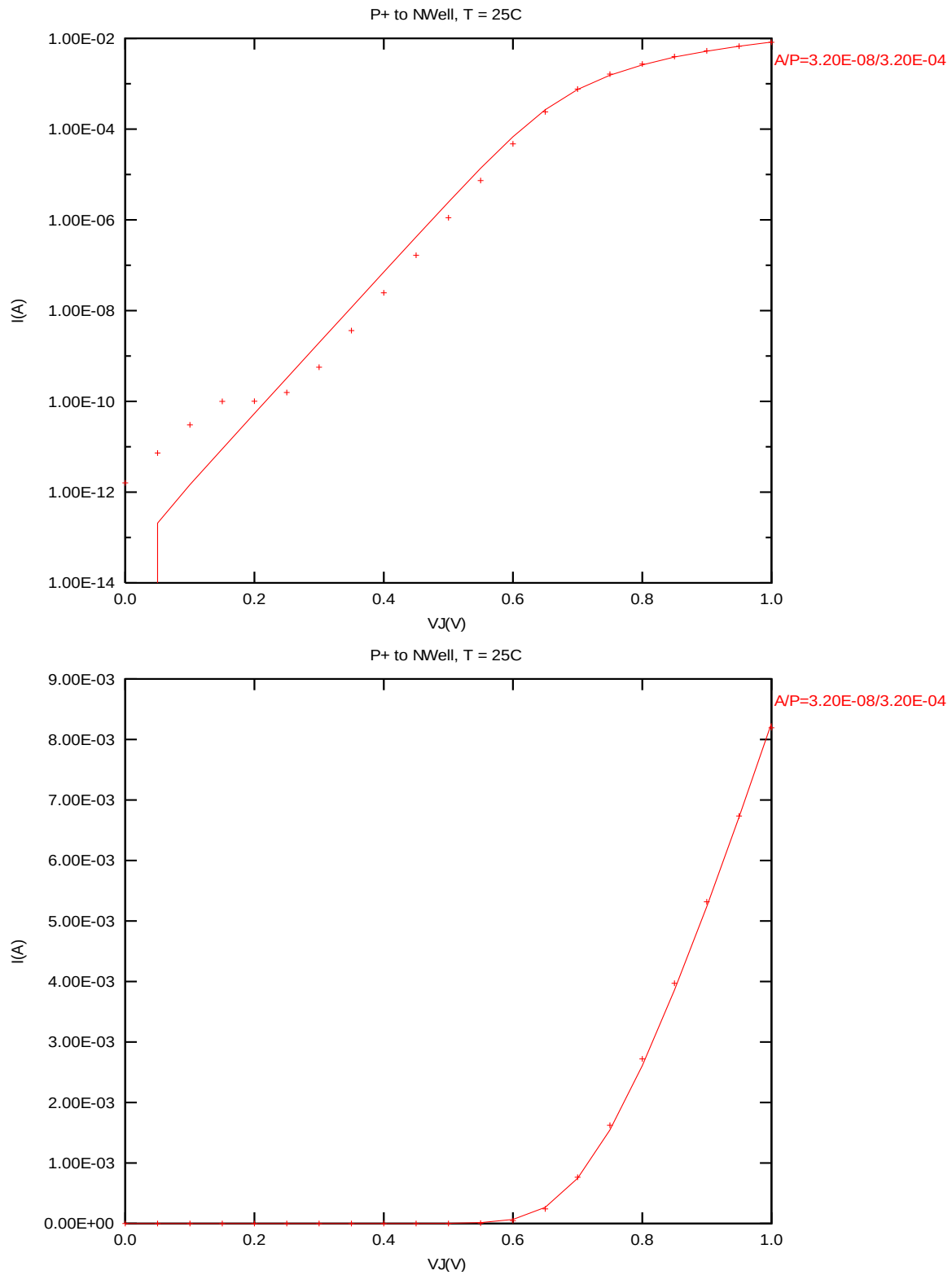


Figure 8-2 Forward IV curve at 25 °C

➤ **Reverse IV curve at 25 °C.**

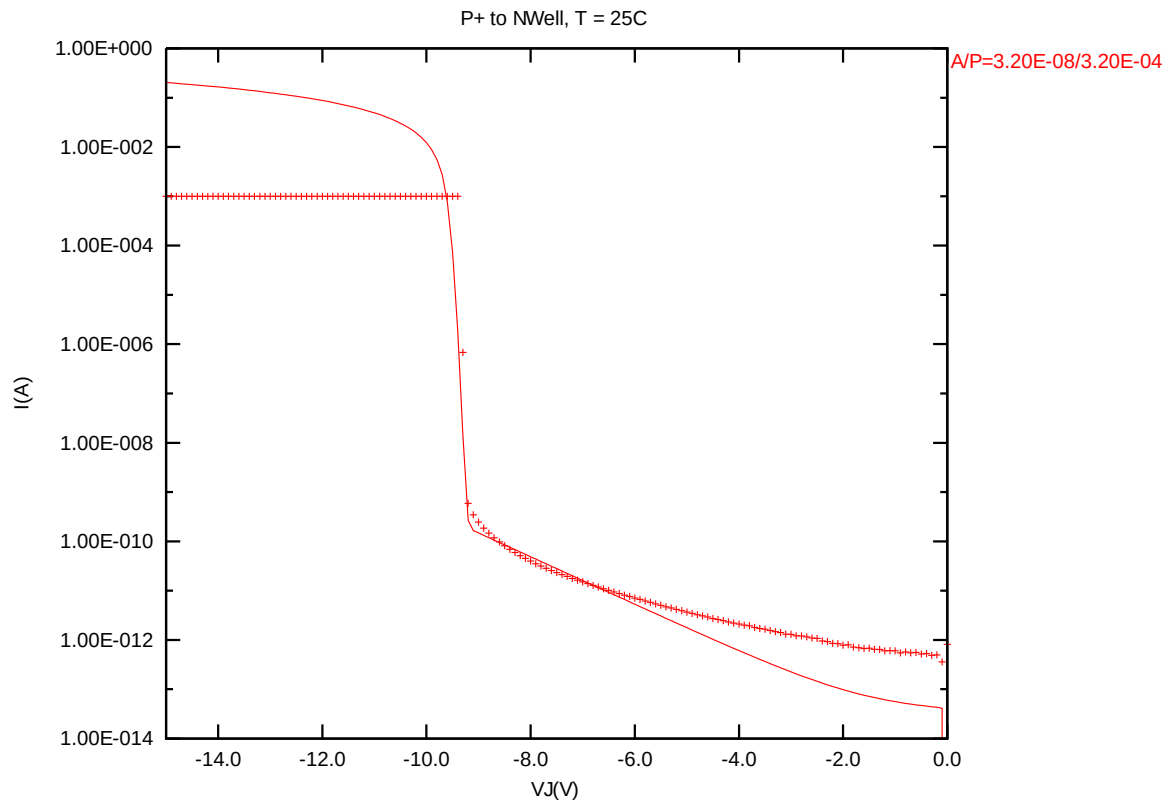


Figure 8-3 Reverse IV curve at 25 °C

8.3.2 DIOPW model verification at 125 °C

➤ *Diode junction CV curve at 125 °C.*

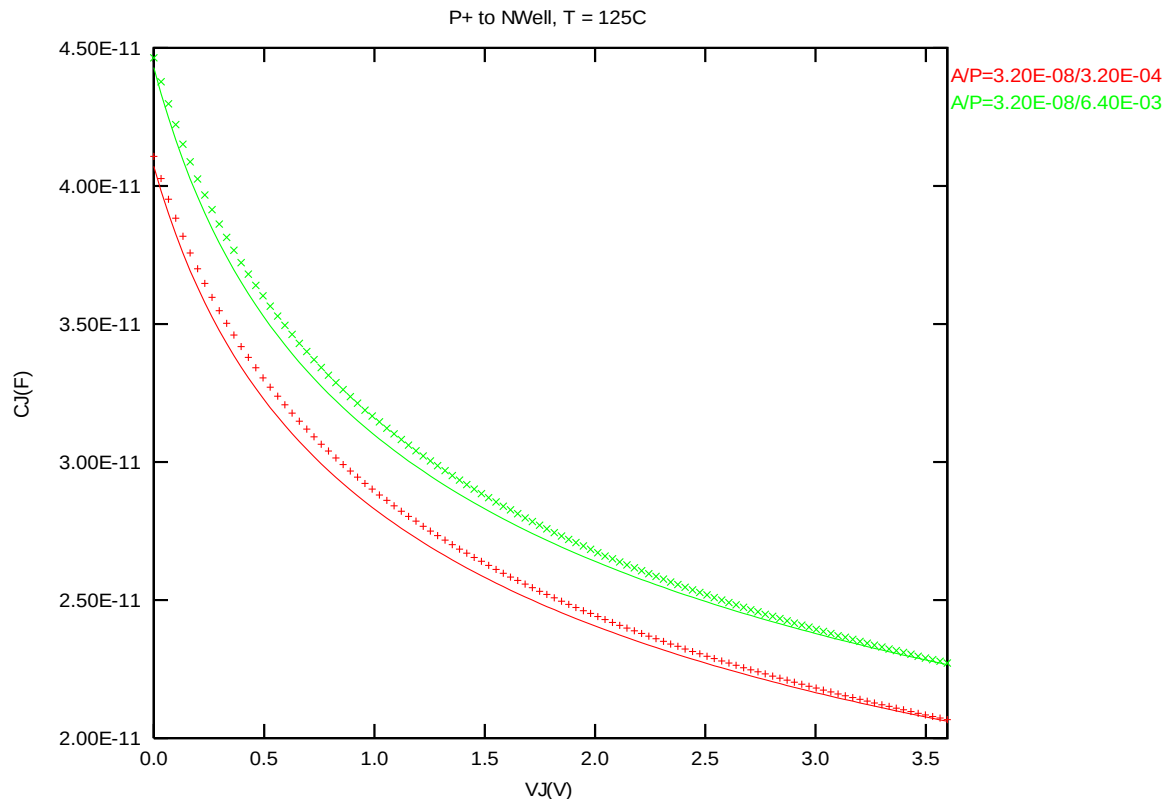


Figure 8-4 Diode junction CV curve at 125 °C

➤ **Forward IV curve at 125 °C.**

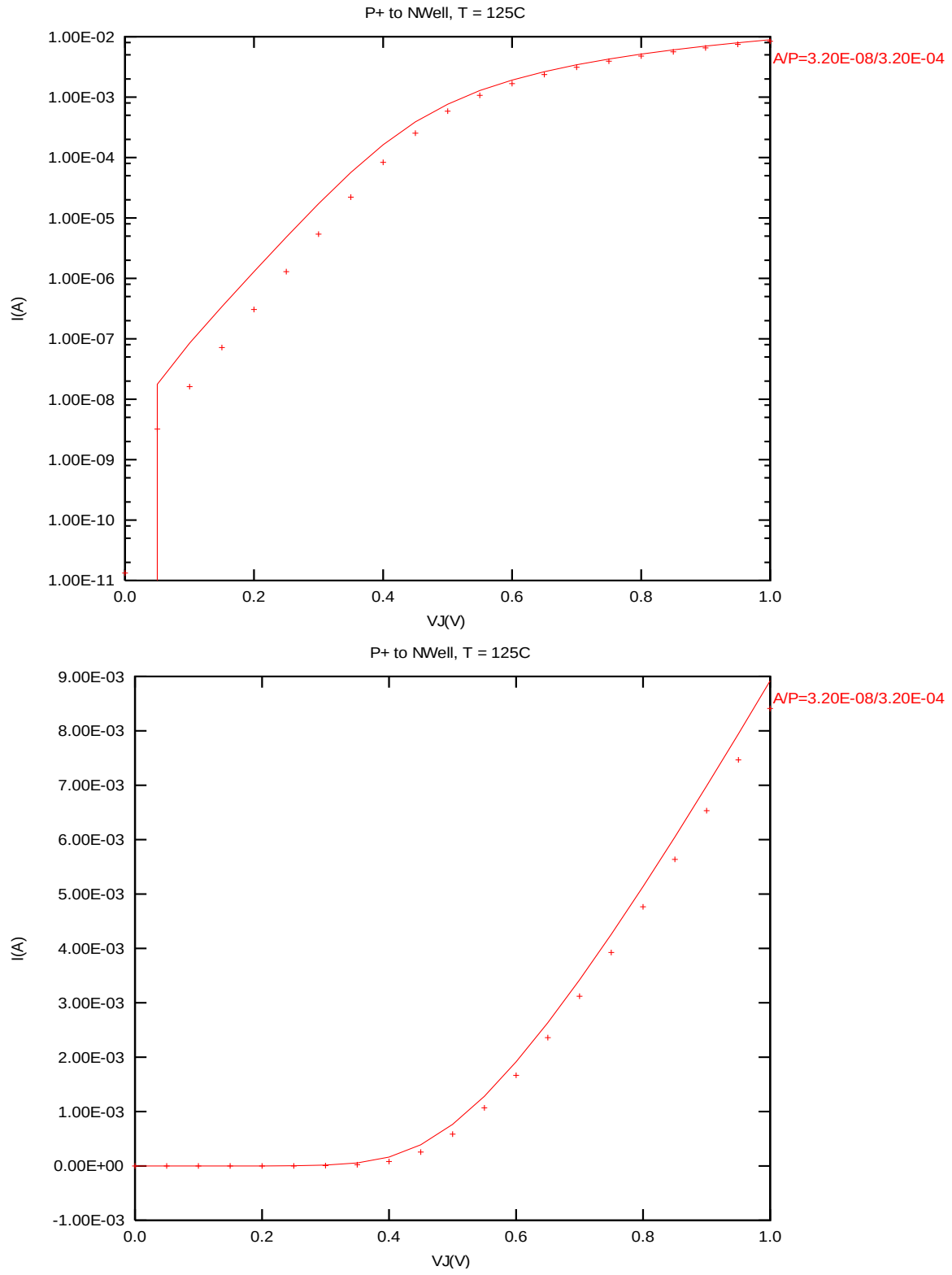


Figure 8-5 Forward IV curve at 125 °C

➤ **Reverse IV curve at 125 °C.**

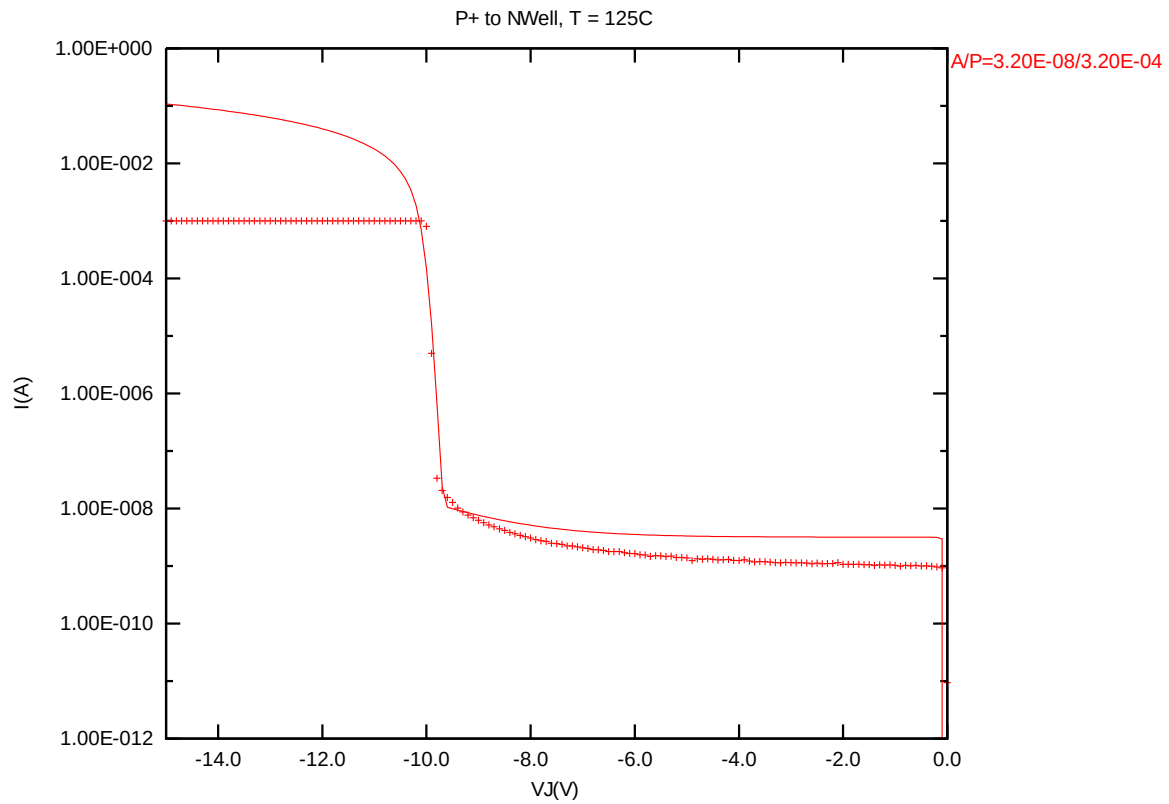


Figure 8-6 Reverse IV curve at 125 °C

8.3.3 DIOPW model verification at -50 °C

➤ *Diode junction CV curve at -50 °C.*

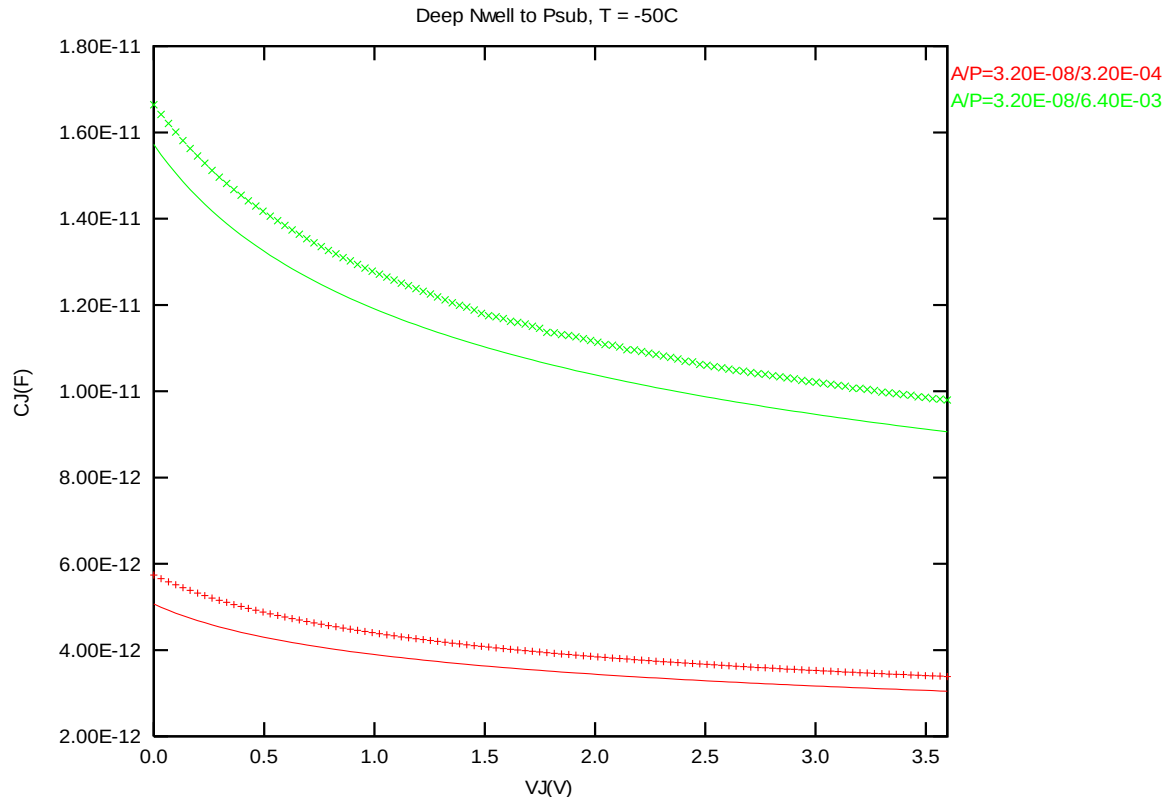
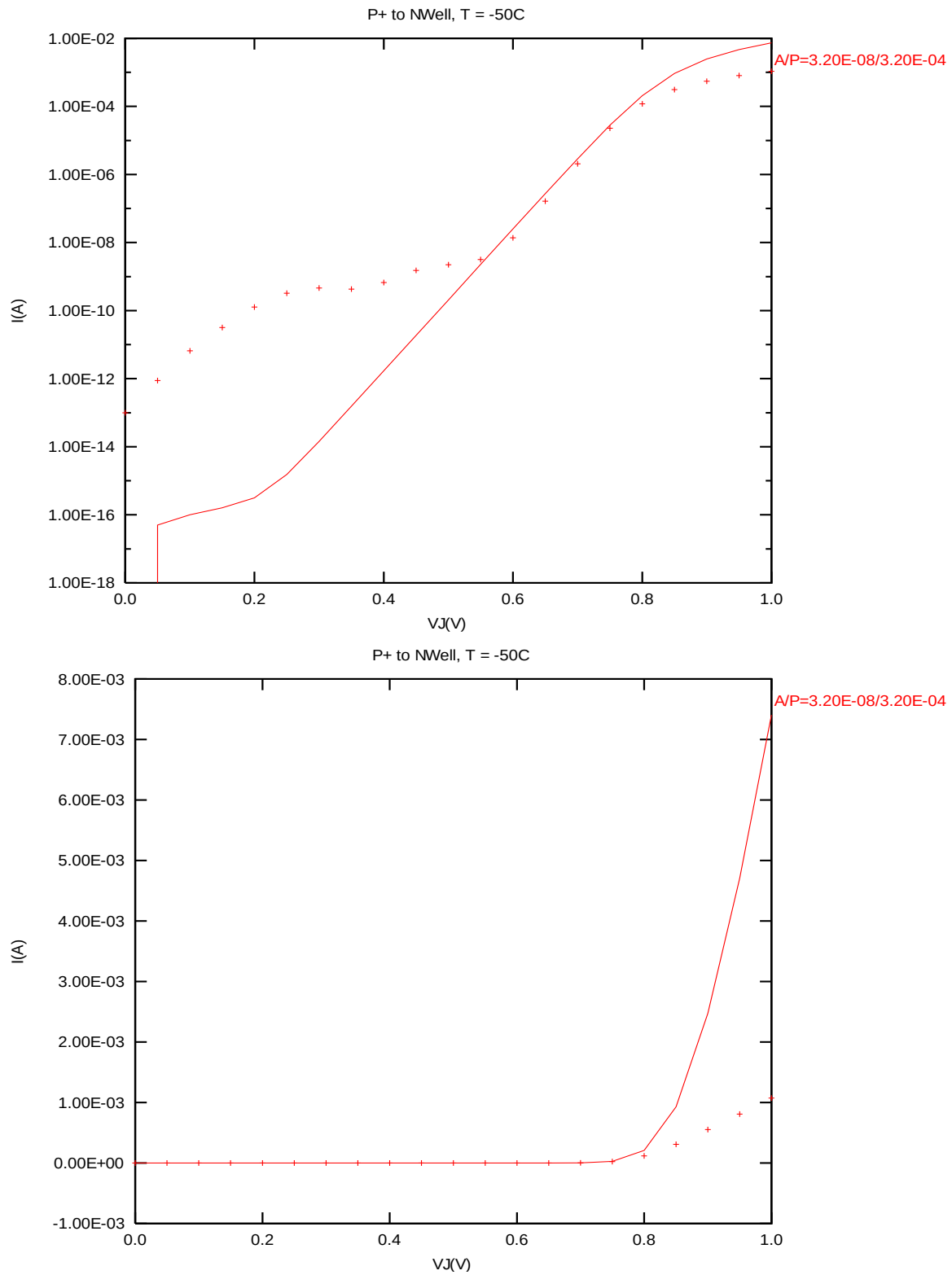


Figure 8-7 Diode junction CV curve at -50 °C

➤ **Forward IV curve at -50 °C.**



➤ **Reverse IV curve at -50 °C.**

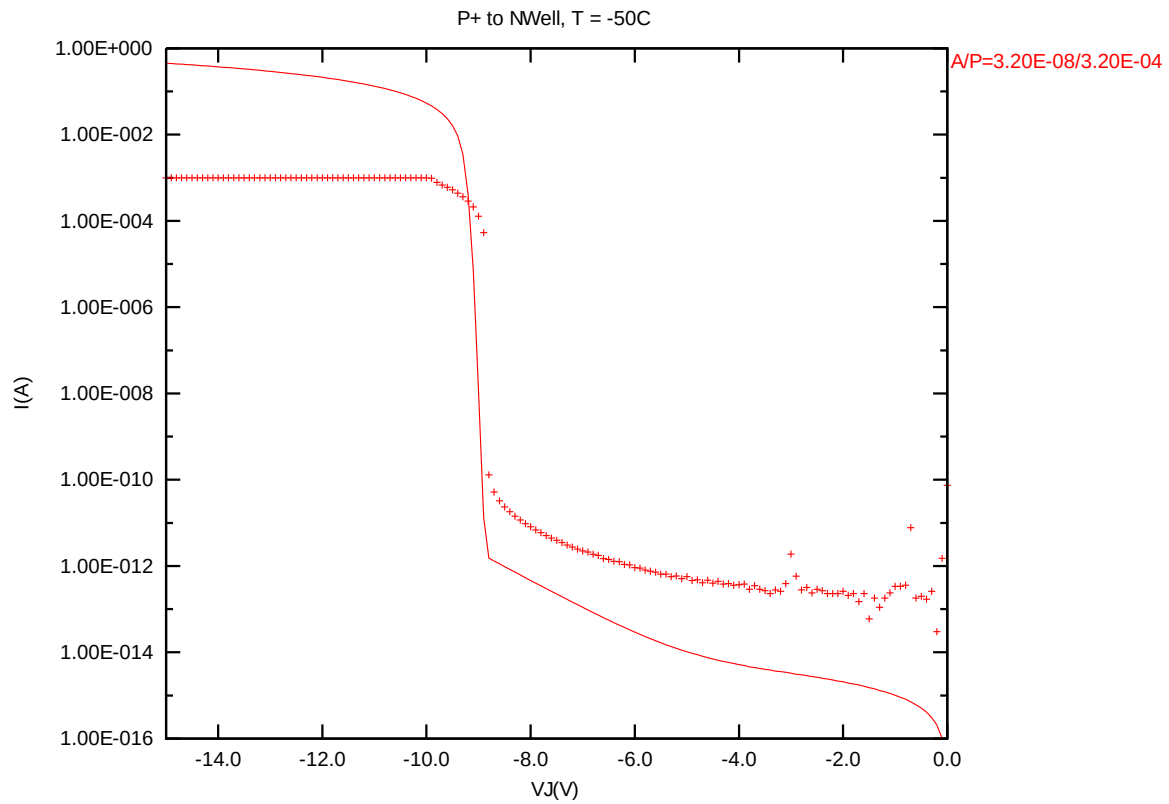


Figure 8-9 Reverse IV curve at -50 °C

8.3.4 DIOPW model verification for temperature effect

➤ *Diode junction CV curve at 25 °C, 125 °C, -50 °C.*

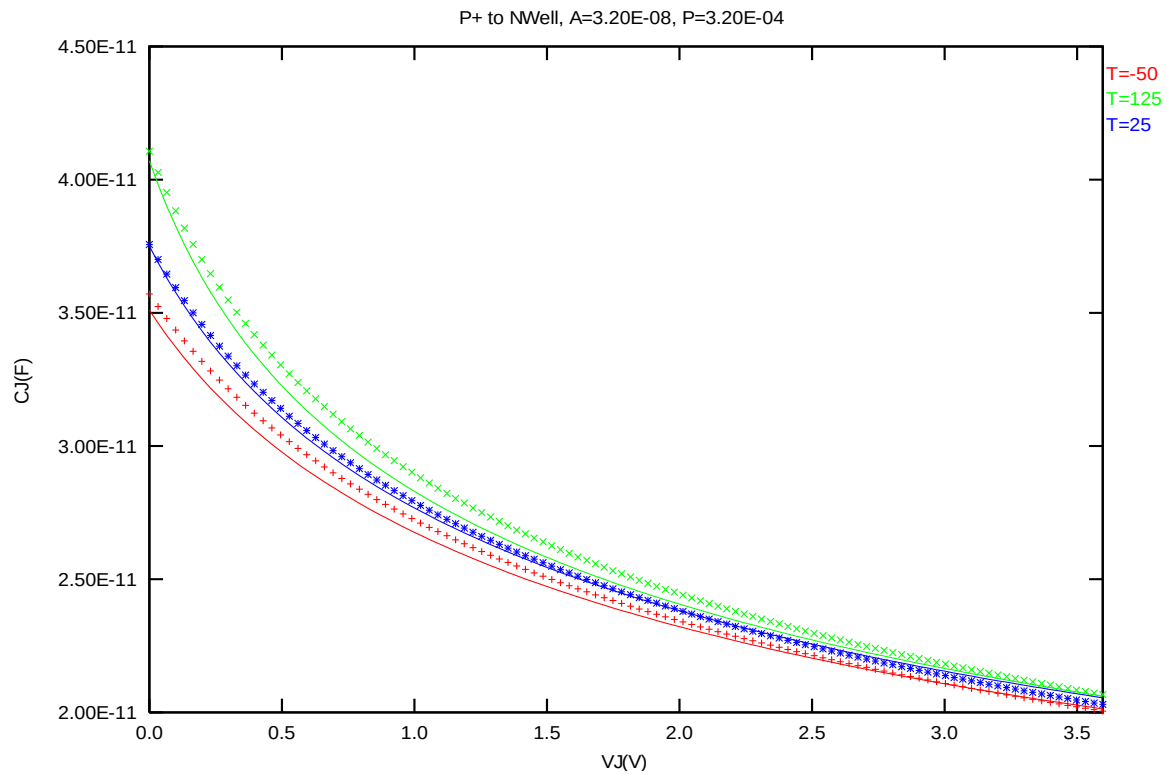


Figure 8-10 Diode junction CV curve at 25 °C, 125 °C, -50 °C.

➤ **Forward IV curve at 25 °C, 125 °C, -50 °C.**

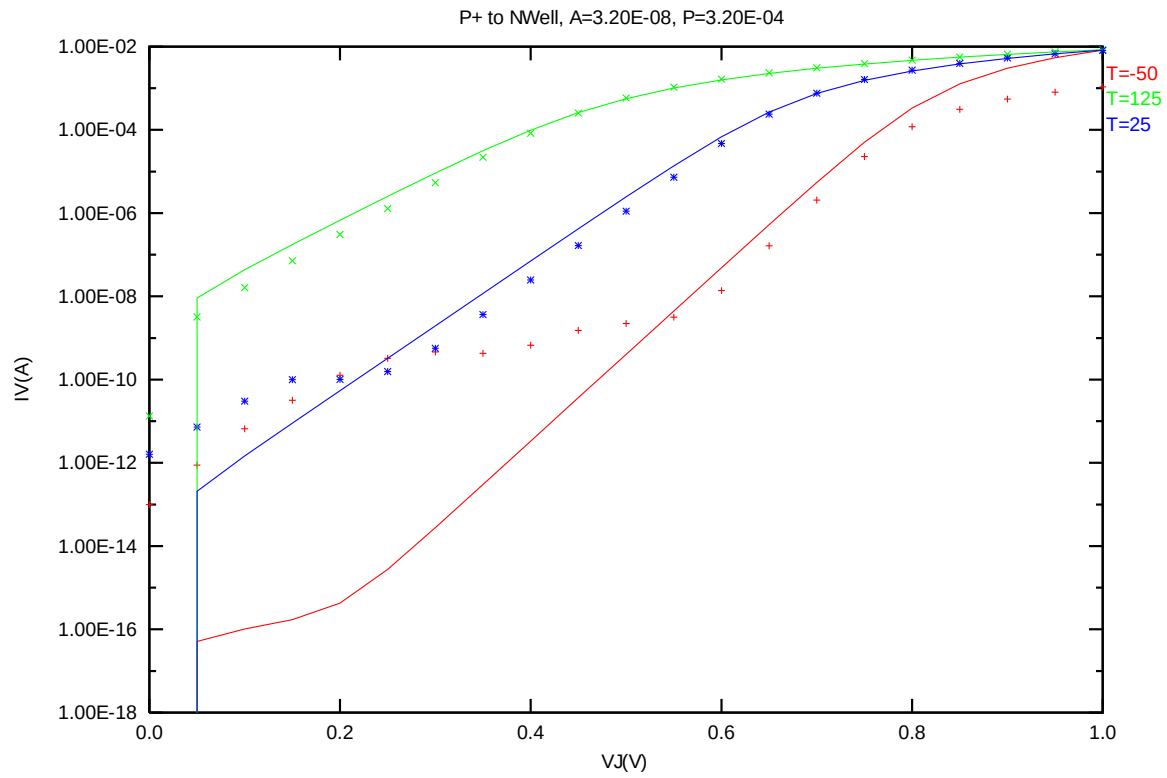


Figure 8-11 Forward IV curve at 25 °C, 125 °C, -50 °C

➤ **Reverse IV curve at 25 °C, 125 °C, -50 °C**

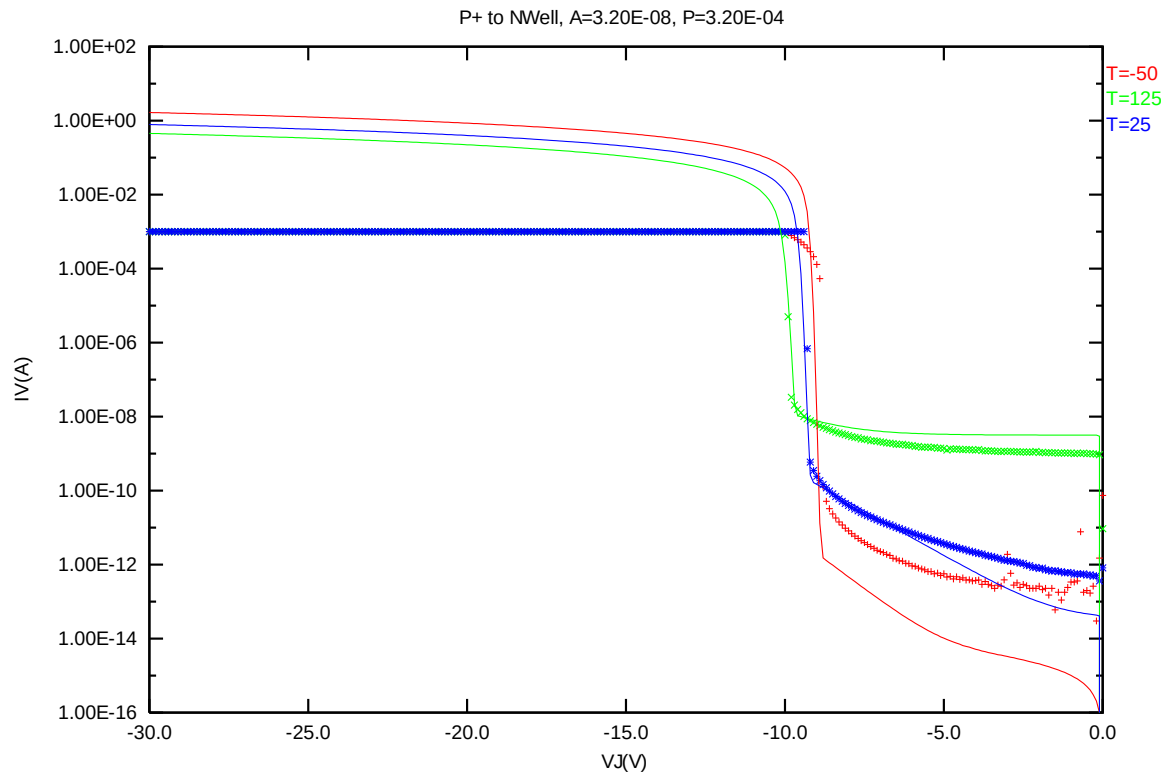


Figure 8-12 Reverse IV curve at 25 °C, 125 °C, -50 °C

9 Appendix

9.1 HSPICE Warning Message

Table 9-1 HSPICE warning message

Item	Warning Message	Explanation
1.	None	None

9.2 ELDO Warning Message

Table 9-2 Eldo warning message

Item	Warning Message	Explanation
1.	None	None

9.3 SPECTRE Warning Message

Table 9-3 Spectre warning message

Item	Warning Message	Explanation
1.	None	None